

1510.06699: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00001	1	0.755	$\{ \}^{\{1,2, \mathrm{a}\}}$	1,2,a	A.N. Lasenby ^{1, 2, a)} and M.P. Hobson ^{1, b)}
1510.06699_F00002	1	0.755	$\{ \}^{\{1, \mathrm{~b}\}}$	1, b	A.N. Lasenby ^{1, 2, a)} and M.P. Hobson ^{1, b)}
1510.06699_F00003	1	0.969	$\{ \}^{\{1\}}$	1)	¹⁾ <i>Astrophysics Group, Cavendish Laboratory, JJ Thomson Avenue, Cambridge CB3 0HE, UK</i>
1510.06699_F00004	1	0.979	$\{ \}^{\{2\}}$	2)	²⁾ <i>Kavli Institute for Cosmology, Madingley Road, Cambridge CB3 0HA, UK</i>
1510.06699_F00005	1	0.947	$\{ \}^{\{\underline{1}\}}$	<u>1</u>	Following an initial attempt by Utiyama ¹ , the idea
1510.06699_F00006	1	0.998	$\mathrm{\varphi}$	φ	(or fields) φ with energy-momentum and, in general,
1510.06699_F00007	1	0.798	$\{ \}^{\{\text{a}\}}$	a)	^{a)} Electronic mail: a.n.lasenby@mrao.cam.ac.uk
1510.06699_F00008	1	0.997	$\{ \}^{\{\text{b}\}}$	b)	^{b)} Electronic mail: mph@mrao.cam.ac.uk
1510.06699_F00009	1	1.000	$S_{\mathrm{M}}=$	$S_{\mathrm{M}}=$	field dynamics are described by a matter action $S_{\mathrm{M}}=$
1510.06699_F00010	1	1.000	$\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$	$\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$	$\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$ that is invariant under global Poincaré
1510.06699_F00011	1	1.000	h_{a}^{μ}	h_a^{μ}	new field variables h_a^{μ} and A_{μ}^{ab} , which are interpreted
1510.06699_F00012	1	1.000	$A^{\mathrm{a} \mathrm{b}}_{\mu}$	A_{μ}^{ab}	new field variables h_a^{μ} and A_{μ}^{ab} , which are interpreted
1510.06699_F00013	1	0.990	$\mathrm{D}_{\mathrm{a}} \varphi$	$\mathcal{D}_a \varphi$	a covariant derivative $\mathcal{D}_a \varphi$, and the matter action in the
1510.06699_F00014	1	1.000	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathrm{D}_{\mathrm{a}} \varphi) d^4 x$	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi) d^4 x$	to obtain $S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi) d^4 x$ (where the factor
1510.06699_F00015	1	1.000	$h \equiv \operatorname{det}(\mathrm{h}_{\mathrm{a}}^{\mu})$	$h \equiv \det(h_a^{\mu})$	containing $h \equiv \det(h_a^{\mu})$ is required to make the inte-
1510.06699_F00016	2	1.000	$\left[\mathrm{D}_{\mathrm{a}}, \mathrm{D}_{\mathrm{b}}\right] \varphi$	$[\mathcal{D}_a, \mathcal{D}_b] \varphi$	tering $[\mathcal{D}_a, \mathcal{D}_b] \varphi$. This procedure yields two field strength
1510.06699_F00017	2	1.000	$\mathcal{R}^{\mathrm{a} \mathrm{b}}_{\mathrm{cd}}(h, A, \partial A)$	$\mathcal{R}_{cd}^{ab}(h, A, \partial A)$	tensors: $\mathcal{R}_{cd}^{ab}(h, A, \partial A)$ corresponding to the gauge field
1510.06699_F00018	2	0.999	$\mathcal{T}^{\mathrm{a}}_{\mathrm{bc}}(h, \partial h, A)$	$\mathcal{T}_{bc}^a(h, \partial h, A)$	Poincaré group; and $\mathcal{T}_{bc}^a(h, \partial h, A)$ corresponding to
1510.06699_F00019	2	1.000	$S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}^{\mathrm{a} \mathrm{b}}_{\mathrm{cd}}, \mathcal{T}^{\mathrm{a}}_{\mathrm{bc}}\right) d^4 x$	$S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}_{cd}^{ab}, \mathcal{T}_{bc}^a\right) d^4 x$	$S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}_{cd}^{ab}, \mathcal{T}_{bc}^a\right) d^4 x$, where the Lagrangian
1510.06699_F00020	2	1.000	L_{G}	L_{G}	L_{G} is a scalar depending on the two field strengths.
1510.06699_F00021	2	1.000	$\tau^{\mathrm{k}}_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_{\mathrm{a}}^{\mu}$	$\tau^k_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_a^{\mu}$	energy-momentum $\tau^k_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_a^{\mu}$ and spin-angular-
1510.06699_F00022	2	1.000	$\sigma_{\mathrm{ab}}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{\mathrm{ab}}$	$\sigma_{ab}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{ab}$	momentum $\sigma_{ab}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{ab}$ of the matter field act
1510.06699_F00023	2	1.000	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathrm{D}_{\mathrm{a}} \varphi)$	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi)$	as sources, where $\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a \varphi)$. In contrast to
1510.06699_F00024	2	1.000	$\mathcal{R} \equiv \mathcal{R}^{\mathrm{ab}}_{\mathrm{ab}}$	$\mathcal{R} \equiv \mathcal{R}_{ab}^{ab}$	derivatives of the gauge fields, namely $\mathcal{R} \equiv \mathcal{R}_{ab}^{ab}$. In-
1510.06699_F00025	2	1.000	$L_{\mathrm{G}} \propto \mathcal{R}$	$L_{\mathrm{G}} \propto \mathcal{R}$	deed, Kibble chose simply to set $L_{\mathrm{G}} \propto \mathcal{R}$, which defines
1510.06699_F00026	2	0.733	$\{ \}^{\{\underline{5}\}}$	<u>5</u>	transformations has been criticised by Hehl et al. ⁵ , who
1510.06699_F00027	2	1.000	$\delta \varphi \equiv \varphi^{\prime}(x^{\prime})-\varphi(x)$	$\delta \varphi \equiv \varphi'(x') - \varphi(x)$	ble considered the ‘total’ variation $\delta \varphi \equiv \varphi'(x') - \varphi(x)$ of
1510.06699_F00028	2	1.000	$\delta_0 \varphi \equiv \varphi^{\prime}(x)-\varphi(x)$	$\delta_0 \varphi \equiv \varphi'(x) - \varphi(x)$	ation $\delta_0 \varphi \equiv \varphi'(x) - \varphi(x)$. The latter approach is closer
1510.06699_F00029	2	1.000	$\delta_0 \varphi$	$\delta_0 \varphi$	matter field transformation $\delta_0 \varphi$, the partial derivative in
1510.06699_F00030	2	0.988	$\{ \}^{\{6,9\}}$	6,9	theory in Minkowski spacetime ^{6,9} , in the same way as the
1510.06699_F00031	2	0.991	$\{ \}^{\{\underline{10}\}}$	<u>10</u>	see Blagojevic ¹⁰ , in which many of the issues above are
1510.06699_F00032	2	1.000	$\mathcal{R}^{\mathrm{ab}}_{\mathrm{cd}}$	$\mathcal{R}_{cd}^{ab}$	sors $\mathcal{R}_{cd}^{ab}$ and $\mathcal{T}_{bc}^a$, and their inclusion in L_{G} cannot
1510.06699_F00033	2	1.000	$\mathcal{T}^{\mathrm{a}}_{\mathrm{bc}}$	$\mathcal{T}_{bc}^a$	sors $\mathcal{R}_{cd}^{ab}$ and $\mathcal{T}_{bc}^a$, and their inclusion in L_{G} cannot
1510.06699_F00034	2	1.000	$\{ \}^{\{11,12\}}$	11,12	adding the linear Hilbert-Einstein term ^{11,12} . Hence, one
1510.06699_F00035	2	1.000	$\kappa=8 \pi G / c^4$	$\kappa=8 \pi G / c^4$	where $\kappa=8 \pi G / c^4$ is Einstein’s gravitational constant,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00036	2	■ 1.000	\Lambda	Λ	Λ is a cosmological constant, a is a dimensionless free
1510.06699_F00037	2	■ 1.000	a	a	Λ is a cosmological constant, a is a dimensionless free
1510.06699_F00038	3	■ 1.000	\alpha_{i}	α_i	in which the α_i and β_i are also dimensionless free parame-
1510.06699_F00039	3	■ 1.000	\beta_{i}	β_i	in which the α_i and β_i are also dimensionless free parame-
1510.06699_F00040	3	■ 0.362	{ }^{\underline{12,14}}-\underline{16}}	$\underline{12,14}-\underline{16}$	investigated previously ^{12,14–16} and may, in principle, be
1510.06699_F00041	3	■ 1.000	\alpha_{1}, \alpha_{3}	α_1, α_3	alised Gauss–Bonnet identity to set one of α_1 , α_3 or α_6
1510.06699_F00042	3	■ 1.000	\alpha_{6}	α_6	alised Gauss–Bonnet identity to set one of α_1 , α_3 or α_6
1510.06699_F00043	3	■ 0.999	c=\hbar=1	$c = \hbar = 1$	in the form (1) so that, in natural units ($c = \hbar = 1$,
1510.06699_F00044	3	■ 0.944	{ }^{-4}	-4	(length) ^{−4} , and hence the corresponding action S_G is di-
1510.06699_F00045	3	■ 0.944	S_{\mathrm{G}}	S_G	(length) ^{−4} , and hence the corresponding action S_G is di-
1510.06699_F00046	3	■ 1.000	\kappa	κ	dimensionless; note that in natural units κ has dimensions
1510.06699_F00047	3	■ 0.986	{ }^{-2}	-2	of (length) ² and Λ has dimensions of (length) ^{−2} .
1510.06699_F00048	3	■ 0.585	{ }^{19,20}	19,20	suffer from Ostrogradsky’s instability ^{19,20} , wherein the
1510.06699_F00049	3	■ 1.000	b^a{}_a{}_{\mu}	$b^a{}_{\mu}$	$b^a{}_{\mu}$ of the inverse h -field) and $A^{ab}{}_{\mu}$, the ‘curvature’ and
1510.06699_F00050	3	■ 1.000	h	h	$b^a{}_{\mu}$ of the inverse h -field) and $A^{ab}{}_{\mu}$, the ‘curvature’ and
1510.06699_F00051	3	■ 1.000	b^a{}_0	$b^a{}_0$	derivatives of $b^a{}_0$ and $A^{ab}{}_0$; thus 10 of the field equa-
1510.06699_F00052	3	■ 1.000	A^a{}_b{}_0	$A^{ab}{}_0$	derivatives of $b^a{}_0$ and $A^{ab}{}_0$; thus 10 of the field equa-
1510.06699_F00053	3	■ 1.000	b^a{}_{\alpha}	$b^a{}_{\alpha}$	remaining 30 variables $b^a{}_{\alpha}$ and $A^{ab}{}_{\alpha}$ ($\alpha = 1, 2, 3$), one
1510.06699_F00054	3	■ 1.000	A^{ab}{}_{\alpha}(\alpha=1,2,3)	$A^{ab}{}_{\alpha}(\alpha = 1, 2, 3)$	remaining 30 variables $b^a{}_{\alpha}$ and $A^{ab}{}_{\alpha}$ ($\alpha = 1, 2, 3$), one
1510.06699_F00055	3	■ 1.000	\mathcal{R}+\mathcal{T}^2	$\mathcal{R} + \mathcal{T}^2$	so-called $\mathcal{R} + \mathcal{T}^2$ theory, for which $L_G = \kappa^{-1}(a\mathcal{R} + L_{\mathcal{T}^2})$.
1510.06699_F00056	3	■ 1.000	L_{\mathrm{G}}=\kappa^{-1}\left(a\mathcal{R}+L_{\mathcal{T}^2}\right)	$L_G = \kappa^{-1}(a\mathcal{R} + L_{\mathcal{T}^2})$	so-called $\mathcal{R} + \mathcal{T}^2$ theory, for which $L_G = \kappa^{-1}(a\mathcal{R} + L_{\mathcal{T}^2})$.
1510.06699_F00057	3	■ 1.000	A	A	namics imposes an algebraic relation between the A -field
1510.06699_F00058	3	■ 1.000	L_{\mathcal{R}^2}	$L_{\mathcal{R}^2}$	in $L_{\mathcal{R}^2}$, in which case the ‘torsion’ becomes a propagating
1510.06699_F00059	3	■ 1.000	\mathcal{R}+\mathcal{R}^2	$\mathcal{R} + \mathcal{R}^2$	field. Consequently, a number of $\mathcal{R} + \mathcal{R}^2$ -type theories
1510.06699_F00060	3	■ 0.970	\mathcal{R}^2+\mathcal{T}^2	$\mathcal{R}^2 + \mathcal{T}^2$	have been investigated, as have $\mathcal{R}^2 + \mathcal{T}^2$ theories ²¹ . It
1510.06699_F00061	3	■ 0.986	\mathcal{R}+\mathcal{R}^2+\mathcal{T}^2	$\mathcal{R} + \mathcal{R}^2 + \mathcal{T}^2$	$\mathcal{R} + \mathcal{R}^2 + \mathcal{T}^2$ theories is given by Obukhov et al. ¹² . The
1510.06699_F00062	3	■ 1.000	\mathcal{R}	$\mathcal{R}$	if L_G consists of $\mathcal{R}$ and $\mathcal{T}^2$ terms, then it contains dimen-
1510.06699_F00063	3	■ 1.000	\mathcal{T}^2	$\mathcal{T}^2$	if L_G consists of $\mathcal{R}$ and $\mathcal{T}^2$ terms, then it contains dimen-
1510.06699_F00064	3	■ 1.000	\mathcal{R}^2	$\mathcal{R}^2$	quantisation. On the other hand, the $\mathcal{R}^2$ terms contain
1510.06699_F00065	3	■ 1.000	{ }^{\underline{25}}	$\underline{25}$	can eliminate such higher derivatives from the theory ²⁵ ,
1510.06699_F00066	3	■ 0.881	{ }^{\underline{26}}	$\underline{26}$	well-defined initial value problem ²⁶ . Nonetheless, the
1510.06699_F00067	3	■ 0.995	a, \alpha_{i}	a, α_i	that the parameters a , α_i and β_i in the Lagrangian L_G
1510.06699_F00068	3	■ 0.924	{ }^{\underline{27}}	$\underline{27}$	obey certain conditions ²⁷ .
1510.06699_F00069	4	■ 1.000	h_a{}^{\mu}	$h_a{}^{\mu}$	fields $h_a{}^{\mu}$ and $A^{ab}{}_{\mu}$ about the trivial vacuum solution to
1510.06699_F00070	4	■ 1.000	L_{\mathrm{G}}^{\left(2\right)}	$L_G^{(2)}$	fields to obtain $L_G^{(2)}$. After transforming to momentum
1510.06699_F00071	4	■ 1.000	k	k	structed using k -space projection operators ²⁸ , which de-
1510.06699_F00072	4	■ 1.000	J^P	J^P	duce to states of definite spin-parity J^P in a ‘rest-frame’
1510.06699_F00073	4	■ 1.000	k^a	k^a	where k^a has only a time component, and are thus ir-
1510.06699_F00074	4	■ 0.998	J^P=2^{+}	$J^P = 2^+$	ton (with spin-parity $J^P = 2^+$) and the decomposition
1510.06699_F00075	4	■ 1.000	J^P=2^{\pm}, 1^{\pm}, 0^{\pm}	$J^P = 2^{\pm}, 1^{\pm}, 0^{\pm}$	(or <i>rotons</i>) with spin-parities of $J^P = 2^{\pm}, 1^{\pm}, 0^{\pm}$, with
1510.06699_F00076	4	■ 1.000	2(5+3+1)=18	$2(5 + 3 + 1) = 18$	$\underline{2(5 + 3 + 1) = 18}$ degrees of freedom. The propagator is
1510.06699_F00077	4	■ 1.000	S	S	field energy, which destroys the unitarity of the S matrix)
1510.06699_F00078	4	■ 1.000	k^2	k^2	k^2 , whereas the condition for no ghosts is that the ma-
1510.06699_F00079	4	■ 1.000	-R /\left(k^2-m^2\right)	$-R/(k^2 - m^2)$	tor must be of the form $-R/(k^2 - m^2)$ with $R > 0$ and
1510.06699_F00080	4	■ 1.000	R>0	$R > 0$	tor must be of the form $-R/(k^2 - m^2)$ with $R > 0$ and
1510.06699_F00081	4	■ 1.000	m^2\geq 0	$m^2 \geq 0$	$m^2 \geq 0$ (assuming a ‘mostly minus’ metric signature).
1510.06699_F00082	4	■ 1.000	{ }^{\underline{18,33}}	18,33	ativity). In this case, the results are not so complete ^{18,33} .
1510.06699_F00083	4	■ 1.000	J^P=1^{-}	$J^P = 1^-$	that propagates (only) a massless $J^P = 1^-$ torsion and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00084	4	■ 1.000	$\sim 1 / k^4$	$\sim 1/k^4$	as $\sim 1/k^4$, whereas those for the tordions should be
1510.06699_F00085	4	■ 1.000	$\sim 1 / k^2$	$\sim 1/k^2$	$\sim 1/k^2$, since torsion vertices are better behaved than
1510.06699_F00086	4	■ 1.000	$\{ \}^{29,31,35}$	29,31,35	those of gravitons ^{29,31,35} . Pure $1/k^4$ poles can occur in
1510.06699_F00087	4	■ 1.000	$1 / k^4$	$1/k^4$	those of gravitons ^{29,31,35} . Pure $1/k^4$ poles can occur in
1510.06699_F00088	4	■ 1.000	$\left(k^2-m_1^2\right)^{-1}\left(k^2-m_2^2\right)^{-1}=\left(k^2-m_1^2\right)^{-1}\left(k^2-m_2^2\right)^{-1}$	$\left(k^2-m_1^2\right)^{-1}\left(k^2-m_2^2\right)^{-1}=\left(k^2-m_1^2\right)^{-1}\left(k^2-m_2^2\right)^{-1}$	pole structures of the form $\left(k^2-m_1^2\right)^{-1}\left(k^2-m_2^2\right)^{-1}=\left(k^2-m_1^2\right)^{-1}\left(k^2-m_2^2\right)^{-1}$
1510.06699_F00089	4	■ 1.000	$\left(m_2^2-m_1^2\right)^{-1}\left[\left(k^2-m_2^2\right)^{-1}-\left(k^2-m_1^2\right)^{-1}\right]$	$\left(m_2^2-m_1^2\right)^{-1}\left[\left(k^2-m_2^2\right)^{-1}-\left(k^2-m_1^2\right)^{-1}\right]$	$\left(m_2^2-m_1^2\right)^{-1}\left[\left(k^2-m_2^2\right)^{-1}-\left(k^2-m_1^2\right)^{-1}\right]$. These can
1510.06699_F00090	4	■ 1.000	$\{ \}^{6,39}$	6,39	have been investigated ^{6,39} . More innovative approaches
1510.06699_F00091	4	■ 0.945	$\{ \}^{40,41}$	40,41	to quantisation of PGT have been pursued ^{40,41} , but are
1510.06699_F00092	4	■ 0.991	$\{ \}^{\underline{31,43}}$	<u>31,43</u>	tary and renormalizable theory ^{31,43} , without having to
1510.06699_F00093	4	■ 0.566	$\{ \}^{44-46}$	44–46	resort to non-local theories ^{44–46} . In particular, perhaps
1510.06699_F00094	4	■ 1.000	$m^2 \neq 0$	$m^2 \neq 0$	either continuous (if $m^2 \neq 0$) or all the masses vanish.
1510.06699_F00095	4	■ 0.838	$\mathrm{C}(1,3)^{\underline{48-50}}$	$\mathrm{C}(1,3)^{48-50}$	of the full conformal group $\mathrm{C}(1,3)^{48-50}$. This may be for-
1510.06699_F00096	5	■ 1.000	$S_{\mathrm{M}}=\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$	$S_{\mathrm{M}}=\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$	ter action $S_{\mathrm{M}}=\int L_{\mathrm{M}}(\varphi, \partial_{\mu} \varphi) d^4 x$ that is invariant under
1510.06699_F00097	5	■ 1.000	B_{μ}	B_{μ}	a new gravitational gauge field B_{μ} corresponding to the
1510.06699_F00098	5	■ 0.870	$\{ \}^{51-53}$	51–53	dilation part of the Weyl group ^{51–53} . Similar to PGT, the
1510.06699_F00099	5	■ 1.000	h_a^{μ}, A_{μ}^{ab}	h_a^{μ}, A_{μ}^{ab}	gravitational gauge fields h_a^{μ}, A_{μ}^{ab} and B_{μ} are used to
1510.06699_F00100	5	■ 0.991	$\mathcal{D}_a^* \varphi$	$\mathcal{D}_a^* \varphi$	assemble a covariant derivative $\mathcal{D}_a^* \varphi$, and the matter ac-
1510.06699_F00101	5	■ 1.000	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi) d^4 x$	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi) d^4 x$	coupling procedure, so that $S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi) d^4 x$.
1510.06699_F00102	5	■ 1.000	L_{M}	L_{M}	cise form of L_{M} . In particular, since h^{-1} has a Weyl
1510.06699_F00103	5	■ 1.000	h^{-1}	h^{-1}	cise form of L_{M} . In particular, since h^{-1} has a Weyl
1510.06699_F00104	5	■ 0.788	$w\left(h^{-1}\right)=4$	$w\left(h^{-1}\right)=4$	(or conformal) weight $w\left(h^{-1}\right)=4$ (see Section II B), the
1510.06699_F00105	5	■ 1.000	$w\left(L_{\mathrm{G}}\right)=-4$	$w\left(L_{\mathrm{G}}\right)=-4$	Lagrangian L_{M} must have a weight $w\left(L_{\mathrm{G}}\right)=-4$.
1510.06699_F00106	5	■ 1.000	$\left[\mathcal{D}_a^*, \mathcal{D}_b^*\right] \varphi$	$\left[\mathcal{D}_a^*, \mathcal{D}_b^*\right] \varphi$	by considering $\left[\mathcal{D}_a^*, \mathcal{D}_b^*\right] \varphi$. One obtains $\mathcal{R}_{cd}^{ab}(h, A, \partial A)$
1510.06699_F00107	5	■ 1.000	$\mathcal{T}_{bc}^{* a}(h, \partial h, A, B)$	$\mathcal{T}_{bc}^{* a}(h, \partial h, A, B)$	and $\mathcal{T}_{bc}^{* a}(h, \partial h, A, B)$ as before (but where the form
1510.06699_F00108	5	■ 1.000	$\mathcal{H}_{ab}(h, \partial B)$	$\mathcal{H}_{ab}(h, \partial B)$	tional field strength $\mathcal{H}_{ab}(h, \partial B)$ corresponding to the new
1510.06699_F00109	5	■ 1.000	$S_{\mathrm{G}}=$	$S_{\mathrm{G}}=$	free gravitational action then has the general form $S_{\mathrm{G}}=$
1510.06699_F00110	5	■ 0.999	$\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}_{cd}^{ab}, \mathcal{T}_{bc}^{* a}, \mathcal{H}_{ab}\right) d^4 x$	$\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}_{cd}^{ab}, \mathcal{T}_{bc}^{* a}, \mathcal{H}_{ab}\right) d^4 x$	$\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}_{cd}^{ab}, \mathcal{T}_{bc}^{* a}, \mathcal{H}_{ab}\right) d^4 x$. As usual, the total ac-
1510.06699_F00111	5	■ 1.000	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}$	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}$	$\sigma_{ab}^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta A_{\mu}^{ab}$ and dilation current $\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}$
1510.06699_F00112	5	■ 1.000	$w\left(\mathcal{R}_{cd}^{ab}\right)=w\left(\mathcal{H}_{ab}\right)=-2$	$w\left(\mathcal{R}_{cd}^{ab}\right)=w\left(\mathcal{H}_{ab}\right)=-2$	$w\left(\mathcal{R}_{cd}^{ab}\right)=w\left(\mathcal{H}_{ab}\right)=-2$ and $w\left(\mathcal{T}_{bc}^{* a}\right)=-1$, which
1510.06699_F00113	5	■ 1.000	$w\left(\mathcal{T}_{bc}^{* a}\right)=-1$	$w\left(\mathcal{T}_{bc}^{* a}\right)=-1$	$w\left(\mathcal{R}_{cd}^{ab}\right)=w\left(\mathcal{H}_{ab}\right)=-2$ and $w\left(\mathcal{T}_{bc}^{* a}\right)=-1$, which
1510.06699_F00114	5	■ 1.000	$\mathcal{H}_{ab}$	$\mathcal{H}_{ab}$	means that L_{G} can be quadratic in $\mathcal{R}_{cd}^{ab}$ and $\mathcal{H}_{ab}$, while
1510.06699_F00115	5	■ 1.000	$\mathcal{T}_{bc}^{* a}$	$\mathcal{T}_{bc}^{* a}$	terms linear in $\mathcal{R} \equiv \mathcal{R}_{ab}^{ab}$ or quadratic in $\mathcal{T}_{bc}^{* a}$ are
1510.06699_F00116	5	■ 1.000	$\mathcal{R}^2+\mathcal{H}^2$	$\mathcal{R}^2+\mathcal{H}^2$	in the first derivatives of gauge fields, is of $\mathcal{R}^2+\mathcal{H}^2$ type:
1510.06699_F00117	5	■ 1.000	ξ	ξ	where the α_i and ξ are dimensionless free parameters.
1510.06699_F00118	5	■ 1.000	$\mathcal{T}^{* 2}$	$\mathcal{T}^{* 2}$	in PGT; essentially the $\mathcal{R}$ and $\mathcal{T}^{* 2}$ terms, which possess
1510.06699_F00119	5	■ 1.000	$w=-4$	$w=-4$	tal Lagrangian must have a Weyl weight $w=-4$ clearly
1510.06699_F00120	5	■ 0.999	ϕ	ϕ	ϕ with Weyl weight $w(\phi)=-1$, often termed the
1510.06699_F00121	5	■ 0.999	$w(\phi)=-1$	$w(\phi)=-1$	ϕ with Weyl weight $w(\phi)=-1$, often termed the
1510.06699_F00122	5	■ 0.785	$(s)^{\underline{10}}$	$(s)^{10}$	compensator(s) ¹⁰ . This opens up possibilities for the in-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00123	5	■ 0.803	$\phi^{\{2\}} \mathcal{R}$	$\phi^2 \mathcal{R}$	ample, terms proportional to $\phi^2 \mathcal{R}$ or $\phi^2 L_{\mathcal{T}^{*2}}$ are Weyl-
1510.06699_F00124	5	■ 0.803	$\phi^{\{2\}} L_{\mathcal{T}^{*2}}$	$\phi^2 L_{\mathcal{T}^{*2}}$	ample, terms proportional to $\phi^2 \mathcal{R}$ or $\phi^2 L_{\mathcal{T}^{*2}}$ are Weyl-
1510.06699_F00125	5	■ 1.000	$\{ \}^{\{54,55\}}$	54,55	been widely investigated ^{54,55} , whereas the phenomeno-
1510.06699_F00126	5	■ 0.999	$\phi^{\{2\}} L_{\mathcal{T}^{*2}}$	$\phi^2 L_{\mathcal{T}^{*2}}$	logical value of the $\phi^2 L_{\mathcal{T}^{*2}}$ terms has received relatively
1510.06699_F00127	5	■ 0.969	$\{ \}^{\{56,57\}}$	56,57	little attention ^{56,57} . In any case, the resulting total La-
1510.06699_F00128	5	■ 1.000	ψ	ψ	tation for a free Dirac field ψ , which has Weyl weight
1510.06699_F00129	5	■ 1.000	$w(\psi)=w(\bar{\psi})=-\frac{3}{2}$	$w(\psi) = w(\bar{\psi}) = -\frac{3}{2}$	$w(\psi) = w(\bar{\psi}) = -\frac{3}{2}$. This action is not scale-invariant
1510.06699_F00130	5	■ 1.000	$m \bar{\psi} \psi$	$m \bar{\psi} \psi$	owing to the mass term $m \bar{\psi} \psi$, so it appears that one re-
1510.06699_F00131	5	■ 0.983	$\{ \}^{\{10\}} \zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$	$^{10} \zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$	vergence of the dilation current ¹⁰ $\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu$. In
1510.06699_F00132	6	■ 1.000	$m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$	$m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$	replacement $m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$ in the Dirac action, where μ
1510.06699_F00133	6	■ 1.000	μ	μ	replacement $m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$ in the Dirac action, where μ
1510.06699_F00134	6	■ 1.000	$\mu \phi$	$\mu \phi$	is a dimensionless parameter but $\mu \phi$ has the dimensions
1510.06699_F00135	6	■ 1.000	G	G	‘constant’ G to vary with cosmic epoch. This idea can
1510.06699_F00136	6	■ 1.000	$\phi = \phi_0$	$\phi = \phi_0$	stein gauge. Setting $\phi = \phi_0$ in the equation of motion
1510.06699_F00137	6	■ 1.000	$m = \mu \phi_0$	$m = \mu \phi_0$	tation as a massive field with $m = \mu \phi_0$. It is usually
1510.06699_F00138	6	■ 1.000	w	w	$h_a{}^\mu$ and $A^a{}_b{}_\mu$ transform covariantly with weights of w ,
1510.06699_F00139	6	■ 1.000	$\mathcal{R}_{abcd}$	$\mathcal{R}_{abcd}$	strength (or ‘curvature’) $\mathcal{R}_{abcd}$ and translational gauge
1510.06699_F00140	6	■ 1.000	$\mathcal{T}_{abc}$	$\mathcal{T}_{abc}$	field strength (or ‘torsion’) $\mathcal{T}_{abc}$ are treated in a more
1510.06699_F00141	6	■ 0.617	$\{ \}^{\{59-62,89\}}$	$^{59-62,89}$	so is the total action) ^{59-62,89} .
1510.06699_F00142	6	■ 1.000	Y_4	Y_4	Weyl–Cartan Y_4 spacetime (although we do give a brief
1510.06699_F00143	7	■ 1.000	$\widehat{Y}_4$	$\widehat{Y}_4$	a new spacetime geometry (that we term $\widehat{Y}_4$), which is
1510.06699_F00144	7	■ 1.000	$\mathcal{M}$	$\mathcal{M}$	We begin with a Minkowski spacetime $\mathcal{M}$, labelled
1510.06699_F00145	7	■ 1.000	x^μ	x^μ	using Cartesian inertial coordinates x^μ , on which the
1510.06699_F00146	7	■ 1.000	$\varphi(x)$	$\varphi(x)$	gravitational) fields $\varphi(x)$ is described by a matter action
1510.06699_F00147	7	■ 0.959	e_μ	e_μ	e_μ , which are tangents to the coordinate curves and for
1510.06699_F00148	7	■ 0.997	$e_\mu \cdot e_\nu = \eta_{\mu\nu}$	$e_\mu \cdot e_\nu = \eta_{\mu\nu}$	Cartesian inertial coordinates satisfy $e_\mu \cdot e_\nu = \eta_{\mu\nu}$, where
1510.06699_F00149	7	■ 1.000	$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$	$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$	$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$. At each point of $\mathcal{M}$, we can
1510.06699_F00150	7	■ 1.000	$\hat{e}_a(x)$	$\hat{e}_a(x)$	mal tetrad $\hat{e}_a(x)$, such that $\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$. In global
1510.06699_F00151	7	■ 1.000	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$	mal tetrad $\hat{e}_a(x)$, such that $\hat{e}_a \cdot \hat{e}_b = \eta_{ab}$. In global
1510.06699_F00152	7	■ 1.000	$e_\mu(x)$	$e_\mu(x)$	$e_\mu(x)$, i.e. $\hat{e}_a = \delta_a^\mu e_\mu$. We will adopt the common con-
1510.06699_F00153	7	■ 1.000	$\hat{e}_a = \delta_a^\mu e_\mu$	$\hat{e}_a = \delta_a^\mu e_\mu$	$e_\mu(x)$, i.e. $\hat{e}_a = \delta_a^\mu e_\mu$. We will adopt the common con-
1510.06699_F00154	7	■ 1.000	$\text{SO}(3, 1)$	$\text{SO}(3, 1)$	$\text{SO}(3, 1)$ by $\text{SL}(2, C)$, so that spinor fields can be
1510.06699_F00155	7	■ 1.000	$\text{SL}(2, C)$	$\text{SL}(2, C)$	$\text{SO}(3, 1)$ by $\text{SL}(2, C)$, so that spinor fields can be
1510.06699_F00156	7	■ 1.000	$\{\varphi^a(x)\}$	$\{\varphi^a(x)\}$	ter field components $\{\varphi^a(x)\}$ simply carry a generic Ro-
1510.06699_F00157	8	■ 0.999	$S(\Lambda)$	$S(\Lambda)$	where $S(\Lambda)$ is the matrix corresponding to the element Λ
1510.06699_F00158	8	■ 1.000	$\Lambda^\mu{}_\nu = \delta^\mu_\nu$	$\Lambda^\mu{}_\nu = \delta^\mu_\nu$	formation, i.e. with $\Lambda^\mu{}_\nu = \delta^\mu_\nu$ and $a^\mu = 0$ in (6) and
1510.06699_F00159	8	■ 1.000	$a^\mu = 0$	$a^\mu = 0$	formation, i.e. with $\Lambda^\mu{}_\nu = \delta^\mu_\nu$ and $a^\mu = 0$ in (6) and
1510.06699_F00160	8	■ 1.000	e^ρ	e^ρ	tor e^ρ . Adopting global inertial coordinates, such that
1510.06699_F00161	8	■ 0.999	$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$	$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$	$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$, the metric must therefore transform
1510.06699_F00162	8	■ 1.000	$\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho} \eta_{\mu\nu}$	$\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho} \eta_{\mu\nu}$	according to $\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho} \eta_{\mu\nu}$. Hence, under this
1510.06699_F00163	8	■ 1.000	$w=2$	$w = 2$	active dilation the metric has a Weyl weight of $w = 2$.
1510.06699_F00164	8	■ 1.000	$x'^\mu = e^\rho x^\mu$	$x'^\mu = e^\rho x^\mu$	ordinate transformation $x'^\mu = e^\rho x^\mu$, which ‘shrinks’ the
1510.06699_F00165	8	■ 1.000	$\gamma'_{\mu\nu} = e^{-2\rho} \gamma_{\mu\nu} = \eta_{\mu\nu}$	$\gamma'_{\mu\nu} = e^{-2\rho} \gamma_{\mu\nu} = \eta_{\mu\nu}$	coordinate transformation, $\gamma'_{\mu\nu} = e^{-2\rho} \gamma_{\mu\nu} = \eta_{\mu\nu}$ and so,
1510.06699_F00166	8	■ 1.000	$e^{4\rho}$	$e^{4\rho}$	transformation (6) (for which the Jacobian equals $e^{4\rho}$), a

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00167	8	■ 1.000	m	m	m , one has the Klein–Gordon Lagrangian
1510.06699_F00168	8	■ 1.000	m=0	$m = 0$	dilation invariant only for $\overline{m=0}$ and provided the Weyl
1510.06699_F00169	8	■ 1.000	$\bar{\psi}\psi$	$\bar{\psi}\psi$	pretation of $\bar{\psi}\psi$ as a number density. As mentioned in the
1510.06699_F00170	8	■ 1.000	A_{μ}	A_{μ}	Finally, we consider the electromagnetic field A_{μ}
1510.06699_F00171	8	■ 1.000	$F_{\mu\nu} \equiv \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}$	$F_{\mu\nu} \equiv \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}$	where $\overline{F_{\mu\nu} \equiv \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}}$ and J^{μ} is some source cur-
1510.06699_F00172	8	■ 1.000	J^{μ}	J^{μ}	where $F_{\mu\nu} \equiv \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}$ and J^{μ} is some source cur-
1510.06699_F00173	8	■ 1.000	w_J	w_J	where w and w_J are the Weyl weights of A_{μ} and J_{μ}
1510.06699_F00174	8	■ 1.000	J_{μ}	J_{μ}	where w and w_J are the Weyl weights of A_{μ} and J_{μ}
1510.06699_F00175	8	■ 1.000	$w(A_{\mu}) = -1$	$w(A_{\mu}) = -1$	(9), provided that $w(A_{\mu}) = -1$ and $w(J_{\mu}) = -3$. It is
1510.06699_F00176	8	■ 1.000	$w(J_{\mu}) = -3$	$w(J_{\mu}) = -3$	(9), provided that $w(A_{\mu}) = -1$ and $w(J_{\mu}) = -3$. It is
1510.06699_F00177	8	■ 1.000	$J^{\mu} \equiv \bar{\psi}\gamma^{\mu}\psi$	$J^{\mu} \equiv \bar{\psi}\gamma^{\mu}\psi$	worth noting that the Dirac current $\overline{J^{\mu} \equiv \bar{\psi}\gamma^{\mu}\psi}$ satisfies
1510.06699_F00178	9	■ 1.000	a^{μ}	a^{μ}	the four translation parameters a^{μ} and the dilation pa-
1510.06699_F00179	9	■ 1.000	ρ	ρ	rameter ρ into eleven independent arbitrary functions of
1510.06699_F00180	9	■ 1.000	x	x	x . This leads to a complete decoupling of the translation
1510.06699_F00181	9	■ 1.000	$x^{\prime}$	$x^{\prime}$	lation parts. Thus, it is customary to view $x^{\prime}$ as arising
1510.06699_F00182	9	■ 1.000	$\Lambda(x)$	$\Lambda(x)$	local Lorentz rotation $\Lambda(x)$ of the orthonormal tetrad
1510.06699_F00183	9	■ 0.997	$\rho(x)$	$\rho(x)$	$\hat{e}_a(x)$, or local (physical) dilation $\rho(x)$, at each point.
1510.06699_F00184	9	■ 1.000	$\Lambda^a{}_b(x)$	$\Lambda^a{}_b(x)$	Lorentz rotations $\Lambda^a{}_b(x)$. This means that any spinor,
1510.06699_F00185	9	■ 1.000	$\partial_{\mu}\varphi$	$\partial_{\mu}\varphi$	field. Consequently, the partial derivative, $\partial_{\mu}\varphi$, is co-
1510.06699_F00186	9	■ 1.000	$\eta_{ab} = \eta^{ab} = \text{diag}(1, -1, -1, -1)$	$\eta_{ab} = \eta^{ab} = \text{diag}(1, -1, -1, -1)$	ric $\overline{\eta_{ab} = \eta^{ab} = \text{diag}(1, -1, -1, -1)}$, whereas Greek in-
1510.06699_F00187	9	■ 1.000	$\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$	$\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$	ric $\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$ and its inverse $\gamma^{\mu\nu}$ corresponding to
1510.06699_F00188	9	■ 1.000	$\gamma^{\mu\nu}$	$\gamma^{\mu\nu}$	ric $\gamma_{\mu\nu} = e_{\mu} \cdot e_{\nu}$ and its inverse $\gamma^{\mu\nu}$ corresponding to
1510.06699_F00189	9	■ 1.000	$\hat{e}^a = \eta^{ab}\hat{e}_b$	$\hat{e}^a = \eta^{ab}\hat{e}_b$	by $\overline{\hat{e}^a = \eta^{ab}\hat{e}_b}$ and $e^{\mu} = \gamma^{\mu\nu}e_{\nu}$, respectively. Moreover,
1510.06699_F00190	9	■ 1.000	$e^{\mu} = \gamma^{\mu\nu}e_{\nu}$	$e^{\mu} = \gamma^{\mu\nu}e_{\nu}$	by $\hat{e}^a = \eta^{ab}\hat{e}_b$ and $\overline{e^{\mu} = \gamma^{\mu\nu}e_{\nu}}$, respectively. Moreover,
1510.06699_F00191	9	■ 1.000	$\gamma_{\mu\nu}$	$\gamma_{\mu\nu}$	metric $\gamma_{\mu\nu}$ of the Minkowski spacetime obeys the trans-
1510.06699_F00192	9	■ 1.000	(Λ, ρ)	(Λ, ρ)	two steps. Firstly, one defines the (Λ, ρ) -covariant deriva-
1510.06699_F00193	9	■ 1.000	$D_{\mu}\varphi(x)$	$D_{\mu}\varphi(x)$	where $D_{\mu}\varphi(x)$ is the Λ -covariant derivative introduced
1510.06699_F00194	9	■ 1.000	$A^{ab}{}_{\mu}(x)$	$A^{ab}{}_{\mu}(x)$	of complex conjugation. Most importantly, $\overline{A^{ab}{}_{\mu}(x)}$ is
1510.06699_F00195	9	■ 1.000	$B_{\mu}(x)$	$B_{\mu}(x)$	tions and $B_{\mu}(x)$ is the gauge-field corresponding to local
1510.06699_F00196	9	■ 1.000	$\Sigma_{ab} = -\Sigma_{ab}$	$\Sigma_{ab} = -\Sigma_{ab}$	dilations. The matrices $\overline{\Sigma_{ab} = -\Sigma_{ab}}$ are the generators of
1510.06699_F00197	9	■ 0.483	$(2, C)$	$(2, C)$	the $\text{SL}(\overline{2, C})$ representation $S(\Lambda)$ to which $\varphi(x)$ belongs,
1510.06699_F00198	9	■ 0.483	(Λ)	(Λ)	the $\text{SL}(2, C)$ representation $S(\Lambda)$ to which $\varphi(x)$ belongs,
1510.06699_F00199	9	■ 1.000	Σ_{ab}	Σ_{ab}	Given the antisymmetry of the generator matrices Σ_{ab} ,
1510.06699_F00200	9	■ 1.000	$A^{ab}{}_{\mu} =$	$A^{ab}{}_{\mu} =$	to share this antisymmetry property, such that $A^{ab}{}_{\mu} =$
1510.06699_F00201	9	■ 0.890	$-A^{ba}{}_{\mu}$	$-A^{ba}{}_{\mu}$	$-A^{ba}{}_{\mu}$. The aim of introducing the A and B gauge fields
1510.06699_F00202	9	■ 0.890	B	B	$-A^{ba}{}_{\mu}$. The aim of introducing the A and B gauge fields
1510.06699_F00203	9	■ 1.000	$w=0$	$w = 0$	local dilations (i.e. has Weyl weight $\overline{w=0}$).
1510.06699_F00204	10	■ 1.000	$\partial_{\mu}^*\varphi(x)$	$\partial_{\mu}^*\varphi(x)$	It is straightforward to show that $\partial_{\mu}^*\varphi(x)$ itself also trans-
1510.06699_F00205	10	■ 0.999	$D_{\mu}^*\varphi$	$D_{\mu}^*\varphi$	derivative, linearly related to $D_{\mu}^*\varphi$, by introducing a new,
1510.06699_F00206	10	■ 1.000	$h_a{}^{\mu}(x)$	$h_a{}^{\mu}(x)$	‘translational’ gauge field $h_a{}^{\mu}(x)$ that transforms via
1510.06699_F00207	10	■ 1.000	$w=-1$	$w = -1$	Lorentz four-vector, and has Weyl weight $\overline{w=-1}$ under
1510.06699_F00208	10	■ 1.000	$b^a{}_{\mu}(x)$	$b^a{}_{\mu}(x)$	$b^a{}_{\mu}(x)$ (which has Weyl weight $w = 1$) and satisfies
1510.06699_F00209	10	■ 1.000	$w=1$	$w = 1$	$b^a{}_{\mu}(x)$ (which has Weyl weight $\overline{w=1}$) and satisfies
1510.06699_F00210	10	■ 1.000	$\partial\varphi$	$\partial\varphi$	mations in the same way as $\partial\varphi$ does under global Weyl
1510.06699_F00211	10	■ 1.000	$x \rightarrow x^{\prime}$	$x \rightarrow x^{\prime}$	$w = -4$. However, since $x \rightarrow x^{\prime}$ is a GCT, we need a

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00212	10	■ 0.997	$L_{\mathrm{M}}(\varphi; \mathcal{D}^*\varphi)$	$L_{\mathrm{M}}(\varphi; \mathcal{D}^*\varphi)$	$L_{\mathrm{M}}(\varphi; \mathcal{D}^*\varphi)$ by a factor which transforms by the appro-
1510.06699_F00213	10	■ 1.000	$w=4$	$w = 4$	weight $w = 4$. From the transformation properties of the
1510.06699_F00214	10	■ 1.000	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi)$	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi)$	by the Lagrangian density $\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* \varphi)$. We
1510.06699_F00215	10	■ 0.999	h, A	h, A	note that the gravitational gauge fields h, A and B con-
1510.06699_F00216	10	■ 0.613	$L_{\mathcal{T}^*2}$	$L_{\mathcal{T}^*2}$	weight $w = -4$ and so may be added to L_{M} (where $L_{\mathcal{T}^*2}$
1510.06699_F00217	10	■ 0.999	D_{μ}^*	D_{μ}^*	choice, one may extend the (Λ, ρ) -covariant derivative D_{μ}^*
1510.06699_F00218	10	■ 1.000	$(\Sigma_{cd}^1)^a{}_b = (\delta_c^a \eta_{db} - \delta_d^a \eta_{cb})$	$(\Sigma_{cd}^1)^a{}_b = (\delta_c^a \eta_{db} - \delta_d^a \eta_{cb})$	where $(\Sigma_{cd}^1)^a{}_b = (\delta_c^a \eta_{db} - \delta_d^a \eta_{cb})$ are the generators of the
1510.06699_F00219	10	■ 0.509	C	C	der GCT, in addition to $\mathrm{SL}(2, C)$ behaviour. In this case,
1510.06699_F00220	11	■ 1.000	${}^0\nabla_{\mu}^* \equiv {}^0\nabla_{\mu} + w B_{\mu}, {}^0\Gamma^{\lambda}{}_{\mu\nu} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} + \partial_{\nu} \gamma_{\mu\rho} -$	${}^0\nabla_{\mu}^* \equiv {}^0\nabla_{\mu} + w B_{\mu}, {}^0\Gamma^{\lambda}{}_{\mu\nu} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} + \partial_{\nu} \gamma_{\mu\rho} -$	where ${}^0\nabla_{\mu}^* \equiv {}^0\nabla_{\mu} + w B_{\mu}, {}^0\Gamma^{\lambda}{}_{\mu\nu} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} + \partial_{\nu} \gamma_{\mu\rho} -$
1510.06699_F00221	11	■ 0.709	$\partial_{\rho} \gamma_{\mu\nu}$	$\partial_{\rho} \gamma_{\mu\nu}$	$\partial_{\rho} \gamma_{\mu\nu}$) is the metric connection corresponding to the
1510.06699_F00222	11	■ 0.991	$X^{\rho}{}_{\sigma}$	$X^{\rho}{}_{\sigma}$	and $X^{\rho}{}_{\sigma}$ are the $\mathrm{GL}(4, R)$ generator matrices appropriate
1510.06699_F00223	11	■ 0.991	$\mathrm{GL}(4, R)$	$\mathrm{GL}(4, R)$	and $X^{\rho}{}_{\sigma}$ are the $\mathrm{GL}(4, R)$ generator matrices appropriate
1510.06699_F00224	11	■ 1.000	${}^0\Gamma^{\sigma}{}_{\rho\mu}$	${}^0\Gamma^{\sigma}{}_{\rho\mu}$	spacetime, the connection coefficients ${}^0\Gamma^{\sigma}{}_{\rho\mu}$ are not dy-
1510.06699_F00225	11	■ 1.000	${}^0\nabla_{\mu}$	${}^0\nabla_{\mu}$	spacetime, meaning that the derivatives ${}^0\nabla_{\mu}$ and ${}^0\nabla_{\nu}$
1510.06699_F00226	11	■ 1.000	${}^0\nabla_{\nu}$	${}^0\nabla_{\nu}$	spacetime, meaning that the derivatives ${}^0\nabla_{\mu}$ and ${}^0\nabla_{\nu}$
1510.06699_F00227	11	■ 1.000	$\boldsymbol{h}$	$\boldsymbol{h}$	rank-2 tensor field, which we will denote by $\boldsymbol{h}$, defined on
1510.06699_F00228	11	■ 1.000	$h_a{}^{\mu} \equiv \boldsymbol{h}(\hat{e}_a, e^{\mu})$	$h_a{}^{\mu} \equiv \boldsymbol{h}(\hat{e}_a, e^{\mu})$	by $h_a{}^{\mu} \equiv \boldsymbol{h}(\hat{e}_a, e^{\mu})$. Similarly, the inverse of $\boldsymbol{h}$, which
1510.06699_F00229	11	■ 1.000	$\boldsymbol{b}$	$\boldsymbol{b}$	we denote by $\boldsymbol{b}$, is also a rank-2 tensor field defined on
1510.06699_F00230	11	■ 1.000	$b^a{}_{\mu} \equiv \boldsymbol{b}(\hat{e}^a, e_{\mu})$	$b^a{}_{\mu} \equiv \boldsymbol{b}(\hat{e}^a, e_{\mu})$	considered so far are given by $b^a{}_{\mu} \equiv \boldsymbol{b}(\hat{e}^a, e_{\mu})$. Other
1510.06699_F00231	11	■ 1.000	$\boldsymbol{J}$	$\boldsymbol{J}$	field $\boldsymbol{J}$ defined on the Minkowski spacetime, which may
1510.06699_F00232	11	■ 1.000	$\boldsymbol{J} = J^{\mu} e_{\mu} = J_{\mu} e^{\mu}$	$\boldsymbol{J} = J^{\mu} e_{\mu} = J_{\mu} e^{\mu}$	basis or its dual as $\boldsymbol{J} = J^{\mu} e_{\mu} = J_{\mu} e^{\mu}$. Replacing the
1510.06699_F00233	11	■ 1.000	$\mathcal{J}_a \equiv h_a{}^{\mu} J_{\mu} = h_{a\mu} J^{\mu}$	$\mathcal{J}_a \equiv h_a{}^{\mu} J_{\mu} = h_{a\mu} J^{\mu}$	$\boldsymbol{h}$, we may obtain the quantities $\mathcal{J}_a \equiv h_a{}^{\mu} J_{\mu} = h_{a\mu} J^{\mu}$ or
1510.06699_F00234	11	■ 1.000	$\mathcal{J}^a \equiv h^a{}_{\mu} J^{\mu} = h^{a\mu} J_{\mu}$	$\mathcal{J}^a \equiv h^a{}_{\mu} J^{\mu} = h^{a\mu} J_{\mu}$	$\mathcal{J}^a \equiv h^a{}_{\mu} J^{\mu} = h^{a\mu} J_{\mu}$. Adopting the viewpoint that $\boldsymbol{h}$ is
1510.06699_F00235	11	■ 0.967	$\mathcal{J} = \boldsymbol{h}(\cdot, \boldsymbol{J})^{67}$	$\mathcal{J} = \boldsymbol{h}(\cdot, \boldsymbol{J})^{67}$	basis of a <i>new</i> vector field $\mathcal{J} = \boldsymbol{h}(\cdot, \boldsymbol{J})^{67}$. It is also worth
1510.06699_F00236	11	■ 0.820	$\boldsymbol{h}(\mathcal{J}, \cdot)$	$\boldsymbol{h}(\mathcal{J}, \cdot)$	the vector $\boldsymbol{h}(\mathcal{J}, \cdot)$, which has contravariant components
1510.06699_F00237	11	■ 1.000	$h_a{}^{\mu} \mathcal{J}^a$	$h_a{}^{\mu} \mathcal{J}^a$	in the coordinate basis given by $h_a{}^{\mu} \mathcal{J}^a$; these are the
1510.06699_F00238	11	■ 1.000	$\mathcal{J}$	$\mathcal{J}$	example, if we place $\mathcal{J}$ into the first ‘slot’ of $\boldsymbol{b}$, we re-
1510.06699_F00239	11	■ 0.690	$\boldsymbol{J} = \boldsymbol{b}(\mathcal{J}, \cdot)$	$\boldsymbol{J} = \boldsymbol{b}(\mathcal{J}, \cdot)$	cover the <i>original</i> vector $\boldsymbol{J} = \boldsymbol{b}(\mathcal{J}, \cdot)$, since its covari-
1510.06699_F00240	11	■ 1.000	$b^a{}_{\mu} \mathcal{J}_a = J_{\mu}$	$b^a{}_{\mu} \mathcal{J}_a = J_{\mu}$	ant components in the coordinate basis are $b^a{}_{\mu} \mathcal{J}_a = J_{\mu}$,
1510.06699_F00241	11	■ 1.000	$J_{\mu} = b_{a\mu} \mathcal{J}^a$	$J_{\mu} = b_{a\mu} \mathcal{J}^a$	$J_{\mu} = b_{a\mu} \mathcal{J}^a$, and their contravariant counterparts as
1510.06699_F00242	11	■ 1.000	$J^{\mu} = b_a{}^{\mu} \mathcal{J}^a = b^{a\mu} \mathcal{J}_a$	$J^{\mu} = b_a{}^{\mu} \mathcal{J}^a = b^{a\mu} \mathcal{J}_a$	$J^{\mu} = b_a{}^{\mu} \mathcal{J}^a = b^{a\mu} \mathcal{J}_a$. In all these cases, the compo-
1510.06699_F00243	11	■ 1.000	$\boldsymbol{b}(\cdot, \boldsymbol{J})$	$\boldsymbol{b}(\cdot, \boldsymbol{J})$	tion: consider, for example, the vector $\boldsymbol{b}(\cdot, \boldsymbol{J})$; this has
1510.06699_F00244	11	■ 1.000	$b^a{}_{\mu} J^{\mu}$	$b^a{}_{\mu} J^{\mu}$	$b^a{}_{\mu} J^{\mu}$, but these are <i>not</i> , in general, equal to $\mathcal{J}^a$.
1510.06699_F00245	11	■ 1.000	$\mathcal{J}^a$	$\mathcal{J}^a$	$b^a{}_{\mu} J^{\mu}$, but these are <i>not</i> , in general, equal to $\mathcal{J}^a$.
1510.06699_F00246	11	■ 0.999	$\boldsymbol{T}$	$\boldsymbol{T}$	indices: for example, if $\boldsymbol{T}$ is a rank-2 tensor field, then
1510.06699_F00247	11	■ 0.968	$\mathcal{T}$	$\mathcal{T}$	one obtains the <i>new</i> rank-2 tensor field $\mathcal{T}$ with covari-
1510.06699_F00248	11	■ 1.000	$\mathcal{T}_{ab} =$	$\mathcal{T}_{ab} =$	ant components (say) in the tetrad basis given by $\mathcal{T}_{ab} =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00249	11	■ 1.000	$\mathrm{h}_{\{a\}\ }^{\{\mu\}}\mathrm{h}_{\{b\}\ }^{\{\nu\}}\mathrm{T}_{\{\mu\}\ \{\nu\}}$	$h_a{}^\mu h_b{}^\nu T_{\mu\nu}$	$h_a{}^\mu h_b{}^\nu T_{\mu\nu}$. Using the reciprocity relations (28), one may
1510.06699_F00250	11	■ 1.000	$\mathrm{T}_{\{\mu\}\ \{\nu\}}=\mathrm{b}^{\{a\}\ }\mathrm{_{\{\mu\}\ }}\mathrm{b}^{\{b\}\ }\mathrm{_{\{\nu\}\ }}\ \mathcal{T}_{\{a\}\ {b\}}$	$T_{\mu\nu} = b^a{}_\mu b^b{}_\nu \mathcal{T}_{ab}$	therefore also write, for example, $T_{\mu\nu} = b^a{}_\mu b^b{}_\nu \mathcal{T}_{ab}$. One
1510.06699_F00251	11	■ 0.999	$\hat{\mathrm{e}}_{\{\mathrm{a}\}}$	$\hat{e}_a$	frame vectors $\hat{e}_a$; in this case, the components $h_a{}^\mu J_\mu$,
1510.06699_F00252	11	■ 0.999	$\mathrm{h}_{\{a\}\ }^{\{\mu\}}\mathrm{J}_{\{\mu\}}$	$h_a{}^\mu J_\mu$	frame vectors $\hat{e}_a$; in this case, the components $h_a{}^\mu J_\mu$,
1510.06699_F00253	11	■ 1.000	$\mathrm{J}_{\{a\}}$	J_a	for example, are interpreted as the components J_a of the
1510.06699_F00254	12	■ 1.000	$\mathrm{R}^{\{a\}\ {b\}}\mathrm{_{\{\mu\}\ }}\mathrm{_{\{\nu\}\ }}$	$R^{ab}{}_{\mu\nu}$	Both $R^{ab}{}_{\mu\nu}$ and $H_{\mu\nu}$ transform covariantly under GCT
1510.06699_F00255	12	■ 1.000	$\mathrm{H}_{\{\mu\}\ \{\nu\}}$	$H_{\mu\nu}$	Both $R^{ab}{}_{\mu\nu}$ and $H_{\mu\nu}$ transform covariantly under GCT
1510.06699_F00256	12	■ 1.000	$\mathcal{D}_{\{a\}\ }^*\varphi=\varphi$	$\mathcal{D}_a^*\varphi =$	two generalised covariant derivatives. Since $\mathcal{D}_a^*\varphi =$
1510.06699_F00257	12	■ 1.000	$\mathrm{h}_{\{a\}\ }^{\{\mu\}}\mathrm{D}_{\{\mu\}\ }^*\varphi=\varphi$	$h_a{}^\mu D_\mu^*\varphi$	$h_a{}^\mu D_\mu^*\varphi$, this commutator differs from (38) by an addi-
1510.06699_F00258	12	■ 1.000	$\mathcal{R}^{\{a\}\ {b\}}\mathrm{_{\{c\}\ }}\mathrm{_{\{d\}\ }}\equiv\mathrm{h}_{\{a\}\ }^{\{\mu\}}\mathrm{h}_{\{b\}\ }^{\{\nu\}}\mathrm{h}_{\{c\}\ }^{\{\lambda\}}\mathrm{h}_{\{d\}\ }^{\{\rho\}}\mathrm{R}^{\{a\}\ {b\}}\mathrm{_{\{\mu\}\ }}\mathrm{_{\{\nu\}\ }}\mathrm{_{\{\lambda\}\ }}\mathrm{_{\{\rho\}\ }}$	$\mathcal{R}^{ab}{}_{cd} \equiv h_a{}^\mu h_b{}^\nu R^{ab}{}_{\mu\nu}$	where $\mathcal{R}^{ab}{}_{cd} \equiv h_a{}^\mu h_b{}^\nu R^{ab}{}_{\mu\nu}$ and $\mathcal{H}_{cd} = h_c{}^\mu h_d{}^\nu H_{\mu\nu}$, and
1510.06699_F00259	12	■ 1.000	$\mathcal{H}_{\{c\}\ {d\}}=\mathrm{h}_{\{c\}\ }^{\{\mu\}}\mathrm{h}_{\{d\}\ }^{\{\nu\}}\mathrm{H}_{\{\mu\}\ \{\nu\}}$	$\mathcal{H}_{cd} = h_c{}^\mu h_d{}^\nu H_{\mu\nu}$	where $\mathcal{R}^{ab}{}_{cd} \equiv h_a{}^\mu h_b{}^\nu R^{ab}{}_{\mu\nu}$ and $\mathcal{H}_{cd} = h_c{}^\mu h_d{}^\nu H_{\mu\nu}$, and
1510.06699_F00260	12	■ 0.669	$\mathcal{B}_{\{a\}}=\mathrm{h}_{\{a\}\ }^{\{\mu\}}\mathrm{B}_{\{\mu\}}$	$\mathcal{B}_a = h_a{}^\mu B_\mu$	where $\mathcal{B}_a = h_a{}^\mu B_\mu$. It is also convenient to define $\mathcal{T}_b^* \equiv$
1510.06699_F00261	12	■ 0.669	$\mathcal{T}_{\{b\}\ }^*\equiv$	$\mathcal{T}_b^* \equiv$	where $\mathcal{B}_a = h_a{}^\mu B_\mu$. It is also convenient to define $\mathcal{T}_b^* \equiv$
1510.06699_F00262	12	■ 1.000	$\mathcal{T}^{\{a\}\ }_{\{b\}\ }=\mathcal{T}_{\{b\}\ }+3\mathcal{B}_{\{b\}}$	$\mathcal{T}^{*a}{}_{ba} = \mathcal{T}_b + 3\mathcal{B}_b$	$\mathcal{T}^{*a}{}_{ba} = \mathcal{T}_b + 3\mathcal{B}_b$, where $\mathcal{T}_b$ is the corresponding PGT
1510.06699_F00263	12	■ 1.000	$\mathcal{T}_{\{b\}}$	$\mathcal{T}_b$	$\mathcal{T}^{*a}{}_{ba} = \mathcal{T}_b + 3\mathcal{B}_b$, where $\mathcal{T}_b$ is the corresponding PGT
1510.06699_F00264	12	■ 1.000	$\mathcal{R}^{\{a\}\ {b\}}\mathrm{_{\{c\}\ }}\mathrm{_{\{d\}\ }},\ \mathcal{H}_{\{c\}\ {d\}}$	$\mathcal{R}^{ab}{}_{cd}, \mathcal{H}_{cd}$	It is easy to show that $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{H}_{cd}$ and $\mathcal{T}^{*a}{}_{cd}$ trans-
1510.06699_F00265	12	■ 1.000	$\mathcal{T}^{\{a\}\ }_{\{c\}\ {d\}}$	$\mathcal{T}^{*a}{}_{cd}$	It is easy to show that $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{H}_{cd}$ and $\mathcal{T}^{*a}{}_{cd}$ trans-
1510.06699_F00266	12	■ 1.000	$w(\mathcal{R}^{\{a\}\ {b\}}\mathrm{_{\{c\}\ }}\mathrm{_{\{d\}\ }})=w(\mathcal{H}_{\{c\}\ {d\}})=-2$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{cd}) = -2$	$w(\mathcal{R}^{ab}{}_{cd}) \equiv w(\mathcal{H}_{cd}) = -2$ and $w(\mathcal{T}^{*a}{}_{cd}) = -1$ respec-
1510.06699_F00267	12	■ 1.000	$w(\mathcal{T}^{\{a\}\ }_{\{c\}\ {d\}})=-1$	$w(\mathcal{T}^{*a}{}_{cd}) = -1$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{cd}) = -2$ and $w(\mathcal{T}^{*a}{}_{cd}) = -1$ respec-
1510.06699_F00268	12	■ 1.000	$\mathcal{R}^{\{a\}\ {b\}}\mathrm{_{\{c\}\ }}\mathrm{_{\{d\}\ }},\ \mathcal{H}_{\{a\}\ {b\}}$	$\mathcal{R}^{ab}{}_{cd}, \mathcal{H}_{ab}$	strengths $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{H}_{ab}$ and $\mathcal{T}^{*a}{}_{bc}$ are related to the <i>curva-</i>
1510.06699_F00269	12	■ 1.000	$\mathcal{A}_{\{a\}\ {b\}\ \{\mu\}}=$	$A_{ab\mu} =$	the relevant relationship is easily found to be $A_{ab\mu} =$
1510.06699_F00270	12	■ 1.000	$\mathrm{b}^{\{c\}\ }\mathrm{_{\{\mu\}\ }}\mathcal{A}_{\{a\}\ {b\}\ {c\}}$	$b^c{}_\mu \mathcal{A}_{abc}$	$b^c{}_\mu \mathcal{A}_{abd}$, where
1510.06699_F00271	12	■ 1.000	$\mathcal{T}_{\{a\}\ {b\}\ {c\}}^*$	$\mathcal{T}_{abc}^*$	equations of motion result in $\mathcal{T}_{abc}^*$ being independent of
1510.06699_F00272	12	■ 1.000	$\mathcal{A}_{\{a\}\ {b\}\ {c\}}$	$\mathcal{A}_{abc}$	$\mathcal{A}_{abc}$, (46) gives an explicit expression for the <i>A</i> -field in
1510.06699_F00273	12	■ 1.000	$\{ \}^{\{0\}}\ \mathcal{A}_{\{a\}\ {b\}\ {c\}}^*\equiv\frac{1}{2}\left(c_{\{a\}\ {b\}\ {c\}}^*+\right.$	${}^0\mathcal{A}_{abc}^* \equiv \frac{1}{2}(c_{abc}^* +$	in which we have defined the quantities ${}^0\mathcal{A}_{abc}^* \equiv \frac{1}{2}(c_{abc}^* +$
1510.06699_F00274	12	■ 0.506	$\left.\mathrm{c}_{\{b\}\ {c\}\ {a\}}^*\right)-\mathrm{c}_{\{c\}\ {a\}\ {b\}}^*\right)$	$c_{bca}^* - c_{cab}^*)$	$c_{bca}^* - c_{cab}^*)$ that, as indicated, are fully determined by the
1510.06699_F00275	12	■ 1.000	$\mathcal{K}_{\{a\}\ {b\}\ {c\}}^*\equiv-\frac{1}{2}\left(\mathcal{T}_{\{a\}\ {b\}\ {c\}}^*+\mathcal{T}_{\{b\}\ {c\}\ {a\}}^*-\right.$	$\mathcal{K}_{abc}^* \equiv -\frac{1}{2}(\mathcal{T}_{abc}^* + \mathcal{T}_{bca}^* - \mathcal{T}_{cab}^*)$	defined $\mathcal{K}_{abc}^* \equiv -\frac{1}{2}(\mathcal{T}_{abc}^* + \mathcal{T}_{bca}^* - \mathcal{T}_{cab}^*)$, which when re-
1510.06699_F00276	12	■ 0.982	$\{ \}^{\{0\}}\ \mathcal{A}_{\{a\}\ {b\}\ {c\}}^*$	${}^0\mathcal{A}_{abc}^*$	that both ${}^0\mathcal{A}_{abc}^*$ and $\mathcal{K}_{abc}^*$ are antisymmetric in their first
1510.06699_F00277	12	■ 0.982	$\mathcal{K}_{\{a\}\ {b\}\ {c\}}^*$	$\mathcal{K}_{abc}^*$	that both ${}^0\mathcal{A}_{abc}^*$ and $\mathcal{K}_{abc}^*$ are antisymmetric in their first
1510.06699_F00278	12	■ 0.991	$\mathcal{A}_{\{a\}\ {b\}\ {c\}}^*$	$\mathcal{A}_{abc}^*$	${}^0\mathcal{A}_{abc}^*$ transform in the same way as $\mathcal{A}_{abc}^*$, whereas $\mathcal{K}_{abc}^*$
1510.06699_F00279	12	■ 0.961	$\{ \}^{\{0\}}\ \mathcal{D}_{\{a\}\ }^*\equiv\mathrm{h}_{\{a\}\ }^{\{\mu\}}\mathrm{D}_{\{\mu\}\ }^*\varphi$	${}^0\mathcal{D}_a^* \equiv h_a{}^\mu D_\mu^*$	covariant derivative ${}^0\mathcal{D}_a^* \equiv h_a{}^\mu D_\mu^*$, which consequently
1510.06699_F00280	12	■ 1.000	$\mathcal{D}_{\{a\}\ }^*$	$\mathcal{D}_a^*$	torsion (and hence contortion) to zero, then $\mathcal{D}_a^*$ reduces
1510.06699_F00281	12	■ 1.000	$\{ \}^{\{0\}}\ \mathcal{D}_{\{a\}\ }^*$	${}^0\mathcal{D}_a^*$	to ${}^0\mathcal{D}_a^*$.
1510.06699_F00282	12	■ 1.000	$\mathcal{R}^{\{a\}\ {b\}}\mathrm{_{\{c\}\ }}\mathrm{_{\{d\}\ }},\ \mathcal{T}^{\{a\}\ }_{\{b\}\ {c\}}$	$\mathcal{R}^{ab}{}_{cd}, \mathcal{T}^{*a}{}_{bc}$	gauge field strengths $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{T}^{*a}{}_{bc}$ and $\mathcal{H}_{ab}$ in WGT.
1510.06699_F00283	13	■ 0.947	$\mathcal{T}_{\{a\}\ {b\}\ {c\}}^*=$	$\mathcal{T}_{abc}^* =$	the ‘torsion’ is totally antisymmetric, such that $\mathcal{T}_{abc}^* =$
1510.06699_F00284	13	■ 0.990	$\epsilon_{\{a\}\ {b\}\ {c\}\ {d\}}\mathrm{t}^{\{d\}}$	$\epsilon_{abcd}t^d$	$\epsilon_{abcd}t^d$ for some vector t^d , then the second term on the
1510.06699_F00285	13	■ 0.990	$\mathrm{t}^{\{d\}}$	t^d	$\epsilon_{abcd}t^d$ for some vector t^d , then the second term on the
1510.06699_F00286	13	■ 0.999	d	d	By contracting over the indices <i>a</i> and <i>d</i> in the ‘ <i>R</i> -
1510.06699_F00287	13	■ 1.000	b	b	Contracting again over <i>b</i> and <i>e</i> , one then finds the twice-
1510.06699_F00288	13	■ 1.000	e	e	Contracting again over <i>b</i> and <i>e</i> , one then finds the twice-
1510.06699_F00289	13	■ 0.770	$\mathcal{T}_{\{a\}\ }^*$	$\mathcal{T}_a^*$	torsion trace $\mathcal{T}_a^*$ vanishes, which clearly includes the spe-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00290	13	■ 1.000	$\mathcal{H}$	$\mathcal{H}$	It is clear that the ‘ $\mathcal{H}$ -identity’ (53) has no non-trivial
1510.06699_F00291	13	■ 1.000	$\mathcal{R}_{abcd}, \mathcal{T}_{abc}^*$	$\mathcal{R}_{abcd}, \mathcal{T}_{abc}^*$	itational gauge fields. From $\mathcal{R}_{abcd}$, $\mathcal{T}_{abc}^*$ and $\mathcal{H}_{ab}$, one can
1510.06699_F00292	13	■ 1.000	$\mathcal{L}_G \equiv h^{-1} L_G$	$\mathcal{L}_G \equiv h^{-1} L_G$	sity $\mathcal{L}_G \equiv h^{-1} L_G$ has weight zero (i.e scale invari-
1510.06699_F00293	13	■ 1.000	$\mathcal{R}_{abcd}, \mathcal{H}_{ab}$	$\mathcal{R}_{abcd}, \mathcal{H}_{ab}$	ant). It is trivial to show that $\mathcal{R}_{abcd}$, $\mathcal{H}_{ab}$ and $\mathcal{T}_{abc}^*$
1510.06699_F00294	13	■ 1.000	$w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$	$w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$	weights $w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}_{abc}^*) = -1$
1510.06699_F00295	13	■ 1.000	$w(\mathcal{T}_{abc}^*) = -1$	$w(\mathcal{T}_{abc}^*) = -1$	weights $w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}_{abc}^*) = -1$
1510.06699_F00296	13	■ 0.988	$\{ \}^{\underline{70}}$	$\overline{70}$	respectively ⁷⁰ . Hence L_G can be quadratic in $\mathcal{R}_{abcd}$ and
1510.06699_F00297	13	■ 0.994	$L_{\mathcal{H}^2}$	$L_{\mathcal{H}^2}$	where the expressions for $L_{\mathcal{R}^2}$ and $L_{\mathcal{H}^2}$ are given in (4).
1510.06699_F00298	13	■ 1.000	$D \leq 4$	$D \leq 4$	contributes a total derivative to the action (in $D \leq 4$
1510.06699_F00299	13	■ 0.994	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^*$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^*$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^*$. However, since these terms depend upon the
1510.06699_F00300	13	■ 1.000	$\mathcal{L}_G$	$\mathcal{L}_G$	sity $\mathcal{L}_G$. As an illustration, in Appendix D we give a
1510.06699_F00301	13	■ 1.000	$L_G = L_{\mathcal{H}^2}$	$L_G = L_{\mathcal{H}^2}$	$L_G = L_{\mathcal{H}^2}$ and L_M contains a term proportional to $\phi^2 \mathcal{R}$.
1510.06699_F00302	13	■ 1.000	S_T	S_T	The total action S_T is simply the sum of the matter and
1510.06699_F00303	13	■ 1.000	$h, A, \partial A$	$h, A, \partial A$	$h, A, \partial A$ and ∂B (suppressing indices for brevity). Note,
1510.06699_F00304	13	■ 1.000	∂B	∂B	$h, A, \partial A$ and ∂B (suppressing indices for brevity). Note,
1510.06699_F00305	13	■ 1.000	$\varphi, \partial \varphi, h, A$	$\varphi, \partial \varphi, h, A$	φ induce a dependence on $\varphi, \partial \varphi, h, A$ and B . We will
1510.06699_F00306	13	■ 1.000	∂A	∂A	$\phi^2 \mathcal{R}$, then this brings an additional dependence on ∂A .
1510.06699_F00307	13	■ 0.901	$\phi^2 L_{\mathcal{T}^* \in}$	$\phi^2 L_{\mathcal{T}^* \in}$	Finally, if one includes terms of the generic form $\phi^2 L_{\mathcal{T}^* 2}$,
1510.06699_F00308	13	■ 1.000	∂h	∂h	then it produces an additional dependence on ∂h . Con-
1510.06699_F00309	14	■ 1.000	$\mathcal{L}_T$	$\mathcal{L}_T$	Since $\mathcal{L}_T$ is at most quadratic in the field strength ten-
1510.06699_F00310	14	■ 1.000	$\mathcal{R}_{abcd}(h, A, \partial A), \mathcal{T}_{abc}^*(h, \partial h, A, B)$	$\mathcal{R}_{abcd}(h, A, \partial A), \mathcal{T}_{abc}^*(h, \partial h, A, B)$	sors $\mathcal{R}_{abcd}(h, A, \partial A)$, $\mathcal{T}_{abc}^*(h, \partial h, A, B)$ and $\mathcal{H}_{ab}(h, \partial B)$,
1510.06699_F00311	14	■ 1.000	$t^a{}_\mu \equiv$	$t^a{}_\mu \equiv$	In the free gravitational sector, we have defined $t^a{}_\mu \equiv$
1510.06699_F00312	14	■ 0.999	$\delta \mathcal{L}_G / \delta h_a{}^\mu = \partial \mathcal{L}_G / \partial h_a{}^\mu, s_{ab}{}^\mu \equiv \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$	$\delta \mathcal{L}_G / \delta h_a{}^\mu = \partial \mathcal{L}_G / \partial h_a{}^\mu, s_{ab}{}^\mu \equiv \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$	$\delta \mathcal{L}_G / \delta h_a{}^\mu = \partial \mathcal{L}_G / \partial h_a{}^\mu, s_{ab}{}^\mu \equiv \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ and
1510.06699_F00313	14	■ 1.000	$j^\mu \equiv \delta \mathcal{L}_G / \delta B_\mu$	$j^\mu \equiv \delta \mathcal{L}_G / \delta B_\mu$	$j^\mu \equiv \delta \mathcal{L}_G / \delta B_\mu$, respectively. In the matter sector,
1510.06699_F00314	14	■ 1.000	$\tau^a{}_\mu \equiv \delta \mathcal{L}_M / \delta h_a{}^\mu$	$\tau^a{}_\mu \equiv \delta \mathcal{L}_M / \delta h_a{}^\mu$	the energy-momentum $\tau^a{}_\mu \equiv \delta \mathcal{L}_M / \delta h_a{}^\mu$, spin-angular-
1510.06699_F00315	14	■ 0.692	$\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu = \partial \mathcal{L}_M / \partial B_\mu$	$\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu = \partial \mathcal{L}_M / \partial B_\mu$	$\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu = \partial \mathcal{L}_M / \partial B_\mu$ act as sources ⁷² . It is easy
1510.06699_F00316	14	■ 0.692	$\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu = \partial \mathcal{L}_M / \partial B_\mu$	$\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu = \partial \mathcal{L}_M / \partial B_\mu$	$\zeta^\mu \equiv \delta \mathcal{L}_M / \delta B_\mu = \partial \mathcal{L}_M / \partial B_\mu$ act as sources ⁷² . It is easy
1510.06699_F00317	14	■ 1.000	$w = 1, w = 0$	$w = 1, w = 0$	$w = 1, w = 0$ and $w = 0$, respectively. It is, in fact, more
1510.06699_F00318	14	■ 1.000	$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu, s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu$	$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu, s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu$	where $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu, s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu$ and $j^a \equiv j^\mu b^a{}_\mu$, and
1510.06699_F00319	14	■ 1.000	$j^a \equiv j^\mu b^a{}_\mu$	$j^a \equiv j^\mu b^a{}_\mu$	where $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu, s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu$ and $j^a \equiv j^\mu b^a{}_\mu$, and
1510.06699_F00320	14	■ 0.992	$\bar{\partial} \mathcal{L}_M / \partial \varphi \equiv [\partial \mathcal{L}_M(\varphi, \mathcal{D}_\mu^* u, \dots) / \partial \varphi]_{u=\varphi}$	$\bar{\partial} \mathcal{L}_M / \partial \varphi \equiv [\partial \mathcal{L}_M(\varphi, \mathcal{D}_\mu^* u, \dots) / \partial \varphi]_{u=\varphi}$	where $\bar{\partial} \mathcal{L}_M / \partial \varphi \equiv [\partial \mathcal{L}_M(\varphi, \mathcal{D}_\mu^* u, \dots) / \partial \varphi]_{u=\varphi}$ so that φ
1510.06699_F00321	14	■ 1.000	$\mathcal{D}_\mu^* \varphi$	$\mathcal{D}_\mu^* \varphi$	and $\mathcal{D}_\mu^* \varphi$ are treated as independent variables, rather
1510.06699_F00322	14	■ 1.000	$\mathcal{L}_M = h^{-1} L_M$	$\mathcal{L}_M = h^{-1} L_M$	where we have defined $\mathcal{L}_M = h^{-1} L_M$ and $\bar{\partial} L_M / \partial \varphi \equiv$
1510.06699_F00323	14	■ 1.000	$\bar{\partial} L_M / \partial \varphi \equiv$	$\bar{\partial} L_M / \partial \varphi \equiv$	where we have defined $\mathcal{L}_M = h^{-1} L_M$ and $\bar{\partial} L_M / \partial \varphi \equiv$
1510.06699_F00324	14	■ 0.974	$[\partial L_M(\varphi, \mathcal{D}_a^* u, \dots) / \partial \varphi]_{u=\varphi}$	$[\partial L_M(\varphi, \mathcal{D}_a^* u, \dots) / \partial \varphi]_{u=\varphi}$	$[\partial L_M(\varphi, \mathcal{D}_a^* u, \dots) / \partial \varphi]_{u=\varphi}$, so that φ and $\mathcal{D}_a^* \varphi$ are treated
1510.06699_F00325	14	■ 1.000	S_M	S_M	the minimal-coupling procedure applied to the action S_M ,
1510.06699_F00326	14	■ 0.999	$\{ \}^{9,74}$	9,74	equations directly ^{9,74} , which would be given by (67a) and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00327	14	■ 0.963	2, c	$2, C$	S_G under (infinitesimal) local Lorentz (or $SL(2, C)$) rota-
1510.06699_F00328	15	■ 0.718	(69 c)	(69c)	(69c)
1510.06699_F00329	15	■ 1.000	$w = -\frac{3}{2}$	$w = -\frac{3}{2}$	$w = -\frac{3}{2}$ and $w = -1$, respectively; we note that the
1510.06699_F00330	15	■ 1.000	$\psi \rightarrow \psi e^{i\alpha}$	$\psi \rightarrow \psi e^{i\alpha}$	mation $\psi \rightarrow \psi e^{i\alpha}$.
1510.06699_F00331	15	■ 1.000	γ	γ	where, on each γ -matrix, one has simply replaced the
1510.06699_F00332	15	■ 1.000	$\bar{\psi}$	$\bar{\psi}$	that, since both ψ and $\bar{\psi}$ have Weyl weight $w = -\frac{3}{2}$, the
1510.06699_F00333	15	■ 1.000	$\mathcal{D}_a$	$\mathcal{D}_a$	term in (71) can be replaced by its PGT counterpart $\mathcal{D}_a$,
1510.06699_F00334	15	■ 1.000	$\mathcal{D}_a^* \psi$	$\mathcal{D}_a^* \psi$	varying the action with respect to ψ . Since $\mathcal{D}_a^* \psi$ and $\mathcal{T}_a^*$
1510.06699_F00335	15	■ 1.000	$w = -\frac{5}{2}$	$w = -\frac{5}{2}$	$w = -\frac{5}{2}$ and $w = -1$ respectively, we see immediately
1510.06699_F00336	15	■ 1.000	$\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$	$\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$	directly by recalling that $\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$ and $\mathcal{T}_a =$
1510.06699_F00337	15	■ 1.000	$\mathcal{T}_a =$	$\mathcal{T}_a =$	directly by recalling that $\mathcal{D}_a \psi = \mathcal{D}_a^* \psi + \frac{3}{2} \mathcal{B}_a \psi$ and $\mathcal{T}_a =$
1510.06699_F00338	15	■ 1.000	$\mathcal{T}_a^* - 3 \mathcal{B}_a$	$\mathcal{T}_a^* - 3 \mathcal{B}_a$	$\mathcal{T}_a^* - 3 \mathcal{B}_a$ to recover (73). We note that (73) and (74) are
1510.06699_F00339	16	■ 1.000	$\mathcal{T}_a$	$\mathcal{T}_a$	density $\mathcal{L}_G$, but the expression for $\mathcal{T}_a$, or $\mathcal{T}_a^*$, will depend
1510.06699_F00340	16	■ 1.000	$\Sigma_{ab} = \frac{1}{4} [\gamma_a, \gamma_b]$	$\Sigma_{ab} = \frac{1}{4} [\gamma_a, \gamma_b]$	given by $\Sigma_{ab} = \frac{1}{4} [\gamma_a, \gamma_b]$, after a short calculation one
1510.06699_F00341	16	■ 0.945	$\mathcal{T}_{[abc]}^* = 0$	$\mathcal{T}_{[abc]}^* = 0$	of the Dirac field ψ . In other words, if $\mathcal{T}_{[abc]}^* = 0$ then
1510.06699_F00342	16	■ 1.000	$\tau = h^{-1} \mu \phi \bar{\psi} \psi$	$\tau = h^{-1} \mu \phi \bar{\psi} \psi$	ric and has the trace $\tau = h^{-1} \mu \phi \bar{\psi} \psi$, whereas the spin-
1510.06699_F00343	16	■ 1.000	$\sigma_{abc} = \sigma_{[abc]}$	$\sigma_{abc} = \sigma_{[abc]}$	that $\sigma_{abc} = \sigma_{[abc]}$. Consequently, all contractions of the
1510.06699_F00344	16	■ 1.000	$\mathcal{D}_a \rightarrow \mathcal{D}_a^*$	$\mathcal{D}_a \rightarrow \mathcal{D}_a^*$	ply making the replacement $\mathcal{D}_a \rightarrow \mathcal{D}_a^*$ throughout; the
1510.06699_F00345	16	■ 1.000	μ, ν, λ	μ, ν, λ	where μ, ν, λ and a are dimensionless constants (usually
1510.06699_F00346	16	■ 0.669	β_1, β_2	β_1, β_2	stants β_1, β_2 and β_3 in $L_{T^{*2}}$. Since none of the additional
1510.06699_F00347	16	■ 0.669	β_3	β_3	stants β_1, β_2 and β_3 in $L_{T^{*2}}$. Since none of the additional
1510.06699_F00348	16	■ 1.000	$F_{\mu\nu} \equiv$	$F_{\mu\nu} \equiv$	where the electromagnetic field strength tensor $F_{\mu\nu} \equiv$
1510.06699_F00349	16	■ 1.000	$\partial_\mu A_\nu - \partial_\nu A_\mu$	$\partial_\mu A_\nu - \partial_\nu A_\mu$	$\partial_\mu A_\nu - \partial_\nu A_\mu$ and J^μ is some source current density. We
1510.06699_F00350	16	■ 1.000	$A_\mu \rightarrow A_\mu + \partial_\mu \chi$	$A_\mu \rightarrow A_\mu + \partial_\mu \chi$	formation $A_\mu \rightarrow A_\mu + \partial_\mu \chi$, where χ is any scalar field,
1510.06699_F00351	16	■ 1.000	χ	χ	formation $A_\mu \rightarrow A_\mu + \partial_\mu \chi$, where χ is any scalar field,
1510.06699_F00352	16	■ 1.000	$\partial_\mu J^\mu = 0$	$\partial_\mu J^\mu = 0$	ity equation $\partial_\mu J^\mu = 0$, which embodies the conservation
1510.06699_F00353	16	■ 1.000	$\mathcal{A}_a = h_a^\mu A_\mu$	$\mathcal{A}_a = h_a^\mu A_\mu$	field $\mathcal{A}_a = h_a^\mu A_\mu$. Similarly, we introduce the covari-
1510.06699_F00354	16	■ 1.000	$\mathcal{J}_a = h_a^\mu J_\mu$	$\mathcal{J}_a = h_a^\mu J_\mu$	ant current density $\mathcal{J}_a = h_a^\mu J_\mu$. This leads us, however,
1510.06699_F00355	16	■ 1.000	$w(\mathcal{A}_a) = -1$	$w(\mathcal{A}_a) = -1$	weights $w(\mathcal{A}_a) = -1$ and $w(\mathcal{J}_a) = -3$; consequently, the
1510.06699_F00356	16	■ 1.000	$w(\mathcal{J}_a) = -3$	$w(\mathcal{J}_a) = -3$	weights $w(\mathcal{A}_a) = -1$ and $w(\mathcal{J}_a) = -3$; consequently, the
1510.06699_F00357	16	■ 1.000	$w(A_\mu) = 0$	$w(A_\mu) = 0$	fields carrying a Greek index have weights $w(A_\mu) = 0$
1510.06699_F00358	16	■ 1.000	$w(J_\mu) = -2$	$w(J_\mu) = -2$	and $w(J_\mu) = -2$.
1510.06699_F00359	16	■ 1.000	$\widehat{\mathcal{F}}_{ab}^* \equiv \mathcal{D}_a^* \mathcal{A}_b - \mathcal{D}_b^* \mathcal{A}_a$	$\widehat{\mathcal{F}}_{ab}^* \equiv \mathcal{D}_a^* \mathcal{A}_b - \mathcal{D}_b^* \mathcal{A}_a$	where we have defined $\widehat{\mathcal{F}}_{ab}^* \equiv \mathcal{D}_a^* \mathcal{A}_b - \mathcal{D}_b^* \mathcal{A}_a$. Since
1510.06699_F00360	16	■ 1.000	$w(\widehat{\mathcal{F}}_{ab}^*) = -2, \mathcal{L}_M$	$w(\widehat{\mathcal{F}}_{ab}^*) = -2, \mathcal{L}_M$	$w(\widehat{\mathcal{F}}_{ab}^*) = -2$, $\mathcal{L}_M$ has an overall Weyl weight of zero,
1510.06699_F00361	17	■ 1.000	$\mathcal{F}_{ab} \equiv h_a^\mu h_b^\nu F_{\mu\nu}$	$\mathcal{F}_{ab} \equiv h_a^\mu h_b^\nu F_{\mu\nu}$	where we have defined $\mathcal{F}_{ab} \equiv h_a^\mu h_b^\nu F_{\mu\nu}$, in which
1510.06699_F00362	17	■ 1.000	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\mathcal{T}^{*c}_{ab}$ is
1510.06699_F00363	17	■ 1.000	$\mathcal{T}^{*c}_{ab}$	$\mathcal{T}^{*c}_{ab}$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\mathcal{T}^{*c}_{ab}$ is
1510.06699_F00364	17	■ 1.000	$\mathcal{F}_{ab}$	$\mathcal{F}_{ab}$	gauge field, as defined in (42). It is apparent that $\mathcal{F}_{ab}$
1510.06699_F00365	17	■ 1.000	$F_{\mu\nu}$	$F_{\mu\nu}$	ance of $F_{\mu\nu}$, whereas the second term on the RHS of (82)
1510.06699_F00366	17	■ 1.000	$\widehat{\mathcal{F}}_{ab}^*$	$\widehat{\mathcal{F}}_{ab}^*$	transformation, and so neither is $\widehat{\mathcal{F}}_{ab}^*$. Moreover, $\mathcal{F}_{ab}$ has
1510.06699_F00367	17	■ 1.000	$w(\mathcal{F}_{ab}) = -2$	$w(\mathcal{F}_{ab}) = -2$	a Weyl weight $w(\mathcal{F}_{ab}) = -2$. Thus, substituting (82)
1510.06699_F00368	17	■ 1.000	$\mathcal{L}_M$	$\mathcal{L}_M$	for $\mathcal{L}_M$ is identical to that obtained in PGT, as should be

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00369	17	■ 0.999	$\delta \mathrm{L} / \delta A_{\mu}=0$	$\delta \mathcal{L}_{\mathrm{EM}} / \delta A_{\mu} = 0$	is given by $\delta \mathcal{L}_{\mathrm{EM}} / \delta A_{\mu} = 0$, which is easily shown to be
1510.06699_F00370	17	■ 0.997	$\delta \mathrm{L} / \delta \mathcal{A}_a=0$	$\delta \mathcal{L}_{\mathrm{EM}} / \delta \mathcal{A}_a = 0$	equivalent to $\delta \mathcal{L}_{\mathrm{EM}} / \delta \mathcal{A}_a = 0$, and in manifestly WGT-
1510.06699_F00371	17	■ 1.000	$\mathrm{D}_{\mathrm{a}} \mathcal{F}^{\mathrm{ac}}=\left(\mathrm{D}_{\mathrm{a}}^{*}+\right.$	$\mathcal{D}_a \mathcal{F}^{ac} = (\mathcal{D}_a^* +$	is easily checked directly by noting that $\mathcal{D}_a \mathcal{F}^{ac} = (\mathcal{D}_a^* +$
1510.06699_F00372	17	■ 1.000	$\left.2 \mathcal{B}_{\mathrm{a}}\right) \mathcal{F}^{\mathrm{ac}}$	$2 \mathcal{B}_a) \mathcal{F}^{ac}$	$2 \mathcal{B}_a) \mathcal{F}^{ac}$ and $\mathcal{T}_a = \mathcal{T}_a^* - 3 \mathcal{B}_a$, which on substitution
1510.06699_F00373	17	■ 1.000	$\mathcal{T}_{\mathrm{a}}=\mathcal{T}^{*}\}_{\mathrm{a}}-3 \mathcal{B}_{\mathrm{a}}$	$\mathcal{T}_a = \mathcal{T}_a^* - 3 \mathcal{B}_a$	$2 \mathcal{B}_a) \mathcal{F}^{ac}$ and $\mathcal{T}_a = \mathcal{T}_a^* - 3 \mathcal{B}_a$, which on substitution
1510.06699_F00374	17	■ 1.000	$\mathcal{J}^{\mathrm{a}}=0$	$\mathcal{J}^a = 0$	angular-momentum tensors. Setting $\mathcal{T}^a \equiv 0$ in (83), one
1510.06699_F00375	17	■ 1.000	$\sigma_{\mathrm{ab}}^{\mathrm{c}}=0$	$\sigma_{ab}^c = 0$	whereas $\sigma_{ab}^c = 0$ immediately. We see that τ^a_b is the
1510.06699_F00376	17	■ 1.000	$\tau^{\mathrm{a}}_{\mathrm{b}}$	τ^a_b	whereas $\sigma_{ab}^c = 0$ immediately. We see that τ^a_b is the
1510.06699_F00377	17	■ 1.000	$\tau_{\mathrm{a} \mathrm{b}}$	τ_{ab}	gravity. Moreover, τ_{ab} is symmetric and traceless, as
1510.06699_F00378	17	■ 1.000	$\mathcal{J}^{\mathrm{a}}=q \bar{\psi} \gamma^{\mathrm{a}} \psi$	$\mathcal{J}^a = q \bar{\psi} \gamma^a \psi$	current density of the Dirac field is simply $\mathcal{T}^a = q \bar{\psi} \gamma^a \psi$,
1510.06699_F00379	17	■ 1.000	$w\left(\mathcal{J}^{\mathrm{a}}\right)=-3$	$w\left(\mathcal{J}^a\right) = -3$	which has the required Weyl weight $w\left(\mathcal{J}^a\right) = -3$. Thus,
1510.06699_F00380	17	■ 1.000	$\phi \neq 0$	$\phi \neq 0$	(provided $\phi \neq 0$), which is known as the Einstein gauge.
1510.06699_F00381	17	■ 1.000	$\phi(x)=1$	$\phi(x) = 1$	Although it is most common simply to set $\phi(x) = 1$, this
1510.06699_F00382	17	■ 1.000	$\phi(x)=\phi_{\mathrm{0}}$	$\phi(x) = \phi_0$	can be misleading and so we instead set $\phi(x) = \phi_0$, where
1510.06699_F00383	17	■ 1.000	ϕ_{0}	ϕ_0	ϕ_0 is a constant with the same units as ϕ . Adopting the
1510.06699_F00384	18	■ 0.638	(91 c)	(91c)	(91c)
1510.06699_F00385	18	■ 1.000	$\widehat{\mathcal{A}}^{\mathrm{ab}}_{\mathrm{c}} \equiv \widehat{h}_{\mathrm{a}}^{\mu} \widehat{A}^{\mathrm{ab}}_{\mu}$	$\hat{\mathcal{A}}^{ab}_c \equiv \hat{h}_a^\mu \hat{A}^{ab}_\mu$	ables $\hat{\mathcal{A}}^{ab}_c \equiv \hat{h}_a^\mu \hat{A}^{ab}_\mu$ and $\hat{\mathcal{B}}_a \equiv \hat{h}_a^\mu \hat{B}_\mu$.
1510.06699_F00386	18	■ 1.000	$\widehat{\mathcal{B}}_{\mathrm{a}} \equiv \widehat{h}_{\mathrm{a}}^{\mu} \widehat{B}_{\mu}$	$\hat{\mathcal{B}}_a \equiv \hat{h}_a^\mu \hat{B}_\mu$	ables $\hat{\mathcal{A}}^{ab}_c \equiv \hat{h}_a^\mu \hat{A}^{ab}_\mu$ and $\hat{\mathcal{B}}_a \equiv \hat{h}_a^\mu \hat{B}_\mu$.
1510.06699_F00387	18	■ 1.000	$\mathcal{D}_{\mathrm{a}}^* \phi=-\left(\phi^2 / \phi_{\mathrm{0}}\right) \widehat{\mathcal{B}}_{\mathrm{a}}$	$\mathcal{D}_a^* \phi = -\left(\phi^2 / \phi_0\right) \hat{\mathcal{B}}_a$	the scalar field ϕ may be written as $\mathcal{D}_a^* \phi = -\left(\phi^2 / \phi_0\right) \hat{\mathcal{B}}_a$,
1510.06699_F00388	18	■ 1.000	$\widehat{\chi}=\left(\phi / \phi_{\mathrm{0}}\right)^{-w} \chi$	$\hat{\chi} = \left(\phi / \phi_0\right)^{-w} \chi$	where $\hat{\chi} = \left(\phi / \phi_0\right)^{-w} \chi$ and we have defined the derivative
1510.06699_F00389	18	■ 0.996	$\widehat{\mathcal{D}}_{\mathrm{a}} \equiv \widehat{h}_{\mathrm{a}}^{\mu} \mathcal{D}_{\mu}$	$\hat{\mathcal{D}}_a \equiv \hat{h}_a^\mu D_\mu$	operators $\hat{\mathcal{D}}_a \equiv \hat{h}_a^\mu D_\mu$ and $\hat{\mathcal{D}}_a^* = \hat{\mathcal{D}}_a + w \hat{\mathcal{B}}_a$, the latter of
1510.06699_F00390	18	■ 0.996	$\widehat{\mathcal{D}}_{\mathrm{a}}^*=\widehat{\mathcal{D}}_{\mathrm{a}}+w \widehat{\mathcal{B}}_{\mathrm{a}}$	$\hat{\mathcal{D}}_a^* = \hat{\mathcal{D}}_a + w \hat{\mathcal{B}}_a$	operators $\hat{\mathcal{D}}_a \equiv \hat{h}_a^\mu D_\mu$ and $\hat{\mathcal{D}}_a^* = \hat{\mathcal{D}}_a + w \hat{\mathcal{B}}_a$, the latter of
1510.06699_F00391	18	■ 1.000	$\left(\phi / \phi_{\mathrm{0}}\right)^{1-w}$	$\left(\phi / \phi_0\right)^{1-w}$	all multiplicative factor $\left(\phi / \phi_0\right)^{1-w}$, the scale-invariant
1510.06699_F00392	18	■ 1.000	$\widehat{\mathcal{D}}_{\mathrm{a}}^* \widehat{\chi}$	$\hat{\mathcal{D}}_a^* \hat{\chi}$	quantity $\hat{\mathcal{D}}_a^* \hat{\chi}$ has the <i>same</i> functional dependency on $\hat{\chi}$,
1510.06699_F00393	18	■ 1.000	$\widehat{\chi}$	$\hat{\chi}$	quantity $\hat{\mathcal{D}}_a^* \hat{\chi}$ has the <i>same</i> functional dependency on $\hat{\chi}$,
1510.06699_F00394	18	■ 0.999	$\widehat{h}_{\mathrm{a}}^{\mu}, \widehat{A}^{\mathrm{ab}}_{\mu}$	$\hat{h}_a^\mu, \hat{A}^{ab}_\mu$	$\hat{h}_a^\mu, \hat{A}^{ab}_\mu$ and $\hat{B}_\mu$, respectively, as $\mathcal{D}_a^* \chi$ does on χ, h_a^μ ,
1510.06699_F00395	18	■ 0.999	$\widehat{B}_{\mu}$	$\hat{B}_\mu$	$\hat{h}_a^\mu, \hat{A}^{ab}_\mu$ and $\hat{B}_\mu$, respectively, as $\mathcal{D}_a^* \chi$ does on χ, h_a^μ ,
1510.06699_F00396	18	■ 0.999	$\mathcal{D}_{\mathrm{a}}^* \chi$	$\mathcal{D}_a^* \chi$	$\hat{h}_a^\mu, \hat{A}^{ab}_\mu$ and $\hat{B}_\mu$, respectively, as $\mathcal{D}_a^* \chi$ does on χ, h_a^μ ,
1510.06699_F00397	18	■ 0.999	$\chi, h_{\mathrm{a}}^{\mu}$	χ, h_a^μ	$\hat{h}_a^\mu, \hat{A}^{ab}_\mu$ and $\hat{B}_\mu$, respectively, as $\mathcal{D}_a^* \chi$ does on χ, h_a^μ ,
1510.06699_F00398	18	■ 1.000	$\mathcal{R}^{\mathrm{ab}}_{\mathrm{cd}}=\left(\phi / \phi_{\mathrm{0}}\right)^2 \widehat{\mathcal{R}}^{\mathrm{ab}}_{\mathrm{cd}}, \mathcal{T}^{* \mathrm{a}}_{\mathrm{bc}}=\left(\phi / \phi_{\mathrm{0}}\right) \widehat{\mathcal{T}}^{* \mathrm{a}}_{\mathrm{bc}}$	$\mathcal{R}^{ab}_{cd} = \left(\phi / \phi_0\right)^2 \hat{\mathcal{R}}^{ab}_{cd}, \mathcal{T}^{*a}_{bc} = \left(\phi / \phi_0\right) \hat{\mathcal{T}}^{*a}_{bc}$	ten as $\mathcal{R}^{ab}_{cd} = \left(\phi / \phi_0\right)^2 \hat{\mathcal{R}}^{ab}_{cd}, \mathcal{T}^{*a}_{bc} = \left(\phi / \phi_0\right) \hat{\mathcal{T}}^{*a}_{bc}$ and
1510.06699_F00399	18	■ 1.000	$\mathcal{H}_{\mathrm{ab}}=\left(\phi / \phi_{\mathrm{0}}\right)^2 \widehat{\mathcal{H}}_{\mathrm{ab}}$	$\mathcal{H}_{ab} = \left(\phi / \phi_0\right)^2 \hat{\mathcal{H}}_{ab}$	$\mathcal{H}_{ab} = \left(\phi / \phi_0\right)^2 \hat{\mathcal{H}}_{ab}$, where each quantity with a caret is
1510.06699_F00400	18	■ 1.000	$h^{-1}=\left(\phi / \phi_{\mathrm{0}}\right)^{-4} \widehat{h}^{-1}$	$h^{-1} = \left(\phi / \phi_0\right)^{-4} \hat{h}^{-1}$	Noting that $\hat{h}^{-1} = \left(\phi / \phi_0\right)^{-4} \hat{h}^{-1}$, one is thus led to the
1510.06699_F00401	18	■ 1.000	$\varphi, h_{\mathrm{a}}^{\mu}, A^{\mathrm{ab}}_{\mu}$	$\varphi, h_a^\mu, A^{ab}_\mu$	Thus, if χ represents $\varphi, h_a^\mu, A^{ab}_\mu$ or B_μ , one may im-
1510.06699_F00402	18	■ 1.000	$\delta \mathcal{L}_{\mathrm{T}} / \delta \widehat{\chi}=0$	$\delta \mathcal{L}_{\mathrm{T}} / \delta \hat{\chi} = 0$	motion $\delta \mathcal{L}_{\mathrm{T}} / \delta \hat{\chi} = 0$ has the <i>same</i> functional form as
1510.06699_F00403	18	■ 1.000	$\delta \mathcal{L}_{\mathrm{T}} / \delta \chi _{\phi=\phi_{\mathrm{0}}}=0$	$\delta \mathcal{L}_{\mathrm{T}} / \delta \chi _{\phi=\phi_0} = 0$	(the corresponding term in) $\delta \mathcal{L}_{\mathrm{T}} / \delta \chi _{\phi=\phi_0} = 0$, but with
1510.06699_F00404	18	■ 1.000	$\delta \mathcal{L}_{\mathrm{T}} / \delta \phi=\delta \mathcal{L}_{\mathrm{M}} / \delta \phi$	$\delta \mathcal{L}_{\mathrm{T}} / \delta \phi = \delta \mathcal{L}_{\mathrm{M}} / \delta \phi$	variables, one begins by noting that $\delta \mathcal{L}_{\mathrm{T}} / \delta \phi = \delta \mathcal{L}_{\mathrm{M}} / \delta \phi$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00405	18	■ 1.000	$\backslash\mathrm{delta}\backslash\mathrm{mathcal{L}}_{\backslash\mathrm{L}}_{\backslash\mathrm{M}} / \backslash\mathrm{delta}\backslash\mathrm{varphi}=\backslash\mathrm{delta}\backslash\mathrm{mathcal{L}}_{\backslash\mathrm{L}}_{\backslash\mathrm{M}} / \backslash\mathrm{delta}\backslash\mathrm{varphi}=0$	$\delta\mathcal{L}_T/\delta\varphi = \delta\mathcal{L}_M/\delta\varphi = 0$	$\delta\mathcal{L}_T/\delta\varphi = \delta\mathcal{L}_M/\delta\varphi = 0$ and set $\phi = \phi_0$ in the result-
1510.06699_F00406	18	■ 1.000	$\backslash\mathrm{partial}\backslash\mathrm{mathcal{L}}_{\backslash\mathrm{L}}_{\backslash\mathrm{M}} / \backslash\mathrm{partial}\backslash\mathrm{widehat{B}}_{\backslash\mathrm{B}}_{\backslash\mathrm{mu}}$	$\partial\mathcal{L}_M/\partial\widehat{B}_\mu$	may replace $\partial\mathcal{L}_M/\partial\widehat{B}_\mu$ by $\delta\mathcal{L}_M/\delta\widehat{B}_\mu$ in the above equa-
1510.06699_F00407	18	■ 1.000	$\backslash\mathrm{delta}\backslash\mathrm{mathcal{L}}_{\backslash\mathrm{L}}_{\backslash\mathrm{M}} / \backslash\mathrm{delta}\backslash\mathrm{widehat{B}}_{\backslash\mathrm{B}}_{\backslash\mathrm{mu}}$	$\delta\mathcal{L}_M/\delta\widehat{B}_\mu$	may replace $\partial\mathcal{L}_M/\partial\widehat{B}_\mu$ by $\delta\mathcal{L}_M/\delta\widehat{B}_\mu$ in the above equa-
1510.06699_F00408	19	■ 1.000	$m\rightarrow\mu\phi$	$m\rightarrow\mu\phi$	making the replacement $m\rightarrow\mu\phi$, as in (70). One begins
1510.06699_F00409	19	■ 1.000	$\frac{1}{2}$	$\frac{1}{2}$	by constructing the appropriate action for a spin- $\frac{1}{2}$ point
1510.06699_F00410	19	■ 1.000	$p_{\backslash\mathrm{mu}}\rightarrow p_a\equiv h_a{}^\mu p_\mu$	$p_\mu\rightarrow p_a\equiv h_a{}^\mu p_\mu$	needs only to make the replacements $p_\mu\rightarrow p_a\equiv h_a{}^\mu p_\mu$,
1510.06699_F00411	19	■ 1.000	$\backslash\mathrm{dot{x}}^{\backslash\mathrm{mu}}\rightarrow v^a\equiv b^a{}_\mu\dot{x}^\mu$	$\dot{x}^\mu\rightarrow v^a\equiv b^a{}_\mu\dot{x}^\mu$	$\dot{x}^\mu\rightarrow v^a\equiv b^a{}_\mu\dot{x}^\mu$ and $m\rightarrow\mu\phi$, which gives ⁷⁵
1510.06699_F00412	19	■ 1.000	$p^a(\backslash\mathrm{lambda})$	$p^a(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00413	19	■ 1.000	$v^a(\backslash\mathrm{lambda})$	$v^a(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00414	19	■ 1.000	$e(\backslash\mathrm{lambda})$	$e(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00415	19	■ 1.000	$\backslash\mathrm{lambda}$	λ	along the worldline, which is parameterised by λ . We
1510.06699_F00416	19	■ 1.000	$w\backslash\mathrm{left}(p^a\backslash\mathrm{right})=-1, w\backslash\mathrm{left}(v^a\backslash\mathrm{right})=0, w(e)=1, w(\backslash\mathrm{lambda})=1$	$w(p^a)=-1, w(v^a)=0, w(e)=1, w(\lambda)=1$	(96) are $w(p^a)=-1, w(v^a)=0, w(e)=1, w(\lambda)=1$ and
1510.06699_F00417	19	■ 1.000	$p^2\equiv p_ap^a$	$p^2\equiv p_ap^a$	where $p^2\equiv p_ap^a$ and the quantities $c^c{}_{ab}$ are the PGT
1510.06699_F00418	19	■ 1.000	$c^c{}_{\backslash\mathrm{a}\backslash\mathrm{b}}$	$c^c{}_{ab}$	where $p^2\equiv p_ap^a$ and the quantities $c^c{}_{ab}$ are the PGT
1510.06699_F00419	19	■ 1.000	$\{ \}^{\backslash\mathrm{0}}\backslash\mathrm{mathcal{D}}_{\backslash\mathrm{c}}^{\backslash\mathrm{*}}$	${}^0\mathcal{D}_c^*$	where ${}^0\mathcal{D}_c^*$ is the reduced WGT covariant derivative, de-
1510.06699_F00420	19	■ 1.000	$e=1/(\backslash\mathrm{mu}\backslash\mathrm{phi})$	$e=1/(\mu\phi)$	one chooses $e=1/(\mu\phi)$, in which case $v^2=1$ and λ cor-
1510.06699_F00421	19	■ 1.000	$v^2=1$	$v^2=1$	one chooses $e=1/(\mu\phi)$, in which case $v^2=1$ and λ cor-
1510.06699_F00422	19	■ 1.000	$\backslash\mathrm{tau}$	τ	responds to the proper time τ along the worldline; in this
1510.06699_F00423	19	■ 0.348	$\{ \}^{\backslash\mathrm{7}\backslash\mathrm{6}\backslash\mathrm{7}\backslash\mathrm{8}}$	${}^{\mathbf{76-78}}$	terms of the reduced PGT-covariant derivative as ⁷⁶⁻⁷⁸
1510.06699_F00424	19	■ 0.996	$\{ \}^{\backslash\mathrm{0}}A^{\backslash\mathrm{a}\backslash\mathrm{b}}_{\backslash\mathrm{mu}}$	${}^0A^{ab}{}_\mu$	Appendix B), the quantities ${}^0A^{ab}{}_\mu$ appearing in ${}^0\mathcal{D}_b$ con-
1510.06699_F00425	19	■ 0.996	$\{ \}^{\backslash\mathrm{0}}\backslash\mathrm{mathcal{D}}_{\backslash\mathrm{b}}$	${}^0\mathcal{D}_b$	Appendix B), the quantities ${}^0A^{ab}{}_\mu$ appearing in ${}^0\mathcal{D}_b$ con-
1510.06699_F00426	19	■ 0.906	$v^{\backslash\mathrm{b}}\backslash\mathrm{mathcal{D}}_{\backslash\mathrm{b}}v_{\backslash\mathrm{a}}=0$	$v^b\mathcal{D}_bv_a=0$	the equation of motion $v^b\mathcal{D}_bv_a=0$). We note that
1510.06699_F00427	19	■ 1.000	$\{ \}^{\backslash\mathrm{79},\backslash\mathrm{80}}$	79,80	this conclusion differs from some previous studies ^{79,80} ,
1510.06699_F00428	19	■ 1.000	$\backslash\mathrm{left}(q_{\backslash\mathrm{W}}/m)\backslash\mathrm{mathcal{H}}_{\backslash\mathrm{a}\backslash\mathrm{b}}v^{\backslash\mathrm{b}}$	$(q_W/m)\mathcal{H}_{ab}v^b$	$(q_W/m)\mathcal{H}_{ab}v^b$, where q_W is the particle's Weyl charge
1510.06699_F00429	19	■ 1.000	$q_{\backslash\mathrm{W}}$	q_W	$(q_W/m)\mathcal{H}_{ab}v^b$, where q_W is the particle's Weyl charge
1510.06699_F00430	19	■ 1.000	$\backslash\mathrm{mu}=0$	$\mu=0$	By setting $\mu=0$ in the action (96) and choosing the
1510.06699_F00431	19	■ 1.000	$e=1$	$e=1$	einbein $e=1$, such that $v^2=0$, one finds that the
1510.06699_F00432	19	■ 1.000	$v^2=0$	$v^2=0$	einbein $e=1$, such that $v^2=0$, one finds that the
1510.06699_F00433	20	■ 1.000	$\backslash\mathrm{widehat{p}}_{\backslash\mathrm{a}}, \backslash\mathrm{widehat{v}}^{\backslash\mathrm{a}}$	$\widehat{p}_a, \widehat{v}^a$	where $\widehat{p}_a, \widehat{v}^a$ and $\widehat{e}$ are to be considered as functions of
1510.06699_F00434	20	■ 1.000	$\backslash\mathrm{widehat{e}}$	$\widehat{e}$	where $\widehat{p}_a, \widehat{v}^a$ and $\widehat{e}$ are to be considered as functions of
1510.06699_F00435	20	■ 0.990	$\backslash\mathrm{widehat{\lambda}}$	$\widehat{\lambda}$	$\widehat{\lambda}$, such that for example $\widehat{p}_a(\widehat{\lambda})\equiv\widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$. It is then
1510.06699_F00436	20	■ 0.990	$\backslash\mathrm{widehat{p}}_{\backslash\mathrm{a}}(\backslash\mathrm{widehat{\lambda}})\equiv\backslash\mathrm{widehat{h}}_{\backslash\mathrm{a}}^{\backslash\mathrm{mu}}p_{\backslash\mathrm{mu}}(\backslash\mathrm{lambda}(\backslash\mathrm{widehat{\lambda}}))$	$\widehat{p}_a(\widehat{\lambda})\equiv\widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$	$\widehat{\lambda}$, such that for example $\widehat{p}_a(\widehat{\lambda})\equiv\widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$. It is then
1510.06699_F00437	20	■ 1.000	$\backslash\mathrm{widehat{\epsilon}}=\backslash\mathrm{left}[1-\backslash\mathrm{widehat{\lambda}}\backslash\mathrm{left}(d\backslash\mathrm{x}^{\backslash\mathrm{mu}}/d\backslash\mathrm{widehat{\lambda}})\backslash\mathrm{partial}_{\backslash\mathrm{mu}}\backslash\mathrm{ln}\backslash\mathrm{left}(\backslash\mathrm{phi}/\backslash\mathrm{phi}_{\backslash\mathrm{0}}\backslash\mathrm{right})\backslash\mathrm{right}]\backslash\mathrm{widehat{e}}$	$\widehat{\epsilon}=\left[1-\widehat{\lambda}\left(dx^\mu/d\widehat{\lambda}\right)\partial_\mu\ln(\phi/\phi_0)\right]\widehat{e}$	bein $\widehat{\epsilon}=\left[1-\widehat{\lambda}(dx^\mu/d\widehat{\lambda})\partial_\mu\ln(\phi/\phi_0)\right]\widehat{e}$. Thus, in terms
1510.06699_F00438	20	■ 1.000	$\backslash\mathrm{widehat{\lambda}}, \backslash\mathrm{widehat{p}}_{\backslash\mathrm{a}}, \backslash\mathrm{widehat{v}}^{\backslash\mathrm{a}}$	$\widehat{\lambda}, \widehat{p}_a, \widehat{v}^a$	of the scale-invariant variables $\widehat{\lambda}, \widehat{p}_a, \widehat{v}^a$ and $\widehat{\epsilon}$, respec-
1510.06699_F00439	20	■ 1.000	$\backslash\mathrm{widehat{\epsilon}}$	$\widehat{\epsilon}$	of the scale-invariant variables $\widehat{\lambda}, \widehat{p}_a, \widehat{v}^a$ and $\widehat{\epsilon}$, respec-
1510.06699_F00440	20	■ 1.000	$\backslash\mathrm{lambda}, p_{\backslash\mathrm{a}}, v^{\backslash\mathrm{a}}$	λ, p_a, v^a	it does in terms of the original variables λ, p_a, v^a and e
1510.06699_F00441	20	■ 0.999	$\backslash\mathrm{widehat{h}}_{\backslash\mathrm{a}}^{\backslash\mathrm{mu}}$	$\widehat{h}_a{}^\mu$	counterpart $\widehat{h}_a{}^\mu$ or $\widehat{b}_\mu^a$. Thus, the calculation presented
1510.06699_F00442	20	■ 0.999	$\backslash\mathrm{widehat{b}}_{\backslash\mathrm{mu}}^{\backslash\mathrm{a}}$	$\widehat{b}_\mu^a$	counterpart $\widehat{h}_a{}^\mu$ or $\widehat{b}_\mu^a$. Thus, the calculation presented
1510.06699_F00443	20	■ 1.000	$\{ \}^{\backslash\mathrm{0}}\backslash\mathrm{widehat{\mathcal{D}}}_{\backslash\mathrm{a}}$	${}^0\widehat{\mathcal{D}}_a\equiv$	where we have defined the derivative operator ${}^0\widehat{\mathcal{D}}_a\equiv$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00444	20	■ 0.977	$\widehat{h}_{\{a\}}^{\{\mu\}} \mathcal{D}_{\{\mu\}}$	$\widehat{h}_a^{\mu 0} \mathcal{D}_\mu$	$\widehat{h}_a^{\mu 0} \mathcal{D}_\mu$.
1510.06699_F00445	20	■ 1.000	$\{ \}^{\{0\}} D_{\{\mu\}}^{\{*\}}$	${}^0 D_\mu^*$	variant derivative operator ${}^0 D_\mu^*$ in (48), to which the
1510.06699_F00446	20	■ 0.837	$\mathcal{T}_{\{a\} \{b\} \{c\}}^{\{*\}} \equiv 0$	$\mathcal{T}_{abc}^* \equiv 0$	$\mathcal{T}_{abc}^* \equiv 0$. Hence such quantities may be written entirely
1510.06699_F00447	20	■ 1.000	$\{ \}^{\{0\}} R^{\{*\} \{a\} \{b\}}_{\{\mu\} \{\nu\}} \left(h, \partial h, \partial^2 h, B, \partial B \right)$	${}^0 R_{\mu\nu}^{*ab} (h, \partial h, \partial^2 h, B, \partial B)$	or ${}^0 R_{\mu\nu}^{*ab} (h, \partial h, \partial^2 h, B, \partial B)$, which is again given by
1510.06699_F00448	20	■ 1.000	$\{ \}^{\{0\}} A^{\{*\} \{a\} \{b\}}_{\{\mu\}}$	${}^0 A_{\mu}^{*ab}$	the formula (39), but with A_{μ}^{ab} replaced by ${}^0 A_{\mu}^{*ab}$, and
1510.06699_F00449	20	■ 0.998	$\{ \}^{\{0\}} \mathcal{R}^{\{*\} \{a\} \{b\}}_{\{c\} \{d\}} h_{\{c\}}^{\{*\}} h_{\{d\}}^{\{\mu\}} \{ \}^{\{0\}} R^{\{*\} \{a\} \{b\}}_{\{\mu\} \{\nu\}}$	${}^0 \mathcal{R}_{cd}^{*ab} = h_c^\mu h_d^{\mu 0} R_{\mu\nu}^{*ab}$	where ${}^0 \mathcal{R}_{cd}^{*ab} = h_c^\mu h_d^{\mu 0} R_{\mu\nu}^{*ab}$. Unlike (41), the com-
1510.06699_F00450	20	■ 1.000	$\{ \}^{\{0\}} \mathcal{T}^{\{*\} \{a\}}_{\{b\} \{c\}} \equiv$	${}^0 \mathcal{T}_{bc}^{*a} \equiv$	lational field strength of the h gauge field, since ${}^0 \mathcal{T}_{bc}^{*a} \equiv$
1510.06699_F00451	20	■ 1.000	$h_{\{b\}}^{\{*\}} h_{\{c\}}^{\{\mu\}} h_{\{d\}}^{\{\nu\}} \left(\{ \}^{\{0\}} D_{\{\mu\}}^{\{*\}} b^{\{a\}}_{\{a\}} \{ \}^{\{0\}} D_{\{\nu\}}^{\{*\}} b^{\{a\}}_{\{a\}} \right) = 0$	$h_b^\mu h_c^\nu ({}^0 D_\mu^* b_{\nu}^a - {}^0 D_\nu^* b_\mu^a) = 0$	$h_b^\mu h_c^\nu ({}^0 D_\mu^* b_{\nu}^a - {}^0 D_\nu^* b_\mu^a) = 0$. As one might expect, the
1510.06699_F00452	20	■ 0.913	$\mathcal{D}_{\{a\}}^{\{*\}} \rightarrow \mathcal{D}_{\{a\}}^{\{*\}}, \mathcal{R}_{\{a\} \{b\} \{c\} \{d\}} \rightarrow \mathcal{R}_{\{a\} \{b\} \{c\} \{d\}}^{\{*\}}$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*, \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}$	with the replacements $\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*, \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}$ and
1510.06699_F00453	20	■ 0.982	$\mathcal{T}_{\{a\} \{b\} \{c\}}^{\{*\}} \rightarrow \{ \}^{\{0\}} \mathcal{T}_{\{a\} \{b\} \{c\}}^{\{*\}} \equiv 0$	$\mathcal{T}_{abc}^* \rightarrow {}^0 \mathcal{T}_{abc}^* \equiv 0$	$\mathcal{T}_{abc}^* \rightarrow {}^0 \mathcal{T}_{abc}^* \equiv 0$.
1510.06699_F00454	20	■ 1.000	$\{ \}^{\{0\}} \mathcal{R}^{\{*\} \{a\} \{b\}}_{\{c\} \{d\}}$	${}^0 \mathcal{R}_{cd}^{*ab}$	${}^0 \mathcal{R}_{cd}^{*ab}$.
1510.06699_F00455	20	■ 1.000	$\mathcal{L}_{\{G\}} = h^{-1} L_G$	$\mathcal{L}_G = h^{-1} L_G$	density $\mathcal{L}_G = h^{-1} L_G$, where by analogy with (58),
1510.06699_F00456	21	■ 0.516	$\{ \}^{\{0\}} \mathcal{R}_{\{abcd\}}^{\{*\}}$	${}^0 \mathcal{R}_{abcd}^*$	dimensions), since ${}^0 \mathcal{R}_{abcd}^*$ and its contractions also sat-
1510.06699_F00457	21	■ 0.999	$\mathcal{R}_{\{a\} \{b\} \{c\} \{d\}} \rightarrow \{ \}^{\{0\}} \mathcal{R}_{\{a\} \{b\} \{c\} \{d\}}^{\{*\}}$	$\mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^*$	$\mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^*$. Thus, one can set to zero any one of the
1510.06699_F00458	21	■ 0.996	$\mathcal{D}_{\{a\}}^{\{*\}} \rightarrow \{ \}^{\{0\}} \mathcal{D}_{\{a\}}^{\{*\}}$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*$ and $\mathcal{T}_a^* \rightarrow {}^0 \mathcal{T}_a^* \equiv 0$. One should also note,
1510.06699_F00459	21	■ 0.996	$\mathcal{T}_{\{a\}}^{\{*\}} \rightarrow \{ \}^{\{0\}} \mathcal{T}_{\{a\}}^{\{*\}} \equiv 0$	$\mathcal{T}_a^* \rightarrow {}^0 \mathcal{T}_a^* \equiv 0$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*$ and $\mathcal{T}_a^* \rightarrow {}^0 \mathcal{T}_a^* \equiv 0$. One should also note,
1510.06699_F00460	21	■ 0.756	$\mathcal{D}_{\{a\}}^{\{*\}} \rightarrow \mathcal{D}_{\{a\}}^{\{*\}}, \mathcal{T}_{\{a\} \{b\} \{c\}}^{\{*\}} \rightarrow \{ \}^{\{0\}} \mathcal{T}_{\{a\} \{b\} \{c\}}^{\{*\}} \equiv 0$	$\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*, \mathcal{T}_{abc}^* \rightarrow {}^0 \mathcal{T}_{abc}^* \equiv 0$	with the replacements $\mathcal{D}_a^* \rightarrow {}^0 \mathcal{D}_a^*, \mathcal{T}_{abc}^* \rightarrow {}^0 \mathcal{T}_{abc}^* \equiv 0$ and
1510.06699_F00461	21	■ 1.000	$s_{\{a\} \{b\} \{c\}} \equiv 0 \equiv \sigma_{\{a\} \{b\} \{c\}}$	$s_{abc} \equiv 0 \equiv \sigma_{abc}$	$s_{abc} \equiv 0 \equiv \sigma_{abc}$. Furthermore, by performing analogous
1510.06699_F00462	21	■ 0.975	$\mathcal{T}_{\{a\} \{b\} \{c\}}^{\{*\}} = 0$	$\mathcal{T}_{abc}^* = 0$	condition $\mathcal{T}_{abc}^* = 0$ in the field equations of WGT. In the
1510.06699_F00463	21	■ 1.000	$\{ \}^{\{82-84\}}$	82–84	only for the Lovelock action ^{82–84} , which coincides with
1510.06699_F00464	21	■ 1.000	$D=4$	$D = 4$	term) in $\mathcal{D}=4$ dimensions.
1510.06699_F00465	21	■ 0.741	$\{ \}^{\{\underline{85}\}}$	$\underline{85}$	frame vectors $e_\mu(x)$ are related by ⁸⁵
1510.06699_F00466	21	■ 0.968	$\widehat{\boldsymbol{e}}^{\{a\}}(x)$	$\hat{e}^a(x)$	sis vectors $\hat{e}^a(x)$ and $e^\mu(x)$ in each set. For any other
1510.06699_F00467	21	■ 0.968	$\boldsymbol{e}^{\{\mu\}}(x)$	$e^\mu(x)$	sis vectors $\hat{e}^a(x)$ and $e^\mu(x)$ in each set. For any other
1510.06699_F00468	21	■ 0.769	$J_{\{\mu\}} \boldsymbol{e}^{\{\mu\}}$	$J_\mu e^\mu$	vector $\boldsymbol{J}$, written in the coordinate basis as (say) $J_\mu e^\mu$,
1510.06699_F00469	21	■ 1.000	$J_{\{a\}} = h_{\{a\}}^{\{\mu\}} J_{\{\mu\}}$	$J_a = h_a^\mu J_\mu$	one then identifies the quantities $J_a = h_a^\mu J_\mu$, for ex-
1510.06699_F00470	21	■ 1.000	$h_{\{a\}}^{\{*\}} J^{\{a\}} = J^{\{\mu\}}$	$h_a^\mu J^a = J^\mu$	that $h_a^\mu J^a = J^\mu$ and $b_{\mu}^a J^\mu = J^a$. Moreover, any con-
1510.06699_F00471	21	■ 1.000	$b^{\{a\}}_{\{\mu\}} J^{\{\mu\}} = J^{\{a\}}$	$b_\mu^a J^\mu = J^a$	that $h_a^\mu J^a = J^\mu$ and $b_{\mu}^a J^\mu = J^a$. Moreover, any con-
1510.06699_F00472	21	■ 1.000	$g_{\{\mu\} \{\nu\}}$	$g_{\mu\nu}$	a more general spacetime with metric $g_{\mu\nu}$. Conversely,
1510.06699_F00473	21	■ 1.000	$h^{-1} = \sqrt{-g}$	$h^{-1} = \sqrt{-g}$	From (115), one also finds that $h^{-1} = \sqrt{-g}$ (where we
1510.06699_F00474	22	■ 1.000	$w \left(g_{\{\mu\} \{\nu\}} \right) = 2$	$w(g_{\mu\nu}) = 2$	have Weyl weights $w(g_{\mu\nu}) = 2$ and $w(h_a^\mu) = -1$ respec-
1510.06699_F00475	22	■ 1.000	$w \left(h_{\{a\}}^{\{\mu\}} \right) = -1$	$w(h_a^\mu) = -1$	have Weyl weights $w(g_{\mu\nu}) = 2$ and $w(h_a^\mu) = -1$ respec-
1510.06699_F00476	22	■ 1.000	$w \left(\eta_{\{a\} \{b\}} \right) = 0$	$w(\eta_{ab}) = 0$	tively, and so (115) and (116) imply that $w(\eta_{ab}) = 0$, as
1510.06699_F00477	22	■ 1.000	$e_{\{a\}}^{\{\mu\}}$	e_a^μ	h_a^μ and b_{μ}^a as e_a^μ and e_{μ}^a , respectively.
1510.06699_F00478	22	■ 1.000	$e^{\{a\}}_{\{\mu\}}$	e^a_{μ}	h_a^μ and b_{μ}^a as e_a^μ and e_{μ}^a , respectively.
1510.06699_F00479	22	■ 1.000	$x + \delta x$	$x + \delta x$	and $x + \delta x$, which is accompanied by a local change in
1510.06699_F00480	22	■ 1.000	$J^{\{a\}}$	J^a	for some vector J^a of weight w is defined as
1510.06699_F00481	22	■ 1.000	$J^{\{a\}}(x)$	$J^a(x)$	which is required to compare vectors $J^a(x)$ and $J^a(x + \delta x)$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00520	23	■ 0.983	$\Gamma^{\sigma}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}_{\nu\mu}$	$\Gamma^{\sigma}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}_{\nu\mu}$	but with $R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}_{\sigma\mu\nu}$ and $\Gamma^{\sigma}_{\nu\mu} \rightarrow {}^0\Gamma^{*\sigma}_{\nu\mu}$, whereas
1510.06699_F00521	23	■ 1.000	$\{ \}^{\{0\}} T^{\{*\} \lambda}_{\mu\nu} = 0$	${}^0T^{*\lambda}_{\mu\nu} = 0$	(126) becomes simply ${}^0T^{*\lambda}_{\mu\nu} = 0$, indicating the ab-
1510.06699_F00522	23	■ 0.941	$\nabla_{\mu} \rightarrow {}^0\nabla_{\mu}^*, R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}_{\sigma\mu\nu}$	$\nabla_{\mu}^* \rightarrow {}^0\nabla_{\mu}^*, R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}_{\sigma\mu\nu}$	also valid, but with $\nabla_{\mu}^* \rightarrow {}^0\nabla_{\mu}^*, R^{\rho}_{\sigma\mu\nu} \rightarrow {}^0R^{*\rho}_{\sigma\mu\nu}$ and
1510.06699_F00523	23	■ 0.976	$T^{\{*\} \sigma}_{\mu\nu} \rightarrow {}^0T^{*\sigma}_{\mu\nu} = 0$	$T^{*\sigma}_{\mu\nu} \rightarrow {}^0T^{*\sigma}_{\mu\nu} = 0$	$T^{*\sigma}_{\mu\nu} \rightarrow {}^0T^{*\sigma}_{\mu\nu} = 0$.
1510.06699_F00524	23	■ 1.000	$P_{\nu} \equiv \partial_{\nu} \rho, \mathcal{P}_a \equiv h_a{}^{\nu} P_{\nu}$	$P_{\nu} \equiv \partial_{\nu} \rho, \mathcal{P}_a \equiv h_a{}^{\nu} P_{\nu}$	where $P_{\nu} \equiv \partial_{\nu} \rho, \mathcal{P}_a \equiv h_a{}^{\nu} P_{\nu}$ and θ is an arbitrary pa-
1510.06699_F00525	23	■ 1.000	θ	θ	where $P_{\nu} \equiv \partial_{\nu} \rho, \mathcal{P}_a \equiv h_a{}^{\nu} P_{\nu}$ and θ is an arbitrary pa-
1510.06699_F00526	23	■ 1.000	$w = -2$	$w = -2$	forms covariantly with weight $w = -2$ and also acts as
1510.06699_F00527	24	■ 1.000	$\mathcal{T}^a_{bc} = 0$	$\mathcal{T}^a_{bc} = 0$	field originally has vanishing PGT torsion $\mathcal{T}^a_{bc} = 0$, the
1510.06699_F00528	24	■ 1.000	$\mathcal{T}^a_{bc} \neq 0$	$\mathcal{T}^a_{bc} \neq 0$	means that the local dilation leads to $\mathcal{T}^a_{bc} \neq 0$.
1510.06699_F00529	24	■ 1.000	$\mathcal{T}^{\dagger a}_{bc}$	$\mathcal{T}^{\dagger a}_{bc}$	Thus, for general values of θ , neither $\mathcal{R}^{ab}_{cd}$ nor $\mathcal{T}^{\dagger a}_{bc}$
1510.06699_F00530	24	■ 1.000	$\theta = 0$	$\theta = 0$	transforms covariantly, but for $\theta = 0$ one recovers the
1510.06699_F00531	24	■ 1.000	$\theta = 1$	$\theta = 1$	curvature under local dilations, whereas for $\theta = 1$ one
1510.06699_F00532	24	■ 1.000	$\{ \}^{\{0\}} \mathcal{R}^{ab}_{cd}$	${}^0\mathcal{R}^{ab}_{cd}$	${}^0\mathcal{R}^{ab}_{cd}$ and its contractions (see Appendix B) are given
1510.06699_F00533	24	■ 0.971	$\{ \}^{\{0\}} \mathcal{T}^a_{bc}$	${}^0\mathcal{T}^a_{bc}$	in Section VB; the ‘reduced’ PGT torsion ${}^0\mathcal{T}^a_{bc}$ vanishes
1510.06699_F00534	24	■ 1.000	$D_{\mu}^{\dagger} \varphi$	$D_{\mu}^{\dagger} \varphi$	tive $D_{\mu}^{\dagger} \varphi$ by introducing a dilation vector gauge field
1510.06699_F00535	24	■ 1.000	V_{μ}	V_{μ}	(called V_{μ} to distinguish it from the dilation gauge field
1510.06699_F00536	24	■ 0.999	$\mathcal{V}_a = h_a{}^{\mu} V_{\mu}$	$\mathcal{V}_a = h_a{}^{\mu} V_{\mu}$	where $\mathcal{V}_a = h_a{}^{\mu} V_{\mu}$. Clearly, by construction, $A^{\dagger ab}_{\mu}$ is
1510.06699_F00537	24	■ 0.999	$A^{\dagger ab}_{\mu}$	$A^{\dagger ab}_{\mu}$	where $\mathcal{V}_a = h_a{}^{\mu} V_{\mu}$. Clearly, by construction, $A^{\dagger ab}_{\mu}$ is
1510.06699_F00538	24	■ 1.000	$\partial_{\mu} \varphi'$	$\partial_{\mu} \varphi'$	covariant derivative to cancel terms arising from $\partial_{\mu} \varphi'$
1510.06699_F00539	24	■ 1.000	$T_{\mu} =$	$T_{\mu} =$	in which w is the Weyl weight of the field φ and $T_{\mu} =$
1510.06699_F00540	24	■ 0.999	$b^a_{\mu} \mathcal{T}_a$	$b^a_{\mu} \mathcal{T}_a$	$b^a_{\mu} \mathcal{T}_a$, where $\mathcal{T}_a$ is the trace of the PGT torsion (see
1510.06699_F00541	24	■ 1.000	T_{μ}	T_{μ}	equation (66). Here we consider T_{μ} merely as a short-
1510.06699_F00542	24	■ 1.000	$T'_{\mu} = T_{\mu} + 3(1 - \theta)P_{\mu}$	$T'_{\mu} = T_{\mu} + 3(1 - \theta)P_{\mu}$	$T'_{\mu} = T_{\mu} + 3(1 - \theta)P_{\mu}$ and so (143) does indeed trans-
1510.06699_F00543	24	■ 0.898	$A^{\dagger ab}_{\mu} = A^{\dagger ab}_{\mu}$	$A^{\dagger ab}_{\mu} = A^{\dagger ab}_{\mu}$	tions (136)–(137), i.e. $A^{\dagger ab}_{\mu} = A^{\dagger ab}_{\mu}$ and so it has a
1510.06699_F00544	24	■ 1.000	V	V	covariant derivative (143) and the V -field transformation
1510.06699_F00545	25	■ 1.000	$V_{\mu} = 0$	$V_{\mu} = 0$	that one can covariantly set $V_{\mu} = 0$, so that $A^{\dagger ab}_{\mu} = A^{ab}_{\mu}$
1510.06699_F00546	25	■ 1.000	$A^{\dagger ab}_{\mu} = A^{ab}_{\mu}$	$A^{\dagger ab}_{\mu} = A^{ab}_{\mu}$	that one can covariantly set $V_{\mu} = 0$, so that $A^{\dagger ab}_{\mu} = A^{ab}_{\mu}$
1510.06699_F00547	25	■ 1.000	$D_{\mu}^{\dagger} \varphi _{\theta=0} = \left(D_{\mu} - \frac{1}{3} w T_{\mu} \right) \varphi$	$D_{\mu}^{\dagger} \varphi _{\theta=0} = \left(D_{\mu} - \frac{1}{3} w T_{\mu} \right) \varphi$	as $D_{\mu}^{\dagger} \varphi _{\theta=0} = (D_{\mu} - \frac{1}{3} w T_{\mu}) \varphi$, in which the role played
1510.06699_F00548	25	■ 1.000	$T_{\mu} = 0$	$T_{\mu} = 0$	variantly set the PGT torsion to zero, and hence $T_{\mu} = 0$,
1510.06699_F00549	25	■ 1.000	$D_{\mu}^{\dagger} \varphi _{\theta=1} \equiv \left(\partial_{\mu} + \frac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab} - w V_{\mu} \right) \varphi$	$D_{\mu}^{\dagger} \varphi _{\theta=1} \equiv \left(\partial_{\mu} + \frac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab} - w V_{\mu} \right) \varphi$	$D_{\mu}^{\dagger} \varphi _{\theta=1} \equiv (\partial_{\mu} + \frac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab} - w V_{\mu}) \varphi$.
1510.06699_F00550	25	■ 1.000	$\partial_{\mu}^{\dagger} \varphi(x)$	$\partial_{\mu}^{\dagger} \varphi(x)$	It is straightforward to show that $\partial_{\mu}^{\dagger} \varphi(x)$ itself also trans-
1510.06699_F00551	25	■ 1.000	$w - 1$	$w - 1$	which transforms covariantly with weight $w - 1$, as de-
1510.06699_F00552	25	■ 0.999	$\mathcal{L}_{\text{M}} \equiv h^{-1} L_{\text{M}}(\varphi, \mathcal{D}^{\dagger} \varphi)$	$\mathcal{L}_{\text{M}} \equiv h^{-1} L_{\text{M}}(\varphi, \mathcal{D}^{\dagger} \varphi)$	action by the Lagrangian density $\mathcal{L}_{\text{M}} \equiv h^{-1} L_{\text{M}}(\varphi, \mathcal{D}^{\dagger} \varphi)$.
1510.06699_F00553	25	■ 1.000	$\phi^2 \mathcal{R}^{\dagger}$	$\phi^2 \mathcal{R}^{\dagger}$	tional to $\phi^2 \mathcal{R}^{\dagger}$ or $\phi^2 L_{\mathcal{T}^{\dagger 2}}$ are extended Weyl-covariant
1510.06699_F00554	25	■ 1.000	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	tional to $\phi^2 \mathcal{R}^{\dagger}$ or $\phi^2 L_{\mathcal{T}^{\dagger 2}}$ are extended Weyl-covariant
1510.06699_F00555	25	■ 0.968	$\text{SU}(2, 2)$	$\text{SU}(2, 2)$	of $\text{SU}(2, 2)$, to which $\text{O}(4, 2)$ is locally isomorphic. In
1510.06699_F00556	25	■ 0.968	$\text{O}(4, 2)$	$\text{O}(4, 2)$	of $\text{SU}(2, 2)$, to which $\text{O}(4, 2)$ is locally isomorphic. In
1510.06699_F00557	26	■ 1.000	$e_{b\mu} b_a - e_{a\mu} b_b$	$e_{b\mu} b_a - e_{a\mu} b_b$	in which a piece $e_{b\mu} b_a - e_{a\mu} b_b$ is added to the spin connec-
1510.06699_F00558	26	■ 1.000	$\omega_{\mu ab}$	$\omega_{\mu ab}$	tion $\omega_{\mu ab}$ (their b -field is the ‘dilatation field’, which in
1510.06699_F00559	26	■ 0.999	$\mathcal{V}$	$\mathcal{V}$	our case corresponds to $\mathcal{V}$, while their $e_{a\mu}$ is a version of

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00560	26	■ 0.999	$e_{\{a \ \mu\}}$	$e_{a\mu}$	our case corresponds to $\mathcal{V}$, while their $e_{a\mu}$ is a version of
1510.06699_F00561	26	■ 1.000	$c_{\{b \ \mu\}} x_{\{a\}} - c_{\{a \ \mu\}} x_{\{b\}}$	$c_{b\mu}x_a - c_{a\mu}x_b$	$c_{b\mu}x_a - c_{a\mu}x_b$, where here $c_{a\mu}$ is the extra ‘vierbein-type’
1510.06699_F00562	26	■ 1.000	$c_{\{a \ \mu\}}$	$c_{a\mu}$	$c_{b\mu}x_a - c_{a\mu}x_b$, where here $c_{a\mu}$ is the extra ‘vierbein-type’
1510.06699_F00563	26	■ 1.000	$x_{\{a\}}$	x_a	tions. Here x_a is a global surrogate for $\mathcal{V}_a$ (noting the fact
1510.06699_F00564	26	■ 1.000	$\mathcal{V}_{\{a\}}$	$\mathcal{V}_a$	tions. Here x_a is a global surrogate for $\mathcal{V}_a$ (noting the fact
1510.06699_F00565	26	■ 1.000	$x \cdot \nabla$	$x \cdot \nabla$	that the generator of global dilations is $x \cdot \nabla$), and so this
1510.06699_F00566	26	■ 1.000	$\{ \}^{\{0\}} \nabla_{\{\mu\}}^{\{\dagger\}} \equiv \{ \}^{\{0\}} \nabla_{\{\mu\}} - w \left(V_{\{\mu\}} + \frac{1}{3} T_{\{\mu\}} \right), \{ \}^{\{0\}} \Gamma_{\{\mu \ \nu\}}^{\{\lambda\}} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} +$	${}^0\nabla_{\mu}^{\dagger} \equiv {}^0\nabla_{\mu} - w \left(V_{\mu} + \frac{1}{3} T_{\mu} \right), {}^0\Gamma_{\mu\nu}^{\lambda} \equiv \frac{1}{2} \gamma^{\lambda\rho} (\partial_{\mu} \gamma_{\nu\rho} +$	where ${}^0\nabla_{\mu}^{\dagger} \equiv {}^0\nabla_{\mu} - w(V_{\mu} + \frac{1}{3}T_{\mu})$, ${}^0\Gamma_{\mu\nu}^{\lambda} \equiv \frac{1}{2}\gamma^{\lambda\rho}(\partial_{\mu}\gamma_{\nu\rho} +$
1510.06699_F00567	26	■ 0.817	$\partial_{\{\nu\}} \gamma_{\{\mu \ \rho\}} - \partial_{\{\rho\}} \gamma_{\{\mu \ \nu\}}$	$\partial_{\nu} \gamma_{\mu\rho} - \partial_{\rho} \gamma_{\mu\nu}$	$\partial_{\nu} \gamma_{\mu\rho} - \partial_{\rho} \gamma_{\mu\nu}$) is the metric connection corresponding to
1510.06699_F00568	26	■ 0.801	$D_{\{\mu\}}$	D_{μ}	$D_{\mu}^{\dagger}$ is applied, and w is its Weyl weight.
1510.06699_F00569	26	■ 1.000	$T^{\{a\}\{ \}_{\{\mu \ \nu\}}}$	$T^a_{\mu\nu}$	vature) in PGT (and WGT), $T^a_{\mu\nu}$ is the PGT transla-
1510.06699_F00570	26	■ 1.000	$D_{\{\mu\}}$	D_{μ}	tional gauge field (torsion), and D_{μ} is the PGT covariant
1510.06699_F00571	26	■ 1.000	$R^{\{\dagger a \ b\}\{ \}_{\{\mu \ \nu\}}}$	$R^{\dagger ab}_{\mu\nu}$	As in WGT, both $R^{\dagger ab}_{\mu\nu}$ and $H^{\dagger}_{\mu\nu}$ transform covariantly
1510.06699_F00572	26	■ 1.000	$H_{\{\mu \ \nu\}}^{\{\dagger\}}$	$H^{\dagger}_{\mu\nu}$	As in WGT, both $R^{\dagger ab}_{\mu\nu}$ and $H^{\dagger}_{\mu\nu}$ transform covariantly
1510.06699_F00573	26	■ 1.000	$\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}} = h_{\{c\}\{ \}^{\{\mu\}} h_{\{d\}\{ \}^{\{\nu\}} R^{\{\dagger a \ b\}\{ \}_{\{\mu \ \nu\}}}$	$\mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu\nu}$	where $\mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu\nu}$ and $\mathcal{H}_{cd}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu\nu}^{\dagger}$,
1510.06699_F00574	26	■ 1.000	$\mathcal{H}_{\{c \ d\}}^{\{\dagger\}} = h_{\{c\}\{ \}^{\{\mu\}} h_{\{d\}\{ \}^{\{\nu\}} H_{\{\mu \ \nu\}}^{\{\dagger\}}$	$\mathcal{H}_{cd}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu\nu}^{\dagger}$	where $\mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\nu} R^{\dagger ab}_{\mu\nu}$ and $\mathcal{H}_{cd}^{\dagger} = h_c^{\mu} h_d^{\nu} H_{\mu\nu}^{\dagger}$,
1510.06699_F00575	27	■ 1.000	$\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}, \mathcal{H}_{\{c \ d\}}^{\{\dagger\}}$	$\mathcal{R}^{\dagger ab}_{cd}, \mathcal{H}_{cd}^{\dagger}$	It is easy to show that, similarly to WGT, $\mathcal{R}^{\dagger ab}_{cd}, \mathcal{H}_{cd}^{\dagger}$
1510.06699_F00576	27	■ 1.000	$\mathcal{T}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}}$	$\mathcal{T}^{\dagger a}_{cd}$	and $\mathcal{T}^{\dagger a}_{cd}$ transform covariantly under extended local
1510.06699_F00577	27	■ 1.000	$w \left(\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}} \right) = w \left(\mathcal{H}_{\{c \ d\}}^{\{\dagger\}} \right) = -2$	$w \left(\mathcal{R}^{\dagger ab}_{cd} \right) = w \left(\mathcal{H}_{cd}^{\dagger} \right) = -2$	dilations with weights $w(\mathcal{R}^{\dagger ab}_{cd}) = w(\mathcal{H}_{cd}^{\dagger}) = -2$ and
1510.06699_F00578	27	■ 0.999	$w \left(\mathcal{T}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}} \right) = -1$	$w \left(\mathcal{T}^{\dagger a}_{cd} \right) = -1$	$w(\mathcal{T}^{\dagger a}_{cd}) = -1$ respectively.
1510.06699_F00579	27	■ 1.000	$\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}}$	$\mathcal{R}^{\dagger ab}_{cd}$	plicit forms of $\mathcal{R}^{\dagger ab}_{cd}$ and its contractions are given in
1510.06699_F00580	27	■ 1.000	$\mathcal{H}_{\{a \ b\}}^{\{\dagger\}}$	$\mathcal{H}_{ab}^{\dagger}$	strength $\mathcal{H}_{ab}^{\dagger}$ is antisymmetric in a and b and hence also
1510.06699_F00581	27	■ 0.725	$\mathcal{R}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}(h, A, \partial A, V, \partial V), \mathcal{T}^{\{\dagger a \ b\}\{ \}_{\{c \ d\}}(h, \partial h, A)$	$\mathcal{R}^{\dagger ab}_{cd}(h, A, \partial A, V, \partial V), \mathcal{T}^{\dagger a}_{bc}(h, \partial h, A)$	tives are $\mathcal{R}^{\dagger ab}_{cd}(h, A, \partial A, V, \partial V), \mathcal{T}^{\dagger a}_{bc}(h, \partial h, A)$ and
1510.06699_F00582	27	■ 1.000	$\mathcal{H}_{\{a \ b\}}^{\{\dagger\}} \left(h, \partial h, \partial^2 h, A, \partial A, \partial V \right)$	$\mathcal{H}_{ab}^{\dagger}(h, \partial h, \partial^2 h, A, \partial A, \partial V)$	$\mathcal{H}_{ab}^{\dagger}(h, \partial h, \partial^2 h, A, \partial A, \partial V)$. It is important to note that
1510.06699_F00583	27	■ 1.000	$\partial h, \partial^2 h, A$	$\partial h, \partial^2 h, A$	strength $\mathcal{H}_{ab}^{\dagger}$, which depends on $\partial h, \partial^2 h, A$ and ∂A , in
1510.06699_F00584	27	■ 1.000	$\partial^2 h$	$\partial^2 h$	tive $\partial^2 h$ is a particularly unusual feature as compared
1510.06699_F00585	27	■ 1.000	$\{ \}^{\{0\}} \mathcal{A}_{\{a \ b \ c\}}^{\{\dagger\}} \equiv \frac{1}{2} \left(c_{\{a \ b \ c\}}^{\{\dagger\}} +$	${}^0\mathcal{A}_{abc}^{\dagger} \equiv \frac{1}{2} \left(c_{abc}^{\dagger} +$	in which we have defined the quantities ${}^0\mathcal{A}_{abc}^{\dagger} \equiv \frac{1}{2}(c_{abc}^{\dagger} +$
1510.06699_F00586	27	■ 0.973	$\left(c_{\{b \ c \ a\}}^{\{\dagger\}} - c_{\{c \ a \ b\}}^{\{\dagger\}} \right)$	$c_{bca}^{\dagger} - c_{cab}^{\dagger}$	$c_{bca}^{\dagger} - c_{cab}^{\dagger}$), where $c^{\dagger c}_{ab} \equiv h_a^{\mu} h_b^{\nu} (\partial_{\mu}^{\dagger} b^c_{\nu} - \partial_{\nu}^{\dagger} b^c_{\mu})$, and
1510.06699_F00587	27	■ 0.973	$c^{\{\dagger c \ a\}\{ \}_{\{a \ b\}} \equiv h_{\{a\}\{ \}^{\{\mu\}} h_{\{b\}\{ \}^{\{\nu\}} \left(\partial_{\mu}^{\{\dagger\}} b^c_{\nu} - \partial_{\nu}^{\{\dagger\}} b^c_{\mu} \right) -$	$c^{\dagger c}_{ab} \equiv h_a^{\mu} h_b^{\nu} (\partial_{\mu}^{\dagger} b^c_{\nu} - \partial_{\nu}^{\dagger} b^c_{\mu})$	$c_{bca}^{\dagger} - c_{cab}^{\dagger}$), where $c^{\dagger c}_{ab} \equiv h_a^{\mu} h_b^{\nu} (\partial_{\mu}^{\dagger} b^c_{\nu} - \partial_{\nu}^{\dagger} b^c_{\mu})$, and
1510.06699_F00588	27	■ 0.981	$\mathcal{K}_{\{a \ b \ c\}}^{\{\dagger\}} \equiv -\frac{1}{2} \left(\mathcal{T}_{\{a \ b \ c\}}^{\{\dagger\}} + \mathcal{T}_{\{b \ c \ a\}}^{\{\dagger\}} - \mathcal{T}_{\{c \ a \ b\}}^{\{\dagger\}} \right)$	$\mathcal{K}_{abc}^{\dagger} \equiv -\frac{1}{2} \left(\mathcal{T}_{abc}^{\dagger} + \mathcal{T}_{bca}^{\dagger} - \mathcal{T}_{cab}^{\dagger} \right)$	$\mathcal{K}_{abc}^{\dagger} \equiv -\frac{1}{2}(\mathcal{T}_{abc}^{\dagger} + \mathcal{T}_{bca}^{\dagger} - \mathcal{T}_{cab}^{\dagger})$. As before, both ${}^0\mathcal{A}_{abc}^{\dagger}$ and
1510.06699_F00589	27	■ 0.981	$\{ \}^{\{0\}} \mathcal{A}_{\{a \ b \ c\}}^{\{\dagger\}}$	${}^0\mathcal{A}_{abc}^{\dagger}$	$\mathcal{K}_{abc}^{\dagger} \equiv -\frac{1}{2}(\mathcal{T}_{abc}^{\dagger} + \mathcal{T}_{bca}^{\dagger} - \mathcal{T}_{cab}^{\dagger})$. As before, both ${}^0\mathcal{A}_{abc}^{\dagger}$ and
1510.06699_F00590	27	■ 0.840	$\mathcal{K}_{\{a \ b \ c\}}^{\{\dagger\}}$	$\mathcal{K}_{abc}^{\dagger}$	$\mathcal{K}_{abc}^{\dagger}$ are antisymmetric in their first two indices. There
1510.06699_F00591	27	■ 0.568	$\{ \}^{\{0\}} \mathcal{A}_{\{\text{abc}\}}^{\{\dagger\}}$	${}^0\mathcal{A}_{abc}^{\dagger}$	${}^0\mathcal{A}_{abc}^{\dagger}$ itself depends on the A gauge field, whereas this

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00592	27	■ 0.551	$\mathcal{K}_{\text{abc}}^{\dagger}$	$\mathcal{K}_{abc}^{\dagger}$	was not the case previously. Less importantly, $\mathcal{K}_{abc}^{\dagger}$ is
1510.06699_F00593	27	■ 0.992	$\mathcal{A}_{\text{abc}}^{\dagger}$	$\mathcal{A}_{abc}^{\dagger}$	quantities ${}^0\mathcal{A}_{abc}^{\dagger}$ transform in the same way as $\mathcal{A}_{abc}^{\dagger}$,
1510.06699_F00594	27	■ 1.000	$\{ \}^{\{0\}} \mathcal{D}_{\{\mu\}}^{\dagger} \varphi$	${}^0D_{\mu}^{\dagger} \varphi$	in eWGT, however, is that ${}^0D_{\mu}^{\dagger} \varphi$ still depends on the
1510.06699_F00595	27	■ 0.523	$\{ \}^{\{0\}} \mathcal{D}_{\text{a}}^{\dagger} \equiv h_a{}^{\mu} {}^0D_{\mu}^{\dagger}$	${}^0\mathcal{D}_a^{\dagger} \equiv h_a{}^{\mu} {}^0D_{\mu}^{\dagger}$	${}^0\mathcal{D}_a^{\dagger} \equiv h_a{}^{\mu} {}^0D_{\mu}^{\dagger}$, which consequently transforms in the
1510.06699_F00596	27	■ 1.000	$\mathcal{D}_{\text{a}}^{\dagger}$	$\mathcal{D}_a^{\dagger}$	same way as $\mathcal{D}_a^{\dagger}$, as required. The ‘full’ generalised co-
1510.06699_F00597	27	■ 1.000	$\{ \}^{\{0\}} \mathcal{D}_{\text{a}}^{\dagger}$	${}^0\mathcal{D}_a^{\dagger}$	to ${}^0\mathcal{D}_a^{\dagger}$.
1510.06699_F00598	27	■ 1.000	$\mathcal{R}^{\dagger ab}{}_{cd}, \mathcal{T}^{\dagger a}{}_{bc}$	$\mathcal{R}^{\dagger ab}{}_{cd}, \mathcal{T}^{\dagger a}{}_{bc}$	the gravitational gauge field strengths $\mathcal{R}^{\dagger ab}{}_{cd}, \mathcal{T}^{\dagger a}{}_{bc}$ and
1510.06699_F00599	28	■ 0.999	$\mathcal{T}_{\text{abc}}^{\dagger} = \epsilon_{abcd} t^d$	$\mathcal{T}_{abc}^{\dagger} = \epsilon_{abcd} t^d$	sion’ is totally antisymmetric, such that $\mathcal{T}_{abc}^{\dagger} = \epsilon_{abcd} t^d$
1510.06699_F00600	28	■ 0.800	$\mathcal{R}_{\text{abc d}}^{\dagger}$	$\mathcal{R}_{abcd}^{\dagger}$	and discussed in Section II G. In principle, from $\mathcal{R}_{abcd}^{\dagger}$,
1510.06699_F00601	28	■ 0.899	$\mathcal{T}_{\text{abc}}^{\dagger}$	$\mathcal{T}_{abc}^{\dagger}$	$\mathcal{T}_{abc}^{\dagger}$ and $\mathcal{H}_{ab}^{\dagger}$, one can construct a free gravitational ac-
1510.06699_F00602	28	■ 1.000	$\mathcal{R}^{\dagger}$	$\mathcal{R}^{\dagger} \equiv$	be quadratic in $\mathcal{R}_{abcd}^{\dagger}$ and $\mathcal{H}_{ab}^{\dagger}$, but terms linear in $\mathcal{R}^{\dagger} \equiv$
1510.06699_F00603	28	■ 1.000	$\mathcal{R}^{\dagger ab}{}_{ab}$	$\mathcal{R}^{\dagger ab}{}_{ab}$	$\mathcal{R}^{\dagger ab}{}_{ab}$ or quadratic in $\mathcal{T}_{abc}^{\dagger}$ are not allowed. Similarly,
1510.06699_F00604	28	■ 1.000	$L_{\mathcal{R}^{\dagger 2}}$	$L_{\mathcal{R}^{\dagger 2}}$	where the expressions for $L_{\mathcal{R}^{\dagger 2}}$ and $L_{\mathcal{H}^{\dagger 2}}$ are given in
1510.06699_F00605	28	■ 1.000	$L_{\mathcal{H}^{\dagger 2}}$	$L_{\mathcal{H}^{\dagger 2}}$	where the expressions for $L_{\mathcal{R}^{\dagger 2}}$ and $L_{\mathcal{H}^{\dagger 2}}$ are given in
1510.06699_F00606	28	■ 1.000	$\mathcal{R}_{\text{abc d}} \rightarrow \mathcal{R}_{\text{abc d}}^{\dagger}$	$\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^{\dagger}$	i.e. making the replacements $\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^{\dagger}$ and $\mathcal{H}_{ab} \rightarrow$
1510.06699_F00607	28	■ 1.000	$\mathcal{H}_{\text{ab}} \rightarrow$	$\mathcal{H}_{ab} \rightarrow$	i.e. making the replacements $\mathcal{R}_{abcd} \rightarrow \mathcal{R}_{abcd}^{\dagger}$ and $\mathcal{H}_{ab} \rightarrow$
1510.06699_F00608	28	■ 0.829	$\{ \}^{\{\underline{71}\}}$	$\underline{71}$	$\mathcal{H}_{ab}^{\dagger}$. Performing a similar derivation to the WGT case ⁷¹ ,
1510.06699_F00609	28	■ 0.556	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	$\phi^2 L_{\mathcal{T}^{\dagger 2}}$	form $\phi^2 \mathcal{R}^{\dagger}$ and $\phi^2 L_{\mathcal{T}^{\dagger 2}}$, where $L_{\mathcal{T}^{\dagger 2}}$ is given in (3) with
1510.06699_F00610	28	■ 0.556	$L_{\mathcal{T}^{\dagger 2}}$	$L_{\mathcal{T}^{\dagger 2}}$	form $\phi^2 \mathcal{R}^{\dagger}$ and $\phi^2 L_{\mathcal{T}^{\dagger 2}}$, where $L_{\mathcal{T}^{\dagger 2}}$ is given in (3) with
1510.06699_F00611	28	■ 0.999	$\mathcal{T}_{\text{abc}} \rightarrow \mathcal{T}_{\text{abc}}^{\dagger}$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^{\dagger}$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^{\dagger}$ (but with no term proportional to β_3 ,
1510.06699_F00612	28	■ 1.000	$\mathcal{H}_{\text{ab}}^{\dagger} \mathcal{H}^{\dagger ab}$	$\mathcal{H}_{ab}^{\dagger} \mathcal{H}^{\dagger ab}$	tional to $\mathcal{H}_{ab}^{\dagger} \mathcal{H}^{\dagger ab}$ in (166). In the matter sector, covari-
1510.06699_F00613	28	■ 1.000	$\varphi, \partial \varphi, h, \partial h, A$	$\varphi, \partial \varphi, h, \partial h, A$	on $\varphi, \partial \varphi, h, \partial h, A$ and V ; the dependence on ∂h is new
1510.06699_F00614	28	■ 0.998	$\mathcal{D}_{\text{a}}^{\dagger} \varphi$	$\mathcal{D}_a^{\dagger} \varphi$	$\mathcal{T}_a$ in $\mathcal{D}_a^{\dagger} \varphi$. We will also consider here the inclusion of
1510.06699_F00615	28	■ 1.000	∂V	∂V	tional dependence on ∂A and ∂V ; the latter is new com-
1510.06699_F00616	29	■ 0.626	$\mathcal{R}_{\text{abc d}}^{\dagger}(h, A, \partial A, V, \partial V), \mathcal{T}_{\text{abc}}^{\dagger}(h, \partial h, A)$	$\mathcal{R}_{abcd}^{\dagger}(h, A, \partial A, V, \partial V), \mathcal{T}_{abc}^{\dagger}(h, \partial h, A)$	strength tensors $\mathcal{R}_{abcd}^{\dagger}(h, A, \partial A, V, \partial V), \mathcal{T}_{abc}^{\dagger}(h, \partial h, A)$
1510.06699_F00617	29	■ 1.000	$\mathcal{H}^{\dagger ab} \mathcal{H}_{ab}^{\dagger}$	$\mathcal{H}^{\dagger ab} \mathcal{H}_{ab}^{\dagger}$	the term proportional to $\mathcal{H}^{\dagger ab} \mathcal{H}_{ab}^{\dagger}$ is included in $\mathcal{L}_G$, then
1510.06699_F00618	29	■ 1.000	$t^a{}_{\mu} \equiv \delta \mathcal{L}_G / \delta h_a{}^{\mu}, s_{ab}{}^{\mu} \equiv \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$	$t^a{}_{\mu} \equiv \delta \mathcal{L}_G / \delta h_a{}^{\mu}, s_{ab}{}^{\mu} \equiv \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$	sector, $t^a{}_{\mu} \equiv \delta \mathcal{L}_G / \delta h_a{}^{\mu}, s_{ab}{}^{\mu} \equiv \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ and $j^{\mu} \equiv$
1510.06699_F00619	29	■ 1.000	$j^{\mu} \equiv$	$j^{\mu} \equiv$	sector, $t^a{}_{\mu} \equiv \delta \mathcal{L}_G / \delta h_a{}^{\mu}, s_{ab}{}^{\mu} \equiv \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ and $j^{\mu} \equiv$
1510.06699_F00620	29	■ 1.000	$\delta \mathcal{L}_G / \delta V_{\mu}$	$\delta \mathcal{L}_G / \delta V_{\mu}$	$\delta \mathcal{L}_G / \delta V_{\mu}$ respectively. In the matter sector, the energy-
1510.06699_F00621	29	■ 1.000	$\zeta^{\mu} \equiv \delta \mathcal{L}_M / \delta V_{\mu}$	$\zeta^{\mu} \equiv \delta \mathcal{L}_M / \delta V_{\mu}$	$\sigma_{ab}{}^{\mu} \equiv \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$ and dilation current $\zeta^{\mu} \equiv \delta \mathcal{L}_M / \delta V_{\mu}$
1510.06699_F00622	29	■ 0.998	$t^a{}_b$	$t^a{}_b$	to WGT, however, the individual quantities $t^a{}_b$ and $\tau^a{}_b$
1510.06699_F00623	29	■ 1.000	$t^a{}_a$	$t^a{}_a$	although it is worth noting that the traces $t^a{}_a$ and $\tau^a{}_a$
1510.06699_F00624	29	■ 1.000	$\tau^a{}_a$	$\tau^a{}_a$	although it is worth noting that the traces $t^a{}_a$ and $\tau^a{}_a$
1510.06699_F00625	29	■ 1.000	t_{ab}	t_{ab}	momentum tensor. Nonetheless, although neither t_{ab} nor
1510.06699_F00626	29	■ 1.000	$\tau^{\dagger a}{}_a = \tau^a{}_a$	$\tau^{\dagger a}{}_a = \tau^a{}_a$	is invariant). It is also worth noting that $\tau^{\dagger a}{}_a = \tau^a{}_a$.
1510.06699_F00627	29	■ 0.999	$t_{ab}^{\dagger}$	$t_{ab}^{\dagger}$	One may construct a covariant $t_{ab}^{\dagger}$ in a similar way. In-
1510.06699_F00628	29	■ 1.000	$\mathcal{T}_{\text{a}}^{\dagger} \equiv 0$	$\mathcal{T}_a^{\dagger} \equiv 0$	ever, since $\mathcal{T}_a^{\dagger} \equiv 0$. Consequently, in eWGT, one finds

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00629	29	■ 1.000	$\bar{\partial}\mathrm{L}_{\mathrm{M}}/\partial\varphi\equiv\left[\partial L_{\mathrm{M}}\left(\varphi,\mathcal{D}_a^\dagger u,\phi,\mathcal{D}_a^\dagger\phi,\dots\right)/\partial\varphi\right]_{u=\varphi}$	$\bar{\partial}L_{\mathrm{M}}/\partial\varphi\equiv\left[\partial L_{\mathrm{M}}\left(\varphi,\mathcal{D}_a^\dagger u,\phi,\mathcal{D}_a^\dagger\phi,\dots\right)/\partial\varphi\right]_{u=\varphi}$	where $\bar{\partial}L_{\mathrm{M}}/\partial\varphi\equiv\left[\partial L_{\mathrm{M}}(\varphi,\mathcal{D}_a^\dagger u,\phi,\mathcal{D}_a^\dagger\phi,\dots)/\partial\varphi\right]_{u=\varphi}$, so
1510.06699_F00630	30	■ 1.000	$H_{\{a\}b}^\dagger$	$H_{ab}^\dagger$	strength tensor $H_{ab}^\dagger$ of the vector gauge field (152), but
1510.06699_F00631	30	■ 1.000	$H_{\{a\}b}=h_{\{a\}\{\}\}^\mu h_{\{b\}\{\}\}^\nu\left(\partial_{\mu}\mathrm{V}_{\nu}-\partial_{\nu}\mathrm{V}_{\mu}\right)$	$H_{ab}=h_a{}^\mu h_b{}^\nu\left(\partial_\mu V_\nu-\partial_\nu V_\mu\right)$	instead the simpler object $\overline{H}_{ab}=h_a{}^\mu h_b{}^\nu\left(\partial_\mu V_\nu-\partial_\nu V_\mu\right)$,
1510.06699_F00632	30	■ 1.000	$j^{\dagger a}$	$j^{\dagger a}$	by $j^{\dagger a}$ rather than simply by j^a . The largest difference
1510.06699_F00633	30	■ 1.000	$j^{\{a\}}$	j^a	by $j^{\dagger a}$ rather than simply by j^a . The largest difference
1510.06699_F00634	30	■ 1.000	$\varphi,\phi,h_{\{a\}}^\mu,A^{\{a\}b}_{\{\}\mu},V_{\{\mu\}}$	$\varphi,\phi,h_a^\mu,A^{ab}_\mu,V_\mu$	from the set $\varphi,\phi,h_a^\mu,A^{ab}_\mu,V_\mu$ and their derivatives,
1510.06699_F00635	30	■ 1.000	φ,ϕ	φ,ϕ	with which we have worked so far, to the new set $\varphi,\phi,$
1510.06699_F00636	30	■ 1.000	$h_{\{a\}}^\mu,A^{\dagger ab}_{\{\}\mu},V_{\{\mu\}}$	$h_a^\mu,A^{\dagger ab}_\mu,V_\mu$	$h_a^\mu,A^{\dagger ab}_\mu,V_\mu$ and their derivatives (i.e. we replace A^{ab}_μ
1510.06699_F00637	30	■ 1.000	$h_{\{a\}}^\mu,A^{\{a\}b}_{\{\}\mu},V_{\{\mu\}}$	h_a^μ,A^{ab}_μ,V_μ	of h_a^μ,A^{ab}_μ,V_μ by its defining relationship (142), rather
1510.06699_F00638	30	■ 1.000	$\mathrm{T}_{\{b\}}^{\natural}\equiv\mathrm{T}^{\natural a}_{ba}$	$\mathcal{T}_b^\natural\equiv\mathcal{T}^\natural a_{ba}$	and the corresponding trace $\mathcal{T}_b^\natural\equiv\mathcal{T}^\natural a_{ba}$. It is straight-
1510.06699_F00639	30	■ 1.000	$\mathrm{T}_{\{b\}}^{\natural}=\mathrm{T}_{\{b\}}+3\mathcal{V}_b$	$\mathcal{T}_b^\natural=\mathcal{T}_b+3\mathcal{V}_b$	forward to show that $\mathcal{T}_b^\natural=\mathcal{T}_b+3\mathcal{V}_b$, where $\mathcal{T}_b$ is the
1510.06699_F00640	31	■ 0.997	$h,\partial h$	$h,\partial h$	which is expressible wholly in terms of the fields $h,\partial h$
1510.06699_F00641	31	■ 1.000	$A^{\dagger}$	$A^\dagger$	and $A^\dagger$ (suppressing indices). Since (182) does not de-
1510.06699_F00642	31	■ 0.566	$\mathrm{quad}\,\mathrm{R}_{\{a\}b\,c\,d}^{\dagger}\mathrm{left}(h,A^{\dagger}\mathrm{right}),\mathrm{quad}\,\mathrm{T}_{\{a\}b\,c}^{\dagger}\mathrm{left}(h,\partial h,A^{\dagger}\mathrm{right})\mathrm{quad}$	$\mathcal{R}_{abcd}^\dagger(h,A^\dagger,\partial A^\dagger),\quad\mathcal{T}_{abc}^\dagger(h,\partial h,A^\dagger)$	tensors $\mathcal{R}_{abcd}^\dagger(h,A^\dagger,\partial A^\dagger),\quad\mathcal{T}_{abc}^\dagger(h,\partial h,A^\dagger)$ and
1510.06699_F00643	31	■ 1.000	$\mathcal{H}_{\{a\}b}^\dagger(h,\partial h,\partial^2 h,A^\dagger,\partial A^\dagger)$	$\mathcal{H}_{ab}^\dagger(h,\partial h,\partial^2 h,A^\dagger,\partial A^\dagger)$	$\mathcal{H}_{ab}^\dagger(h,\partial h,\partial^2 h,A^\dagger,\partial A^\dagger)$, where once again the last
1510.06699_F00644	31	■ 1.000	$\mathcal{L}$	$\mathcal{L}$	derivative of a Lagrangian $\mathcal{L}$ with respect to any one
1510.06699_F00645	31	■ 1.000	$(\delta\mathcal{L}/\delta\chi)_{\dagger}$	$(\delta\mathcal{L}/\delta\chi)_{\dagger}$	of the fields χ by $\overline{(\delta\mathcal{L}/\delta\chi)}$ to distinguish it from the
1510.06699_F00646	31	■ 1.000	$\delta\mathcal{L}/\delta\chi$	$\delta\mathcal{L}/\delta\chi$	variational derivative $\delta\mathcal{L}/\delta\chi$ obtained previously using
1510.06699_F00647	31	■ 1.000	$\sigma_{ab}{}^\mu$	$\sigma_{ab}{}^\mu$	ther that the spin-angular momentum tensor $\sigma_{ab}{}^\mu$ and
1510.06699_F00648	31	■ 1.000	$\zeta^{\dagger\mu}=0$	$\zeta^{\dagger\mu}=0$	gauge field V_μ or its derivative, then $\zeta^{\dagger\mu}=0$ and so
1510.06699_F00649	31	■ 1.000	$\phi,h_{\{a\}\{\}\}^\mu,A^{\{a\}b}_{\{\}\mu},V_{\{\mu\}}$	$\phi,h_a{}^\mu,A^{ab}_\mu,V_\mu$	$\overline{\phi},h_a{}^\mu,A^{ab}_\mu,V_\mu$ and their derivatives, but we will use the
1510.06699_F00650	32	■ 1.000	$\mathcal{D}_{\{a\}}^*\rightarrow\mathcal{D}_{\{a\}}^\dagger$	$\mathcal{D}_a^*\rightarrow\mathcal{D}_a^\dagger$	with $\mathcal{D}_a^*\rightarrow\mathcal{D}_a^\dagger$. Similar to WGT, since both ψ and $\bar{\psi}$ have
1510.06699_F00651	32	■ 1.000	$\gamma^a\mathrm{left}(\mathrm{D}_{\{a\}}+\frac{1}{2})\mathrm{right}\,\mathrm{psi}=\gamma^a\mathrm{D}_{\{a\}}^\dagger\mathrm{psi}$	$\gamma^a\left(\mathcal{D}_a+\frac{1}{2}\mathcal{T}_a\right)\psi=\gamma^a\mathcal{D}_a^\dagger\psi$	noting that $\overline{\gamma^a(\mathcal{D}_a+\frac{1}{2}\mathcal{T}_a)\psi}=\overline{\gamma^a\mathcal{D}_a^\dagger\psi}$, where we have
1510.06699_F00652	32	■ 1.000	$\gamma^a\Sigma_{ba}=\frac{1}{4}\gamma^a\left[\gamma_b,\gamma_a\right]=-\frac{3}{2}\gamma_b$	$\gamma^a\Sigma_{ba}=\frac{1}{4}\gamma^a\left[\gamma_b,\gamma_a\right]=-\frac{3}{2}\gamma_b$	$\overline{\gamma^a\Sigma_{ba}}=\frac{1}{4}\overline{\gamma^a\left[\gamma_b,\gamma_a\right]}=-\frac{3}{2}\overline{\gamma_b}$. We also note that (188)
1510.06699_F00653	32	■ 0.997	$\mathrm{T}_{[abc]}=0$	$\mathcal{T}_{[abc]}=0$	presence of torsion (even in the case $\overline{\mathcal{T}_{[abc]}}=\overline{0}$) will still
1510.06699_F00654	32	■ 1.000	$\{\}^{\{0\}}\mathrm{D}_{\{a\}}\rightarrow\{\}^{\{0\}}\mathrm{D}_{\{a\}}^\dagger$	${}^0\mathcal{D}_a\rightarrow{}^0\mathcal{D}_a^\dagger$	again satisfied, but with the replacements ${}^0\mathcal{D}_a\rightarrow{}^0\mathcal{D}_a^\dagger$
1510.06699_F00655	32	■ 1.000	$\mathrm{T}_{[abc]}\rightarrow\mathrm{T}_{[abc]}^\dagger$	$\mathcal{T}_{[abc]}\rightarrow\mathcal{T}_{[abc]}^\dagger$	and $\overline{\mathcal{T}_{[abc]}}\rightarrow\overline{\mathcal{T}_{[abc]}^\dagger}$. Thus, only the total antisymmetric
1510.06699_F00656	32	■ 1.000	$\sigma_{ab}{}^c$	$\sigma_{ab}{}^c$	$\sigma_{ab}{}^c$ is already manifestly eWGT covariant. For the
1510.06699_F00657	32	■ 1.000	$\sigma_{ab}{}^b=0$	$\sigma_{ab}{}^b=0$	Recalling that $\sigma_{ab}{}^b=0$ for Dirac matter, the transfor-
1510.06699_F00658	32	■ 0.444	β_1	β_1	stants β_1 and β_2 in $L_{\mathcal{T}+2}$. It is worth noting that $\mu\phi$ has
1510.06699_F00659	32	■ 0.444	β_2	β_2	stants β_1 and β_2 in $L_{\mathcal{T}+2}$. It is worth noting that $\mu\phi$ has
1510.06699_F00660	32	■ 0.444	$\mathrm{L}_{\mathrm{T}^+}$	$L_{\mathcal{T}+\epsilon}$	stants β_1 and β_2 in $L_{\mathcal{T}+2}$. It is worth noting that $\mu\phi$ has

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00661	32	■ 0.998	$\widehat{\mathcal{F}}_{ab}^* \rightarrow \widehat{\mathcal{F}}_{ab}^\dagger, \mathcal{T}_{ab}^{*c} \rightarrow \mathcal{T}_{ab}^{\dagger c}$		$\widehat{\mathcal{F}}_{ab}^* \rightarrow \widehat{\mathcal{F}}_{ab}^\dagger, \mathcal{T}_{ab}^{*c} \rightarrow \mathcal{T}_{ab}^{\dagger c}$ and $\mathcal{D}_a^* \rightarrow \mathcal{D}_a^\dagger$, and recalling
1510.06699_F00662	33	■ 1.000	$\tau^{\dagger a}{}_b = \tau^a{}_b$	$\tau^{\dagger a}{}_b = \tau^a{}_b$	deed, from (170), one sees immediately that $\tau^{\dagger a}{}_b = \tau^a{}_b$,
1510.06699_F00663	33	■ 1.000	φ, ϕ, h_a^μ	φ, ϕ, h_a^μ	set of field variables φ, ϕ, h_a^μ and $A^{\dagger ab}{}_\mu$ discussed in
1510.06699_F00664	33	■ 0.607	$\widehat{\mathcal{A}}^{\dagger ab}{}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{\dagger ab}{}_\mu$	$\widehat{\mathcal{A}}^{\dagger ab}{}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{\dagger ab}{}_\mu$	variables $\widehat{\mathcal{A}}^{\dagger ab}{}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{\dagger ab}{}_\mu$
1510.06699_F00665	33	■ 1.000	$\mathcal{D}_a^\dagger \chi = \left(\mathcal{D}_a^\natural - \frac{1}{3} w \mathcal{T}_a^\natural \right) \chi$	$\mathcal{D}_a^\dagger \chi = \left(\mathcal{D}_a^\natural - \frac{1}{3} w \mathcal{T}_a^\natural \right) \chi$	written in the form $\mathcal{D}_a^\dagger \chi = (\mathcal{D}_a^\natural - \frac{1}{3} w \mathcal{T}_a^\natural) \chi$, where $\mathcal{D}_a^\natural$
1510.06699_F00666	33	■ 1.000	$\mathcal{D}_a^\natural$	$\mathcal{D}_a^\natural$	written in the form $\mathcal{D}_a^\dagger \chi = (\mathcal{D}_a^\natural - \frac{1}{3} w \mathcal{T}_a^\natural) \chi$, where $\mathcal{D}_a^\natural$
1510.06699_F00667	33	■ 1.000	$\mathcal{T}_a^\natural$	$\mathcal{T}_a^\natural$	and $\mathcal{T}_a^\natural$ are defined in (180) and (181), respectively. In
1510.06699_F00668	33	■ 1.000	$\widehat{\mathcal{D}}_a^\natural$	$\widehat{\mathcal{D}}_a^\natural$	an analogous manner, we may define the quantities $\widehat{\mathcal{D}}_a^\natural$
1510.06699_F00669	33	■ 1.000	$\widehat{\mathcal{T}}_a^\natural$	$\widehat{\mathcal{T}}_a^\natural$	and $\widehat{\mathcal{T}}_a^\natural$, for which each occurrence of $h_a{}^\mu$ and $A^{\dagger ab}{}_\mu$ in
1510.06699_F00670	33	■ 1.000	$\widehat{\mathcal{A}}^{\dagger ab}{}_\mu$	$\widehat{\mathcal{A}}^{\dagger ab}{}_\mu$	terpart $\widehat{h}_a{}^\mu$ and $\widehat{A}^{\dagger ab}{}_\mu$, respectively. Thus, $\widehat{\mathcal{D}}_a^\natural = \widehat{h}_a{}^\mu \mathcal{D}_\mu^\natural$
1510.06699_F00671	33	■ 1.000	$\widehat{\mathcal{D}}_a^\natural = \widehat{h}_a{}^\mu \mathcal{D}_\mu^\natural$	$\widehat{\mathcal{D}}_a^\natural = \widehat{h}_a{}^\mu \mathcal{D}_\mu^\natural$	terpart $\widehat{h}_a{}^\mu$ and $\widehat{A}^{\dagger ab}{}_\mu$, respectively. Thus, $\widehat{\mathcal{D}}_a^\natural = \widehat{h}_a{}^\mu \mathcal{D}_\mu^\natural$
1510.06699_F00672	33	■ 1.000	$\mathcal{D}_a^\dagger \phi = \frac{1}{3} (\phi^2 / \phi_0) \widehat{\mathcal{T}}_a^\natural$	$\mathcal{D}_a^\dagger \phi = \frac{1}{3} (\phi^2 / \phi_0) \widehat{\mathcal{T}}_a^\natural$	be written as $\mathcal{D}_a^\dagger \phi = \frac{1}{3} (\phi^2 / \phi_0) \widehat{\mathcal{T}}_a^\natural$, whereas the covariant
1510.06699_F00673	33	■ 0.996	$\widehat{\mathcal{D}}_a^\dagger = \widehat{\mathcal{D}}_a^\natural - \frac{1}{3} w \widehat{\mathcal{T}}_a^\natural$	$\widehat{\mathcal{D}}_a^\dagger = \widehat{\mathcal{D}}_a^\natural - \frac{1}{3} w \widehat{\mathcal{T}}_a^\natural$	tive operator $\widehat{\mathcal{D}}_a^\dagger = \widehat{\mathcal{D}}_a^\natural - \frac{1}{3} w \widehat{\mathcal{T}}_a^\natural$, which preserves the Weyl
1510.06699_F00674	33	■ 1.000	$\widehat{\mathcal{D}}_a^\dagger \widehat{\chi}$	$\widehat{\mathcal{D}}_a^\dagger \widehat{\chi}$	tor $(\phi / \phi_0)^{1-w}$, the scale-invariant quantity $\widehat{\mathcal{D}}_a^\dagger \widehat{\chi}$ has the
1510.06699_F00675	33	■ 1.000	$\widehat{\chi}, \widehat{h}_a{}^\mu$	$\widehat{\chi}, \widehat{h}_a{}^\mu$	<i>same</i> functional dependency on $\widehat{\chi}, \widehat{h}_a{}^\mu$, and $\widehat{A}^{\dagger ab}{}_\mu$, respec-
1510.06699_F00676	33	■ 1.000	$\mathcal{D}_a^\dagger \chi$	$\mathcal{D}_a^\dagger \chi$	tively, as $\mathcal{D}_a^\dagger \chi$ does on $\chi, h_a{}^\mu$ and $A^{\dagger ab}{}_\mu$. Similarly, one
1510.06699_F00677	33	■ 0.999	$\mathcal{R}^{\dagger ab}{}_{cd} = (\phi / \phi_0)^2 \widehat{\mathcal{R}}^{\dagger ab}{}_{cd}$	$\mathcal{R}^{\dagger ab}{}_{cd} = (\phi / \phi_0)^2 \widehat{\mathcal{R}}^{\dagger ab}{}_{cd}$	defined in (41) may be written as $\mathcal{R}^{\dagger ab}{}_{cd} = (\phi / \phi_0)^2 \widehat{\mathcal{R}}^{\dagger ab}{}_{cd}$,
1510.06699_F00678	33	■ 1.000	$\mathcal{T}^{\dagger a}{}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{\dagger a}{}_{bc}$	$\mathcal{T}^{\dagger a}{}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{\dagger a}{}_{bc}$	$\mathcal{T}^{\dagger a}{}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{\dagger a}{}_{bc}$ and $\mathcal{H}_{ab}^\dagger = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}^\dagger$, where each
1510.06699_F00679	33	■ 1.000	$\mathcal{H}_{ab}^\dagger = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}^\dagger$	$\mathcal{H}_{ab}^\dagger = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}^\dagger$	$\mathcal{T}^{\dagger a}{}_{bc} = (\phi / \phi_0) \widehat{\mathcal{T}}^{\dagger a}{}_{bc}$ and $\mathcal{H}_{ab}^\dagger = (\phi / \phi_0)^2 \widehat{\mathcal{H}}_{ab}^\dagger$, where each
1510.06699_F00680	33	■ 1.000	$\varphi, h_a{}^\mu$	$\varphi, h_a{}^\mu$	if χ represents $\varphi, h_a{}^\mu$, or $A^{\dagger ab}{}_\mu$, one may immediately
1510.06699_F00681	34	■ 1.000	φ, h	φ, h	for the φ, h and A (or $A^\dagger$) fields are the only three <i>in-</i>
1510.06699_F00682	34	■ 1.000	$\varphi, \phi, h_a^\mu, A^{ab}{}_\mu$	$\varphi, \phi, h_a^\mu, A^{ab}{}_\mu$	ables $\varphi, \phi, h_a^\mu, A^{ab}{}_\mu$ and V_μ , rather than the alternative
1510.06699_F00683	34	■ 1.000	$\widehat{V}_\mu = \frac{1}{3} \widehat{\mathcal{T}}_\mu^\natural$	$\widehat{V}_\mu = \frac{1}{3} \widehat{\mathcal{T}}_\mu^\natural$	where it is clear from the last definition that $\widehat{V}_\mu = \frac{1}{3} \widehat{\mathcal{T}}_\mu^\natural$.
1510.06699_F00684	34	■ 1.000	$\widehat{\mathcal{V}}_a \equiv \widehat{h}_a{}^\mu \widehat{V}_\mu$	$\widehat{\mathcal{V}}_a \equiv \widehat{h}_a{}^\mu \widehat{V}_\mu$	variables $\widehat{\mathcal{A}}^{\dagger ab}{}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{\dagger ab}{}_\mu$ and $\widehat{\mathcal{V}}_a \equiv \widehat{h}_a{}^\mu \widehat{V}_\mu$. The situa-
1510.06699_F00685	34	■ 1.000	φ, h_a^μ	φ, h_a^μ	Specifically, the equations of motion in terms of φ, h_a^μ ,
1510.06699_F00686	34	■ 1.000	$V_\mu + \frac{1}{3} T_\mu$	$V_\mu + \frac{1}{3} T_\mu$	tions $A^{\dagger ab}{}_\mu$ and $V_\mu + \frac{1}{3} T_\mu$ (which will always be possible).
1510.06699_F00687	34	■ 1.000	$\widehat{\varphi}, \widehat{h}_a{}^\mu, \widehat{A}^{\dagger ab}{}_\mu$	$\widehat{\varphi}, \widehat{h}_a{}^\mu, \widehat{A}^{\dagger ab}{}_\mu$	$\widehat{\varphi}, \widehat{h}_a{}^\mu, \widehat{A}^{\dagger ab}{}_\mu$ and $\widehat{V}_\mu$, respectively.
1510.06699_F00688	34	■ 1.000	$\widehat{V}_\mu$	$\widehat{V}_\mu$	$\widehat{\varphi}, \widehat{h}_a{}^\mu, \widehat{A}^{\dagger ab}{}_\mu$ and $\widehat{V}_\mu$, respectively.
1510.06699_F00689	34	■ 1.000	$v^c (\mathcal{D}_c^\dagger p_a -$	$v^c (\mathcal{D}_c^\dagger p_a -$	in a manifestly eWGT-covariant manner as $v^c (\mathcal{D}_c^\dagger p_a -$
1510.06699_F00690	34	■ 0.971	$\left. \mathcal{T}_{cab}^\dagger p^b \right) = e \mu^2 \phi \mathcal{D}_a^\dagger \phi$	$\mathcal{T}_{cab}^\dagger p^b \Big) = e \mu^2 \phi \mathcal{D}_a^\dagger \phi$	$\mathcal{T}_{cab}^\dagger p^b) = e \mu^2 \phi \mathcal{D}_a^\dagger \phi$. By analogy with our treatment in

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00691	34	■ 0.997	$v^{\{c\}\ \}^{\{0\}} \mathcal{D}_{c\}^{\dagger} p_a = e \mu^2 \phi^0 \mathcal{D}_a^{\dagger} \phi$ $\phi^{\{0\}} \mathcal{D}_{\}^{\dagger} \phi$	$v^{c0} \mathcal{D}_c^{\dagger} p_a = e \mu^2 \phi^0 \mathcal{D}_a^{\dagger} \phi$	$v^c {}^0 \mathcal{D}_c^{\dagger} p_a = e \mu^2 \phi^0 \mathcal{D}_a^{\dagger} \phi$, which is again manifestly eWGT-
1510.06699_F00692	34	■ 1.000	$\{ \}^{\{0\}} \mathcal{D}_{\mu}^{\dagger}$	${}^0 \mathcal{D}_{\mu}^{\dagger}$	covariant derivative operator ${}^0 \mathcal{D}_{\mu}^{\dagger}$ in (159), to which the
1510.06699_F00693	34	■ 0.718	$\mathcal{T}_{\{a\} \{b\} \{c\}}^{\dagger} \equiv 0$	$\mathcal{T}_{abc}^{\dagger} \equiv 0$	with $\mathcal{T}_{abc}^{\dagger} \equiv 0$ (and hence $\mathcal{K}_{abc}^{\dagger} \equiv 0$). In contrast to
1510.06699_F00694	34	■ 0.718	$\left(\mathcal{K}_{\{a\} \{b\} \{c\}}^{\dagger} \equiv 0 \right)$	$\mathcal{K}_{abc}^{\dagger} \equiv 0$	with $\mathcal{T}_{abc}^{\dagger} \equiv 0$ (and hence $\mathcal{K}_{abc}^{\dagger} \equiv 0$). In contrast to
1510.06699_F00695	35	■ 1.000	$\{ \}^{\{0\}} R^{\dagger} \{a\} \{b\} \{ \}_{\mu} \left(h, \partial h, \partial^2 h, A, \partial A, V, \partial V \right)$	${}^0 R^{\dagger ab}_{\mu\nu} (h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$	${}^0 R^{\dagger ab}_{\mu\nu} (h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$, which is again given by
1510.06699_F00696	35	■ 1.000	$\{ \}^{\{0\}} A^{\dagger} \{a\} \{b\} \{ \}_{\mu}$	${}^0 A^{\dagger ab}_{\mu}$	the formula (39), but with A^{ab}_{μ} replaced by ${}^0 A^{\dagger ab}_{\mu}$.
1510.06699_F00697	35	■ 1.000	$\{ \}^{\{0\}} R^{\dagger} \{a\} \{b\} \{ \}_{\mu} \{ \nu \}$	${}^0 R^{\dagger ab}_{\mu\nu}$	Unlike in WGT, however, ${}^0 R^{\dagger ab}_{\mu\nu}$ depends on all the
1510.06699_F00698	35	■ 1.000	$R^{\dagger} \{a\} \{b\} \{ \}_{\mu} \{ \nu \} (h, A, \partial A, V, \partial V)$	$R^{\dagger ab}_{\mu\nu} (h, A, \partial A, V, \partial V)$	$R^{\dagger ab}_{\mu\nu} (h, A, \partial A, V, \partial V)$. Considering instead the com-
1510.06699_F00699	35	■ 1.000	$\{ \}^{\{0\}} \mathcal{R}^{\dagger} \{a\} \{b\} \{ \}_{cd} = h_{\{c\} \{d\}} \{ \}^{\{ \mu \}} h_{\{d\} \{ \}^{\{ \mu \}} \{ \}^{\{0\}} R^{\dagger} \{a\} \{b\} \{ \}_{\mu} \{ \nu \}$	${}^0 \mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\mu 0} R^{\dagger ab}_{\mu\nu}$	where ${}^0 \mathcal{R}^{\dagger ab}_{cd} = h_c^{\mu} h_d^{\mu 0} R^{\dagger ab}_{\mu\nu}$, and the commuta-
1510.06699_F00700	35	■ 1.000	$\{ \}^{\{0\}} \mathcal{T}_{\{a\} \{b\} \{c\}}^{\dagger} \equiv$	${}^0 \mathcal{T}_{bc}^{\dagger a} \equiv$	strength of the h gauge field (or ‘torsion’), since ${}^0 \mathcal{T}_{bc}^{\dagger a} \equiv$
1510.06699_F00701	35	■ 1.000	$h_{\{b\} \{ \}^{\{ \mu \}} h_{\{c\} \{ \}^{\{ \nu \}} \left(\{ \}^{\{0\}} \mathcal{D}_{\mu}^{\dagger} b^a_{\nu} - {}^0 \mathcal{D}_{\nu}^{\dagger} b^a_{\mu} \right) = 0$	$h_b^{\mu} h_c^{\nu} ({}^0 \mathcal{D}_{\mu}^{\dagger} b^a_{\nu} - {}^0 \mathcal{D}_{\nu}^{\dagger} b^a_{\mu}) = 0$	$h_b^{\mu} h_c^{\nu} ({}^0 \mathcal{D}_{\mu}^{\dagger} b^a_{\nu} - {}^0 \mathcal{D}_{\nu}^{\dagger} b^a_{\mu}) = 0$. As one might expect, the
1510.06699_F00702	35	■ 0.785	$\mathcal{D}_{\}^{\dagger} \{a\}^{\dagger} \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$	$\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$	with the replacements $\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$ and
1510.06699_F00703	35	■ 0.831	$\mathcal{T}_{\{a\} \{b\} \{c\}}^{\dagger} \rightarrow \mathcal{T}_{\{a\} \{b\} \{c\}}^{\dagger} = 0$	$\mathcal{T}_{abc}^{\dagger} \rightarrow {}^0 \mathcal{T}_{abc}^{\dagger} = 0$	$\mathcal{T}_{abc}^{\dagger} \rightarrow {}^0 \mathcal{T}_{abc}^{\dagger} = 0$.
1510.06699_F00704	35	■ 1.000	$\{ \}^{\{0\}} \mathcal{R}^{\dagger} \{a\} \{b\} \{ \}_{cd} \rightarrow$	${}^0 \mathcal{R}^{\dagger ab}_{cd} \rightarrow$	(110c), but with the replacements on the RHS ${}^0 \mathcal{R}^{\dagger ab}_{cd} \rightarrow$
1510.06699_F00705	35	■ 1.000	$\{ \}^{\{0\}} \mathcal{R}^{\dagger} \{a\} \{b\} \{ \}_{cd}, \{ \}^{\{0\}} \mathcal{D}_{\}^{\dagger} \{a\} \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{K}_{abc} \rightarrow \mathcal{K}_{abc}^{\dagger}$	${}^0 \mathcal{R}^{\dagger ab}_{cd}, {}^0 \mathcal{D}_a \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{K}_{abc} \rightarrow \mathcal{K}_{abc}^{\dagger}$	${}^0 \mathcal{R}^{\dagger ab}_{cd}, {}^0 \mathcal{D}_a \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{K}_{abc} \rightarrow \mathcal{K}_{abc}^{\dagger}$ and $\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^{\dagger}$, and
1510.06699_F00706	35	■ 1.000	$\mathcal{R}^{\dagger} \{a\} \{b\} \{ \}_{cd} \rightarrow \mathcal{R}^{\dagger ab}_{cd}$	$\mathcal{R}^{\dagger ab}_{cd} \rightarrow {}^0 \mathcal{R}^{\dagger ab}_{cd}$	which is based on (4) with $\mathcal{R}^{\dagger ab}_{cd} \rightarrow {}^0 \mathcal{R}^{\dagger ab}_{cd}$ and $\mathcal{H}_{ab} \rightarrow$
1510.06699_F00707	35	■ 1.000	$\{ \}^{\{0\}} \mathcal{R}^{\dagger} \{a\} \{b\}$	${}^0 \mathcal{R}^{\dagger}_{ab}$	${}^0 \mathcal{R}^{\dagger}_{ab}$. As was the case for WGT, one can simplify (203)
1510.06699_F00708	35	■ 0.999	$\{ \}^{\{0\}} \mathcal{R}_{\{a\} \{b\} \{c\} \{d\}}^{\dagger}$	${}^0 \mathcal{R}_{abcd}^{\dagger}$	still further (in $D \leq 4$ dimensions), since ${}^0 \mathcal{R}_{abcd}^{\dagger}$ and its
1510.06699_F00709	35	■ 1.000	$\mathcal{R}_{\{a\} \{b\} \{c\} \{d\}} \rightarrow \mathcal{R}_{\{a\} \{b\} \{c\} \{d\}}^{\dagger}$	$\mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$	form (59), but with $\mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$. Thus, one can set to
1510.06699_F00710	35	■ 0.535	$\mathcal{D}_{\}^{\dagger} \{a\}^{\dagger} \mathcal{D}_{\}^{\dagger}$	$\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}$	ment $\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}$. As in WGT, however, the gravitational
1510.06699_F00711	35	■ 0.720	$\phi^2 \mathcal{R}^{\dagger}$	$\phi^{20} \mathcal{R}^{\dagger}$	$\mathcal{L}_G$ and the term proportional to $\phi^2 {}^0 \mathcal{R}^{\dagger}$ in $\mathcal{L}_M$ depends
1510.06699_F00712	35	■ 0.806	$\mathcal{D}_{\}^{\dagger} \{a\}^{\dagger} \mathcal{R}_{abcd} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$	$\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{R}_{abcd}^{\dagger} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$	replacements $\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{R}_{abcd}^{\dagger} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$ and $\mathcal{T}_{abc}^{\dagger} \rightarrow$
1510.06699_F00713	35	■ 0.806	$\mathcal{T}_{\{a\} \{b\} \{c\}}^{\dagger} \rightarrow$	$\mathcal{T}_{abc}^{\dagger} \rightarrow$	replacements $\mathcal{D}_a^{\dagger} \rightarrow {}^0 \mathcal{D}_a^{\dagger}, \mathcal{R}_{abcd}^{\dagger} \rightarrow {}^0 \mathcal{R}_{abcd}^{\dagger}$ and $\mathcal{T}_{abc}^{\dagger} \rightarrow$
1510.06699_F00714	35	■ 0.961	$\{ \}^{\{0\}} \mathcal{T}_{\{a\} \{b\} \{c\}}^{\dagger} \equiv 0$	${}^0 \mathcal{T}_{abc}^{\dagger} \equiv 0$	${}^0 \mathcal{T}_{abc}^{\dagger} \equiv 0$. Furthermore, by performing analogous calcu-
1510.06699_F00715	35	■ 0.891	$\mathcal{T}_{\{a\} \{b\} \{c\}}^{\dagger} = 0$	$\mathcal{T}_{abc}^{\dagger} = 0$	erly eWGT-covariant) condition $\mathcal{T}_{abc}^{\dagger} = 0$ in the field
1510.06699_F00716	36	■ 1.000	$b^{\{b\} \{ \}_{\nu}}$	b^b_{ν}	notate h_a^{μ} and b^b_{ν} by e_a^{μ} and e^b_{ν} , respectively.
1510.06699_F00717	36	■ 1.000	$e^{\{b\} \{ \}_{\nu}}$	e^b_{ν}	notate h_a^{μ} and b^b_{ν} by e_a^{μ} and e^b_{ν} , respectively.
1510.06699_F00718	36	■ 1.000	$e_{\{a\} \{ \}_{\mu}}, A^{\{a\} \{b\} \{ \}_{\mu}}$	e_a^{μ}, A^{ab}_{μ}	(142) of the three gauge fields e_a^{μ} , A^{ab}_{μ} , and V_{μ} that
1510.06699_F00719	36	■ 1.000	$\partial_{\mu}^{\dagger}$	$\partial_{\mu}^{\dagger}$	where we have used the partial derivative operator $\partial_{\mu}^{\dagger}$
1510.06699_F00720	36	■ 1.000	$A^{\dagger} \{a\} \{b\} \{ \}_{\mu} = -A^{\dagger ba}_{\mu}$	$A^{\dagger ab}_{\mu} = -A^{\dagger ba}_{\mu}$	$A^{\dagger ab}_{\mu} = -A^{\dagger ba}_{\mu}$, as previously.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00721	36	■ 0.545	$\nabla_{\mu}^{\dagger}=\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}$ $\nabla_{\mu}^{\dagger}=\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}$	$\nabla_{\mu}^{\dagger}=\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}$	where $\nabla_{\mu}^{\dagger}=\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}$ and $X^{\rho}{}_{\sigma}$ are the $\text{GL}(4, R)$ gen-
1510.06699_F00722	36	■ 0.545	4, R	4, R	where $\nabla_{\mu}^{\dagger}=\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}$ and $X^{\rho}{}_{\sigma}$ are the $\text{GL}(4, R)$ gen-
1510.06699_F00723	36	■ 1.000	$\Delta_{\mu}^{\dagger}$	$\Delta_{\mu}^{\dagger}$	of the field to which $\Delta_{\mu}^{\dagger}$ is applied. If a field ψ carries
1510.06699_F00724	36	■ 1.000	$\nabla_{\mu}^{\dagger}\psi=\partial_{\mu}^{\dagger}\psi$	$\nabla_{\mu}^{\dagger}\psi=\partial_{\mu}^{\dagger}\psi$	only Latin indices, then $\nabla_{\mu}^{\dagger}\psi=\partial_{\mu}^{\dagger}\psi$ and so $\Delta_{\mu}^{\dagger}\psi=D_{\mu}^{\dagger}\psi$;
1510.06699_F00725	36	■ 1.000	$\Delta_{\mu}^{\dagger}\psi=D_{\mu}^{\dagger}\psi$	$\Delta_{\mu}^{\dagger}\psi=D_{\mu}^{\dagger}\psi$	only Latin indices, then $\nabla_{\mu}^{\dagger}\psi=\partial_{\mu}^{\dagger}\psi$ and so $\Delta_{\mu}^{\dagger}\psi=D_{\mu}^{\dagger}\psi$;
1510.06699_F00726	36	■ 1.000	$D_{\mu}^{\dagger}\psi=\partial_{\mu}^{\dagger}\psi$	$D_{\mu}^{\dagger}\psi=\partial_{\mu}^{\dagger}\psi$	$D_{\mu}^{\dagger}\psi=\partial_{\mu}^{\dagger}\psi$ and so $\Delta_{\mu}^{\dagger}\psi=\nabla_{\mu}^{\dagger}\psi$. When acting on an ob-
1510.06699_F00727	36	■ 1.000	$\Delta_{\mu}^{\dagger}\psi=\nabla_{\mu}^{\dagger}\psi$	$\Delta_{\mu}^{\dagger}\psi=\nabla_{\mu}^{\dagger}\psi$	$D_{\mu}^{\dagger}\psi=\partial_{\mu}^{\dagger}\psi$ and so $\Delta_{\mu}^{\dagger}\psi=\nabla_{\mu}^{\dagger}\psi$. When acting on an ob-
1510.06699_F00728	36	■ 1.000	T	T	which relates A and Γ (and V and T). Comparing this
1510.06699_F00729	36	■ 1.000	$\partial_{\mu}^*\rightarrow\partial_{\mu}^{\dagger}$	$\partial_{\mu}^*\rightarrow\partial_{\mu}^{\dagger}$	$\partial_{\mu}^*\rightarrow\partial_{\mu}^{\dagger}$ and $A_{\mu}^{ab}\rightarrow A^{\dagger ab}{}_{\mu}$. Consequently, the results
1510.06699_F00730	36	■ 1.000	$A_{\mu}^{ab}\rightarrow A^{\dagger ab}{}_{\mu}$	$A_{\mu}^{ab}\rightarrow A^{\dagger ab}{}_{\mu}$	$\partial_{\mu}^*\rightarrow\partial_{\mu}^{\dagger}$ and $A_{\mu}^{ab}\rightarrow A^{\dagger ab}{}_{\mu}$. Consequently, the results
1510.06699_F00731	36	■ 1.000	$\nabla_{\sigma}^{\dagger}g_{\mu\nu}=0$	$\nabla_{\sigma}^{\dagger}g_{\mu\nu}=0$	$\nabla_{\sigma}^{\dagger}g_{\mu\nu}=0$, and so this derivative operator commutes
1510.06699_F00732	36	■ 1.000	$R^{\dagger\rho}{}_{\sigma\mu\nu}=e_a{}^{\rho}e^b{}_{\sigma}R^{\dagger a}{}_{b\mu\nu}$	$R^{\dagger\rho}{}_{\sigma\mu\nu}=e_a{}^{\rho}e^b{}_{\sigma}R^{\dagger a}{}_{b\mu\nu}$	where $R^{\dagger\rho}{}_{\sigma\mu\nu}=e_a{}^{\rho}e^b{}_{\sigma}R^{\dagger a}{}_{b\mu\nu}$ and $T^{\dagger\lambda}{}_{\mu\nu}=e_a{}^{\lambda}T^{\dagger a}{}_{\mu\nu}$.
1510.06699_F00733	36	■ 1.000	$T^{\dagger\lambda}{}_{\mu\nu}=e_a{}^{\lambda}T^{\dagger a}{}_{\mu\nu}$	$T^{\dagger\lambda}{}_{\mu\nu}=e_a{}^{\lambda}T^{\dagger a}{}_{\mu\nu}$	where $R^{\dagger\rho}{}_{\sigma\mu\nu}=e_a{}^{\rho}e^b{}_{\sigma}R^{\dagger a}{}_{b\mu\nu}$ and $T^{\dagger\lambda}{}_{\mu\nu}=e_a{}^{\lambda}T^{\dagger a}{}_{\mu\nu}$.
1510.06699_F00734	37	■ 1.000	$T^{\dagger\lambda}{}_{\mu\lambda}=0$	$T^{\dagger\lambda}{}_{\mu\lambda}=0$	torsion vanishes, $T^{\dagger\lambda}{}_{\mu\lambda}=0$, so that the affine connection
1510.06699_F00735	37	■ 1.000	$R^{\dagger\rho}{}_{\sigma\mu\nu}$	$R^{\dagger\rho}{}_{\sigma\mu\nu}$	From (212), we see that $R^{\dagger\rho}{}_{\sigma\mu\nu}$ is not simply its Riemann
1510.06699_F00736	37	■ 0.999	${}^0\Gamma^{\dagger\sigma}{}_{\rho\mu}$	${}^0\Gamma^{\dagger\sigma}{}_{\rho\mu}$	nection ${}^0\Gamma^{\dagger\sigma}{}_{\rho\mu}$ represent the same geometrical object in
1510.06699_F00737	37	■ 1.000	$\widehat{R}_{\rho\sigma\mu\nu}$	$\widehat{R}_{\rho\sigma\mu\nu}$	One should note that, although $\widehat{R}_{\rho\sigma\mu\nu}$ is antisymmet-
1510.06699_F00738	37	■ 1.000	$\widehat{R}_{(\rho\sigma)\mu\nu}=-g_{\rho\sigma}H_{\mu\nu}^{\dagger}$	$\widehat{R}_{(\rho\sigma)\mu\nu}=-g_{\rho\sigma}H_{\mu\nu}^{\dagger}$	$\widehat{R}_{(\rho\sigma)\mu\nu}=-g_{\rho\sigma}H_{\mu\nu}^{\dagger}$. Indeed, it does not satisfy the
1510.06699_F00739	37	■ 1.000	$\widehat{R}^{\rho}{}_{\sigma\mu\nu}$	$\widehat{R}^{\rho}{}_{\sigma\mu\nu}$	that, with the given arrangements of indices, both $\widehat{R}^{\rho}{}_{\sigma\mu\nu}$
1510.06699_F00740	37	■ 1.000	$T^{\dagger\lambda}{}_{\mu\nu}$	$T^{\dagger\lambda}{}_{\mu\nu}$	(or $R^{\dagger\rho}{}_{\sigma\mu\nu}$) and $T^{\dagger\lambda}{}_{\mu\nu}$ transform covariantly with weight
1510.06699_F00741	37	■ 1.000	$\widehat{R}_{\mu\nu}\equiv\widehat{R}_{\mu\lambda\nu}{}^{\lambda}=R_{\mu\nu}^{\dagger}+H_{\mu\nu}^{\dagger}$	$\widehat{R}_{\mu\nu}\equiv\widehat{R}_{\mu\lambda\nu}{}^{\lambda}=R_{\mu\nu}^{\dagger}+H_{\mu\nu}^{\dagger}$	$\widehat{R}_{\mu\nu}\equiv\widehat{R}_{\mu\lambda\nu}{}^{\lambda}=R_{\mu\nu}^{\dagger}+H_{\mu\nu}^{\dagger}$ and $\widehat{R}\equiv\widehat{R}^{\mu}{}_{\mu}=R^{\dagger}$. In a
1510.06699_F00742	37	■ 1.000	$\widehat{R}\equiv\widehat{R}^{\mu}{}_{\mu}=R^{\dagger}$	$\widehat{R}\equiv\widehat{R}^{\mu}{}_{\mu}=R^{\dagger}$	$\widehat{R}_{\mu\nu}\equiv\widehat{R}_{\mu\lambda\nu}{}^{\lambda}=R_{\mu\nu}^{\dagger}+H_{\mu\nu}^{\dagger}$ and $\widehat{R}\equiv\widehat{R}^{\mu}{}_{\mu}=R^{\dagger}$. In a
1510.06699_F00743	37	■ 1.000	${}^0\Delta_{\mu}^{\dagger}e^a{}_{\nu}\equiv\partial_{\mu}^{\dagger}e^a{}_{\nu}-{}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}e^a{}_{\sigma}+{}^0A^{\dagger a}{}_{b\mu}e^b{}_{\nu}=0$	${}^0\Delta_{\mu}^{\dagger}e^a{}_{\nu}\equiv\partial_{\mu}^{\dagger}e^a{}_{\nu}-{}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}e^a{}_{\sigma}+{}^0A^{\dagger a}{}_{b\mu}e^b{}_{\nu}=0$	${}^0\Delta_{\mu}^{\dagger}e^a{}_{\nu}\equiv\partial_{\mu}^{\dagger}e^a{}_{\nu}-{}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}e^a{}_{\sigma}+{}^0A^{\dagger a}{}_{b\mu}e^b{}_{\nu}=0$.
1510.06699_F00744	37	■ 0.992	$\partial_{\mu}^*\rightarrow\partial_{\mu}^{\dagger},\Gamma^{\sigma}{}_{\nu\mu}\rightarrow$ $\Gamma^{\sigma}{}_{\nu\mu}\rightarrow{}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$	$\partial_{\mu}^*\rightarrow\partial_{\mu}^{\dagger},\Gamma^{\sigma}{}_{\nu\mu}\rightarrow$ $\Gamma^{\sigma}{}_{\nu\mu}\rightarrow{}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$	again hold with the replacements $\partial_{\mu}^*\rightarrow\partial_{\mu}^{\dagger}$, $\Gamma^{\sigma}{}_{\nu\mu}\rightarrow$
1510.06699_F00745	37	■ 1.000	$A^a{}_{b\mu}\rightarrow{}^0A^{\dagger a}{}_{b\mu}$	$A^a{}_{b\mu}\rightarrow{}^0A^{\dagger a}{}_{b\mu}$	${}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$ and $A^a{}_{b\mu}\rightarrow{}^0A^{\dagger a}{}_{b\mu}$, from which one can di-
1510.06699_F00746	37	■ 1.000	${}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$	${}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$	${}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$ and $A^a{}_{b\mu}\rightarrow{}^0A^{\dagger a}{}_{b\mu}$, from which one can di-
1510.06699_F00747	37	■ 1.000	${}^0\nabla_{\sigma}^{\dagger}g_{\mu\nu}=0$	${}^0\nabla_{\sigma}^{\dagger}g_{\mu\nu}=0$	dition ${}^0\nabla_{\sigma}^{\dagger}g_{\mu\nu}=0$. Finally, the expression (212) for
1510.06699_F00748	37	■ 1.000	$R^{\dagger\rho}{}_{\sigma\mu\nu}\rightarrow{}^0R^{\dagger\rho}{}_{\sigma\mu\nu}$	$R^{\dagger\rho}{}_{\sigma\mu\nu}\rightarrow{}^0R^{\dagger\rho}{}_{\sigma\mu\nu}$	the curvature also holds, but with $R^{\dagger\rho}{}_{\sigma\mu\nu}\rightarrow{}^0R^{\dagger\rho}{}_{\sigma\mu\nu}$
1510.06699_F00749	37	■ 0.999	$\Gamma^{\sigma}{}_{\nu\mu}\rightarrow{}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$	$\Gamma^{\sigma}{}_{\nu\mu}\rightarrow{}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$	and $\Gamma^{\sigma}{}_{\nu\mu}\rightarrow{}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$, whereas (213) becomes simply
1510.06699_F00750	37	■ 1.000	${}^0T^{\dagger\lambda}{}_{\mu\nu}=0$	${}^0T^{\dagger\lambda}{}_{\mu\nu}=0$	${}^0T^{\dagger\lambda}{}_{\mu\nu}=0$, indicating the absence of torsion, as ex-
1510.06699_F00751	37	■ 0.986	$\nabla_{\mu}^{\dagger}\rightarrow{}^0\nabla_{\mu}^{\dagger},R^{\rho}{}_{\sigma\mu\nu}\rightarrow{}^0R^{\dagger\rho}{}_{\sigma\mu\nu}$	$\nabla_{\mu}^{\dagger}\rightarrow{}^0\nabla_{\mu}^{\dagger},R^{\rho}{}_{\sigma\mu\nu}\rightarrow{}^0R^{\dagger\rho}{}_{\sigma\mu\nu}$	$\nabla_{\mu}^{\dagger}\rightarrow{}^0\nabla_{\mu}^{\dagger},R^{\rho}{}_{\sigma\mu\nu}\rightarrow{}^0R^{\dagger\rho}{}_{\sigma\mu\nu}$ and $T^{\dagger\sigma}{}_{\mu\nu}\rightarrow{}^0T^{\dagger\sigma}{}_{\mu\nu}=0$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00752	37	■ 0.986	$T^{\dagger}\{\mathrm{\sigma}\}_{\{\mu\nu\}}\rightarrow{}^0T^{\dagger}\{\mathrm{\sigma}\}_{\{\mu\nu\}}=0$	$T^{\dagger\sigma}_{\mu\nu}\rightarrow{}^0T^{\dagger\sigma}_{\mu\nu}=0$	$\hat{\nabla}_{\mu}^{\dagger}\rightarrow{}^0\nabla_{\mu}^{\dagger}$, $R^{\rho}_{\sigma\mu\nu}\rightarrow{}^0R^{\dagger\rho}_{\sigma\mu\nu}$ and $T^{\dagger\sigma}_{\mu\nu}\rightarrow{}^0T^{\dagger\sigma}_{\mu\nu}=\mathbb{0}$.
1510.06699_F00753	37	■ 1.000	$L_{\{\mathrm{T}\}}=$	$L_{\mathrm{T}}=$	we now consider in detail has a total Lagrangian $L_{\mathrm{T}}=$
1510.06699_F00754	37	■ 1.000	$L_{\{\mathrm{G}\}}+L_{\{\mathrm{M}\}}$	$L_{\mathrm{G}}+L_{\mathrm{M}}$	$L_{\mathrm{G}}+L_{\mathrm{M}}$ corresponding to the general eWGT form (167).
1510.06699_F00755	37	■ 1.000	$\{\}^{\{0\}}\Gamma^{\lambda}_{\{\mu\nu\}}$	${}^0\Gamma^{\lambda}_{\mu\nu}$	in which ${}^0\Gamma^{\lambda}_{\mu\nu}$ is the standard metric (Christoffel) con-
1510.06699_F00756	37	■ 1.000	$K^{\dagger}\{\mathrm{\lambda}\}_{\{\mu\nu\}}$	$K^{\dagger\lambda}_{\mu\nu}$	nection and $K^{\dagger\lambda}_{\mu\nu}$ is the $\hat{Y}_4$ contortion tensor
1510.06699_F00757	37	■ 1.000	$L_{\{\mathrm{G}\}}=L_{\{\mathrm{mathcal{R}}\}^{\dagger}2}+L_{\{\mathrm{mathcal{H}}\}^{\dagger}2}$	$L_G=L_{\mathcal{R}^{\dagger}2}+L_{\mathcal{H}^{\dagger}2}$	namely $L_G=L_{\mathcal{R}^{\dagger}2}+L_{\mathcal{H}^{\dagger}2}$, where
1510.06699_F00758	37	■ 1.000	$L_{\{\mathrm{M}\}}=$	$L_{\mathrm{M}}=$	of gravity. In particular, we adopt the form $L_{\mathrm{M}}=$
1510.06699_F00759	38	■ 1.000	ν, λ, a	ν, λ, a	in which ν, λ, a and β_i are again dimensionless parame-
1510.06699_F00760	38	■ 1.000	ν, λ	ν, λ	ters (with ν, λ and a usually positive) and some factors
1510.06699_F00761	38	■ 1.000	$L_{\{\varphi\}}$	L_{φ}	the moment we allow the form of L_{φ} for the matter field
1510.06699_F00762	38	■ 1.000	$\left(\tau_{\{\varphi\}}^{\dagger}\right)^a_{\{\mathrm{b}\}}=h_{\{\mathrm{b}\}}^{\mu}\left(\delta L_{\varphi}/\delta h_a^{\mu}\right)_{\dagger}$ $\left\{\right\}^{\{\mu\}}\left\{\right\}\left(\delta L_{\{\varphi\}}/\delta h_{\{\mathrm{a}\}}^{\{\mu\}}\right)_{\dagger}$	$\left(\tau_{\varphi}^{\dagger}\right)^a_b=h_b^{\mu}\left(\delta L_{\varphi}/\delta h_a^{\mu}\right)_{\dagger}$	and in the matter sector, in addition to $\left(\tau_{\varphi}^{\dagger}\right)^a_b=h_b^{\mu}\left(\delta L_{\varphi}/\delta h_a^{\mu}\right)_{\dagger}$, one has
1510.06699_F00763	38	■ 1.000	$\left(t_{\{\mathrm{mathcal{H}}\}^{\dagger}2}\right)^a_{\{\mathrm{b}\}}$	$\left(t_{\mathcal{H}^2}^{\dagger}\right)^a_b$	In particular, we note that $\left(t_{\mathcal{H}^2}^{\dagger}\right)^a_b$ in (230) is linear
1510.06699_F00764	39	■ 1.000	$\left(\sigma_{\{\varphi\}}^{\dagger}\right)_{ab}^c=$	$\left(\sigma_{\varphi}^{\dagger}\right)_{ab}^c=$	and in the matter sector, in addition to $\left(\sigma_{\varphi}^{\dagger}\right)_{ab}^c=$
1510.06699_F00765	39	■ 1.000	$b^c_{\{\mu\}}\delta L_{\varphi}/\delta A^{ab}_{\mu}$	$b^c_{\mu}\delta L_{\varphi}/\delta A^{ab}_{\mu}$	—
1510.06699_F00766	39	■ 0.860	$\left(s_{\{\mathrm{mathcal{H}}\}^{\dagger}2}\right)_{ab}^c$	$\left(s_{\mathcal{H}^2}\right)_{ab}^c$	We note that $\left(s_{\mathcal{H}^2}\right)_{ab}^c$ in (236) is linear in third-order
1510.06699_F00767	39	■ 0.554	$\{\mathrm{T}\}_{\{\mathrm{c}\}\{\mathrm{a}\}\{\mathrm{b}\}}^{\dagger}+2\{\mathrm{T}\}_{\{\{\mathrm{a}\}\{\mathrm{b}\}\}\{\mathrm{c}\}}^{\dagger}\{\mathrm{c}\}^{\dagger}\{\mathrm{a}\}\{\mathrm{b}\}-3\{\mathrm{T}\}_{\{\{\mathrm{c}\}\{\mathrm{a}\}\}\{\mathrm{b}\}}^{\dagger}=0$	$\mathcal{T}_{cab}^{\dagger}+2\mathcal{T}_{[ab]c}^{\dagger}-3\mathcal{T}_{[cab]}^{\dagger}=0$	strength satisfies the identity $\mathcal{T}_{cab}^{\dagger}+2\mathcal{T}_{[ab]c}^{\dagger}-3\mathcal{T}_{[cab]}^{\dagger}=\mathbb{0}$.
1510.06699_F00768	39	■ 1.000	α_4	α_4	in the action (namely those in (235) proportional to α_4 ,
1510.06699_F00769	39	■ 1.000	α_5	α_5	α_5 and α_6 , respectively) have a form similar to the LHS
1510.06699_F00770	39	■ 0.606	$\overline{\{\mathrm{mathcal{R}}\}_{\{\mathrm{a}\}\{\mathrm{b}\}\{\mathrm{c}\}\{\mathrm{d}\}}^{\dagger}}\equiv\{\mathrm{mathcal{R}}\}_{\{\{\mathrm{c}\}\{\mathrm{d}\}\{\mathrm{a}\}\{\mathrm{b}\}}^{\dagger}}$	$\overline{\mathcal{R}}_{abcd}^{\dagger}\equiv\mathcal{R}_{cdab}^{\dagger}$	$\overline{\mathcal{R}}_{abcd}^{\dagger}\equiv\mathcal{R}_{cdab}^{\dagger}$. Since $\overline{\mathcal{R}}^{\dagger a}_b=\mathcal{R}_b^{\dagger a}$, one also sees that
1510.06699_F00771	39	■ 0.606	$\overline{\{\mathrm{mathcal{R}}\}_{\{\mathrm{b}\}}^{\dagger}}=\{\mathrm{mathcal{R}}\}_{\{\mathrm{b}\}}^{\dagger}\{\mathrm{a}\}^{\dagger}$	$\overline{\mathcal{R}}^{\dagger a}_b=\mathcal{R}_b^{\dagger a}$	$\overline{\mathcal{R}}_{abcd}^{\dagger}\equiv\mathcal{R}_{cdab}^{\dagger}$. Since $\overline{\mathcal{R}}^{\dagger a}_b=\mathcal{R}_b^{\dagger a}$, one also sees that
1510.06699_F00772	39	■ 1.000	α_2	α_2	terms in (235) proportional to α_2 and α_3 , respectively,
1510.06699_F00773	39	■ 1.000	α_3	α_3	terms in (235) proportional to α_2 and α_3 , respectively,
1510.06699_F00774	39	■ 1.000	α_1	α_1	proportional to α_1 , since $\overline{\mathcal{R}}^{\dagger}=\mathcal{R}^{\dagger}$. The term propor-
1510.06699_F00775	39	■ 1.000	$\overline{\{\mathrm{mathcal{R}}\}_{\{\mathrm{a}\}}^{\dagger}}=\{\mathrm{mathcal{R}}\}_{\{\mathrm{a}\}}^{\dagger}$	$\overline{\mathcal{R}}^{\dagger}=\mathcal{R}^{\dagger}$	proportional to α_1 , since $\overline{\mathcal{R}}^{\dagger}=\mathcal{R}^{\dagger}$. The term propor-
1510.06699_F00776	39	■ 1.000	$\left(\zeta_{\{\varphi\}}^{\dagger}\right)_a=$	$\left(\zeta_{\varphi}^{\dagger}\right)_a=$	—
1510.06699_F00777	39	■ 1.000	$b_{\{\mathrm{a}\}\{\mu\}}\delta L_{\varphi}/\delta V_{\mu}$	$b_{a\mu}\delta L_{\varphi}/\delta V_{\mu}$	$b_{a\mu}\delta L_{\varphi}/\delta V_{\mu}$,
1510.06699_F00778	39	■ 1.000	c	c	equation (234) contracted on the indices b and c , in ac-
1510.06699_F00779	39	■ 1.000	$j_{\{\mathrm{a}\}}^{\dagger}$	$j_a^{\dagger}$	so $j_a^{\dagger}$ and $\zeta_a^{\dagger}$ vanish identically for each part of the La-
1510.06699_F00780	39	■ 1.000	$\zeta_{\{\mathrm{a}\}}^{\dagger}$	$\zeta_a^{\dagger}$	so $j_a^{\dagger}$ and $\zeta_a^{\dagger}$ vanish identically for each part of the La-
1510.06699_F00781	39	■ 1.000	$\partial_{\phi}\equiv\partial/\partial\phi$	$\partial_{\phi}\equiv\partial/\partial\phi$	where $\partial_{\phi}\equiv\partial/\partial\phi$ and $\delta_{\phi}\equiv\delta/\delta\phi$, since only L_{ϕ} depends
1510.06699_F00782	39	■ 1.000	$\delta_{\phi}\equiv\delta/\delta\phi$	$\delta_{\phi}\equiv\delta/\delta\phi$	where $\partial_{\phi}\equiv\partial/\partial\phi$ and $\delta_{\phi}\equiv\delta/\delta\phi$, since only L_{ϕ} depends
1510.06699_F00783	39	■ 1.000	$L_{\{\phi\}}$	L_{ϕ}	where $\partial_{\phi}\equiv\partial/\partial\phi$ and $\delta_{\phi}\equiv\delta/\delta\phi$, since only L_{ϕ} depends
1510.06699_F00784	40	■ 1.000	$-\phi$	$-\phi$	by $-\phi$, in accordance with the general results (177) and
1510.06699_F00785	40	■ 0.999	$\delta_{\varphi}L_{\varphi}=0$	$\delta_{\varphi}L_{\varphi}=0$	(178d), assuming the φ -equation $\delta_{\varphi}L_{\varphi}=\mathbb{0}$ is satisfied.
1510.06699_F00786	40	■ 1.000	$\xi=0$	$\xi=0$	tional Lagrangian (by setting $\xi=0$). Indeed, one can
1510.06699_F00787	40	■ 1.000	$\left.\mathcal{D}_{\{\mathrm{a}\}}^{\dagger}\right _{\phi=\phi_0}=\phi_0\left(\mathcal{V}_{\{\mathrm{a}\}}+\frac{1}{3}\mathcal{T}_{\{\mathrm{a}\}}\right)$	$\mathcal{D}_a^{\dagger}\big _{\phi=\phi_0}=\phi_0\left(\mathcal{V}_a+\frac{1}{3}\mathcal{T}_a\right)$	the replacement $\mathcal{D}_a^{\dagger}\big _{\phi=\phi_0}=\phi_0(\mathcal{V}_a+\frac{1}{3}\mathcal{T}_a)$ throughout,
1510.06699_F00788	40	■ 1.000	$\widehat{\{\varphi\}},\widehat{\{\mathrm{h}\}_{\{\mathrm{a}\}}^{\{\mu\}}}$	$\widehat{\varphi},\widehat{h}_a^{\mu}$	expressed in terms of the scale-invariant variables $\widehat{\varphi},\widehat{h}_a^{\mu}$,
1510.06699_F00789	41	■ 1.000	$h^{\prime\{\mu\}}_{\{\mathrm{a}\}}\{\}^{\{\mu\}}=e^{\{-\rho\}}h_{\{\mathrm{a}\}}^{\{\mu\}}$	$h^{\prime a\mu}_a=e^{-\rho}h_a^{\mu}$	fields are given by $h^{\prime a\mu}_a=e^{-\rho}h_a^{\mu}$ and $A^{\prime ab}_{\mu}=A^{ab}_{\mu}+$
1510.06699_F00790	41	■ 1.000	$A^{\{\mu\}}_{\{\mathrm{a}\}\{\mathrm{b}\}}=A^{\{\mu\}}_{\{\mathrm{a}\}\{\mathrm{b}\}}+$	$A^{\prime ab}_{\mu}=A^{ab}_{\mu}+$	fields are given by $h^{\prime a\mu}_a=e^{-\rho}h_a^{\mu}$ and $A^{\prime ab}_{\mu}=A^{ab}_{\mu}+$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00791	41	■ 1.000	$\theta\left(b^{\mathrm{a}}_{\mu}\mathcal{P}^{\mathrm{b}}-b^{\mathrm{b}}_{\mu}\mathcal{P}^{\mathrm{a}}\right)$	$\theta\left(b^a_{\mu}\mathcal{P}^b-b^b_{\mu}\mathcal{P}^a\right)$	$\theta(b^a_{\mu}\mathcal{P}^b-\tilde{b}^b_{\mu}\mathcal{P}^a)$, respectively, where $\mathcal{P}_a=h_a{}^{\mu}\partial_{\mu}\rho$. Here
1510.06699_F00792	41	■ 1.000	$\mathcal{P}_{\mathrm{a}}=h_{\mathrm{a}}{}^{\mu}\partial_{\mu}\rho$	$\mathcal{P}_a=h_a{}^{\mu}\partial_{\mu}\rho$	$\theta(b^a_{\mu}\mathcal{P}^b-\tilde{b}^b_{\mu}\mathcal{P}^a)$, respectively, where $\mathcal{P}_a=h_a{}^{\mu}\partial_{\mu}\rho$. Here
1510.06699_F00793	41	■ 1.000	$\mathcal{P}^2\equiv\mathcal{P}_{\mathrm{a}}\mathcal{P}^{\mathrm{a}}$	$\mathcal{P}^2\equiv\mathcal{P}_a\mathcal{P}^a$	where $\mathcal{P}^2\equiv\mathcal{P}_a\mathcal{P}^a$, and the translational gauge field
1510.06699_F00794	41	■ 1.000	$(\theta=0)$	$(\theta=0)$	is covariant with weight $w=-2$ under ‘normal’ ($\theta=0$)
1510.06699_F00795	41	■ 1.000	$(\theta=1)$	$(\theta=1)$	$w=-1$ under ‘special’ transformations ($\theta=1$).
1510.06699_F00796	41	■ 1.000	$\theta\neq 0$	$\theta\neq 0$	extended transformation law for the A -field, with $\theta\neq 0$,
1510.06699_F00797	41	■ 1.000	$\alpha_{\mathrm{i}}=0$	$\alpha_i=0$	forms covariantly (aside from the trivial case $\underline{\alpha_i=0}$ for
1510.06699_F00798	41	■ 0.996	i	i	all i) and so L_{G} must vanish. Nonetheless, by contrast,
1510.06699_F00799	41	■ 0.998	$L_{\mathrm{\mathcal{T}}^2}$	$L_{\mathcal{T}^2}$	parameters β_1 , β_2 and β_3 in $L_{\mathcal{T}^2}$.
1510.06699_F00800	42	■ 0.989	$\mathcal{L}_{\mathrm{L}}_{\mathrm{\mathstrut\mathrm{M}}}\equiv\mathcal{L}_{\mathrm{L}}_{\mathrm{\mathstrut\psi}}+\mathcal{L}_{\mathrm{L}}_{\mathrm{\mathstrut\phi}}+\phi^2\mathcal{L}_{\mathrm{R}}+\phi^2\mathcal{L}_{\mathrm{T}}$	$\mathcal{L}_{\mathrm{M}}\equiv\mathcal{L}_{\psi}+\mathcal{L}_{\phi}+\phi^2\mathcal{L}_{\mathcal{R}}+\phi^2\mathcal{L}_{\mathcal{T}}$	Let us denote (256) by $\underline{\mathcal{L}_{\mathrm{M}}\equiv\mathcal{L}_{\psi}+\mathcal{L}_{\phi}+\phi^2\mathcal{L}_{\mathcal{R}}+\phi^2\mathcal{L}_{\mathcal{T}}}$.
1510.06699_F00801	42	■ 0.990	$\mathcal{L}_{\mathrm{L}}_{\mathrm{\mathstrut\psi}}$	$\mathcal{L}_{\psi}$	As discussed in Section III K, $\mathcal{L}_{\psi}$ is invariant under ex-
1510.06699_F00802	42	■ 1.000	$\nu,\ a$	ν,a	rameters ν , a and β_i ($i=1,2,3$) satisfy the conditions
1510.06699_F00803	42	■ 1.000	$\beta_{\mathrm{i}}(i=1,2,3)$	$\beta_i(i=1,2,3)$	rameters ν , a and β_i ($i=1,2,3$) satisfy the conditions
1510.06699_F00804	42	■ 1.000	ν	ν	the parameters ν and a must be of opposite sign, which
1510.06699_F00805	42	■ 1.000	$\nu=0=2\beta_1+\beta_2+3\beta_3$	$\nu=0=2\beta_1+\beta_2+3\beta_3$	$\nu=0=2\beta_1+\beta_2+3\beta_3$, and a ‘special’ transforma-
1510.06699_F00806	42	■ 0.997	$\nu=0=a$	$\nu=0=a$	tion ($\theta=1$), for which $\nu=0=a$. Thus, in both special
1510.06699_F00807	42	■ 1.000	$\nu=\beta_1=\beta_2=\beta_3=0$	$\nu=\beta_1=\beta_2=\beta_3=0$	Obukhov ⁸⁹ , who set $\underline{\nu=\beta_1=\beta_2=\beta_3=0}$. Adopting
1510.06699_F00808	42	■ 1.000	$2\beta_1+\beta_2+3\beta_3=0$	$2\beta_1+\beta_2+3\beta_3=0$	with $\underline{2\beta_1+\beta_2+3\beta_3=0}$ in $L_{\mathcal{T}^{+2}}$, which is merely a special
1510.06699_F00809	42	■ 0.706	$\mathcal{D}_{\mathrm{a}}\phi\rightarrow\left(\mathcal{D}_{\mathrm{a}}+\frac{1}{3}\mathcal{T}_{\mathrm{a}}\right)\phi$	$\mathcal{D}_a\phi\rightarrow\left(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a\right)\phi$	$\mathcal{D}_a\phi\rightarrow(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a)\phi$ (and similarly for $\tilde{\mathcal{D}}^a\phi$) in the kinetic
1510.06699_F00810	42	■ 0.706	$\mathcal{D}^{\mathrm{a}}\phi$	$\mathcal{D}^a\phi$	$\mathcal{D}_a\phi\rightarrow(\mathcal{D}_a+\frac{1}{3}\mathcal{T}_a)\phi$ (and similarly for $\tilde{\mathcal{D}}^a\phi$) in the kinetic
1510.06699_F00811	42	■ 1.000	$\frac{1}{2}\nu+6a=0$	$\frac{1}{2}\nu+6a=0$	vided only that $\frac{1}{2}\nu+6a=0$ and $2\beta_1+\beta_2+3\beta_3=0$
1510.06699_F00812	42	■ 0.994	$2\beta_1+\beta_2=0$	$2\beta_1+\beta_2=0$	with $2\beta_1+\beta_2=0$ in $L_{\mathcal{T}^{+2}}$ (in which there is no term
1510.06699_F00813	42	■ 0.994	$L_{\mathrm{\mathstrut\mathcal{T}}^{+2}}$	$L_{\mathcal{T}^{+2}}$	with $2\beta_1+\beta_2=0$ in $L_{\mathcal{T}^{+2}}$ (in which there is no term
1510.06699_F00814	43	■ 1.000	$\{ \ }^{\mathrm{0}}\mathcal{D}_{\mathrm{a}}$	${}^0\mathcal{D}_a$	derivative ${}^0\mathcal{D}_a$, to which the full PGT covariant deriva-
1510.06699_F00815	43	■ 1.000	$\{ \ }^{\mathrm{0}}\mathcal{T}^{\mathrm{a}}_{\mathrm{bc}}\equiv h_{\mathrm{b}}{}^{\mu}h_{\mathrm{c}}{}^{\nu}\left({}^0D_{\mu}b^{\mathrm{a}}{}_{\nu}-{}^0D_{\nu}b^{\mathrm{a}}{}_{\mu}\right)=0$	${}^0\mathcal{T}^a_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}\left({}^0D_{\mu}b^a{}_{\nu}-{}^0D_{\nu}b^a{}_{\mu}\right)=0$	$\underline{{}^0\mathcal{T}^a_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}\left({}^0D_{\mu}b^a{}_{\nu}-{}^0D_{\nu}b^a{}_{\mu}\right)=0}$. Thus, reduced
1510.06699_F00816	43	■ 1.000	$\{ \ }^{\mathrm{0}}A^{\prime\mathrm{ab}}{}_{\mu}={}^0A^{\mathrm{ab}}{}_{\mu}+b^{\mathrm{a}}{}_{\mu}\mathcal{P}^{\mathrm{b}}-b^{\mathrm{b}}{}_{\mu}\mathcal{P}^{\mathrm{a}}$	${}^0A^{\prime ab}{}_{\mu}={}^0A^{ab}{}_{\mu}+b^a{}_{\mu}\mathcal{P}^b-b^b{}_{\mu}\mathcal{P}^a$	$\underline{{}^0A^{\prime ab}{}_{\mu}={}^0A^{ab}{}_{\mu}+b^a{}_{\mu}\mathcal{P}^b-b^b{}_{\mu}\mathcal{P}^a}$, which is independent
1510.06699_F00817	43	■ 1.000	$\{ \ }^{\mathrm{0}}A^{\mathrm{ab}}{}_{\mu},\{ \ }^{\mathrm{0}}\mathcal{D}_{\mathrm{c}}\varphi$	${}^0A^{ab}{}_{\mu},{}^0\mathcal{D}_c\varphi$	${}^0\tilde{A}^{ab}{}_{\mu},{}^0\mathcal{D}_c\varphi$ and ${}^0\mathcal{R}^{ab}{}_{cd}$ all transform independently of
1510.06699_F00818	43	■ 0.999	$A^{\mathrm{ab}}{}_{\mu},\mathcal{D}_{\mathrm{c}}\varphi$	$A^{ab}{}_{\mu},\mathcal{D}_c\varphi$	$A^{ab}{}_{\mu},\mathcal{D}_c\varphi$ and $\mathcal{R}^{ab}{}_{cd}$ do assuming the ‘special’ transfor-
1510.06699_F00819	43	■ 1.000	$\mathcal{D}_{\mathrm{c}}\rightarrow{}^0\mathcal{D}_{\mathrm{c}}$	$\mathcal{D}_c\rightarrow{}^0\mathcal{D}_c$	replacements $\mathcal{D}_c\rightarrow{}^0\mathcal{D}_c$ and $\mathcal{T}^a{}_{bc}\rightarrow{}^0\mathcal{T}^a{}_{bc}\equiv 0$.
1510.06699_F00820	43	■ 1.000	$\mathcal{T}^{\mathrm{a}}{}_{\mathrm{bc}}\rightarrow{}^0\mathcal{T}^{\mathrm{a}}{}_{\mathrm{bc}}\equiv 0$	$\mathcal{T}^a{}_{bc}\rightarrow{}^0\mathcal{T}^a{}_{bc}\equiv 0$	replacements $\mathcal{D}_c\rightarrow{}^0\mathcal{D}_c$ and $\underline{\mathcal{T}^a{}_{bc}\rightarrow{}^0\mathcal{T}^a{}_{bc}\equiv 0}$.
1510.06699_F00821	43	■ 0.915	L_{G}	L_G	Following the discussion in Appendix B, if L_G contains
1510.06699_F00822	43	■ 1.000	$2\alpha_3+\alpha_2=0$	$2\alpha_3+\alpha_2=0$	$2\alpha_3+\alpha_2=0$ and $3\alpha_1-\alpha_3=0$. Letting $\alpha=\alpha_3$ for con-
1510.06699_F00823	43	■ 1.000	$3\alpha_1-\alpha_3=0$	$3\alpha_1-\alpha_3=0$	$2\alpha_3+\alpha_2=0$ and $3\alpha_1-\alpha_3=0$. Letting $\alpha=\alpha_3$ for con-
1510.06699_F00824	43	■ 1.000	$\alpha=\alpha_3$	$\alpha=\alpha_3$	$2\alpha_3+\alpha_2=0$ and $3\alpha_1-\alpha_3=0$. Letting $\alpha=\alpha_3$ for con-
1510.06699_F00825	43	■ 1.000	$\{ \ }^{\mathrm{0}}\mathcal{R}_{\mathrm{abcd}}{}^0\mathcal{R}^{\mathrm{abcd}}$	${}^0\mathcal{R}_{abcd}{}^0\mathcal{R}^{abcd}$	proportional to $\underline{{}^0\mathcal{R}_{abcd}{}^0\mathcal{R}^{abcd}}$ to obtain
1510.06699_F00826	43	■ 0.899	$\{ \ }^{\mathrm{0}}\underline{\mathcal{L}}_{101}$	$\underline{\mathcal{L}}_{101}$	theory ¹⁰¹ . Indeed, as one might suspect, one can easily
1510.06699_F00827	43	■ 0.999	$L_{\mathrm{W}}={}^0\mathcal{C}_{\mathrm{abcd}}{}^0\mathcal{C}^{\mathrm{abcd}}$	$L_{\mathrm{W}}={}^0\mathcal{C}_{abcd}{}^0\mathcal{C}^{abcd}$	grangian $L_{\mathrm{W}}={}^0\mathcal{C}_{abcd}{}^0\mathcal{C}^{abcd}$, where ${}^0\mathcal{C}_{abcd}$ is the gauge
1510.06699_F00828	43	■ 0.999	$\{ \ }^{\mathrm{0}}\mathcal{C}_{\mathrm{abcd}}$	${}^0\mathcal{C}_{abcd}$	grangian $L_{\mathrm{W}}={}^0\mathcal{C}_{abcd}{}^0\mathcal{C}^{abcd}$, where ${}^0\mathcal{C}_{abcd}$ is the gauge
1510.06699_F00829	44	■ 1.000	$\{ \ }^{\mathrm{0}}\mathcal{R}_{\mathrm{abcd}}$	${}^0\mathcal{R}_{abcd}$	This tensor represents the traceless part of ${}^0\mathcal{R}_{abcd}$; it has

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00864	47	■ 1.000	$\gamma^5=\gamma_5=i\gamma^0\gamma^1\gamma^2\gamma^3$	$\gamma^5 = \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$	where $\gamma^5 = \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$ (and
1510.06699_F00865	47	■ 1.000	$\sigma^{\nu\rho}=\frac{i}{2}\left[\gamma^\nu,\gamma^\rho\right]$	$\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$	where $\gamma^5 = \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$ (and
1510.06699_F00866	47	■ 1.000	$\rho\equiv\bar{\psi}\psi$	$\rho \equiv \bar{\psi}\psi$	It is convenient to define the real scalar $\rho \equiv \bar{\psi}\psi$ and
1510.06699_F00867	47	■ 0.671	$\beta\equiv\bar{\psi}i\gamma^5\psi$	$\beta \equiv \bar{\psi}i\gamma^5\psi$	real pseudoscalar $\beta \equiv \bar{\psi}i\gamma^5\psi$, so that (A4) reads $\bar{\psi}\not{J} =$
1510.06699_F00868	47	■ 0.671	$\bar{\psi}/\hbar=$	$\bar{\psi}/\hbar =$	real pseudoscalar $\beta \equiv \bar{\psi}i\gamma^5\psi$, so that (A4) reads $\bar{\psi}\not{J} =$
1510.06699_F00869	47	■ 1.000	$\bar{\psi}(\rho+\beta i\gamma^5)$	$\bar{\psi}(\rho + \beta i\gamma^5)$	$\bar{\psi}(\rho + \beta i\gamma^5)$. Postmultiplying this expression by $(\rho - \beta i\gamma^5)$
1510.06699_F00870	47	■ 1.000	$(\rho-\beta i\gamma^5)$	$(\rho - \beta i\gamma^5)$	$\bar{\psi}(\rho + \beta i\gamma^5)$. Postmultiplying this expression by $(\rho - \beta i\gamma^5)$
1510.06699_F00871	47	■ 0.982	$\nexists\partial\psi$	$\nexists\partial\psi$	In particular, we note that the quantity $\not{J}\not{\partial}\psi$ appearing
1510.06699_F00872	47	■ 1.000	$J^\mu\partial_\mu\psi$	$J^\mu\partial_\mu\psi$	The quantity $J^\mu\partial_\mu\psi$ is clearly interpreted as a derivative
1510.06699_F00873	47	■ 1.000	$n^\nu\equiv-i\sigma^{\mu\nu}J_\mu$	$n^\nu \equiv -i\sigma^{\mu\nu}J_\mu$	the direction $\bar{n}^\nu \equiv -i\sigma^{\mu\nu}J_\mu$, which is orthogonal to J^μ ,
1510.06699_F00874	47	■ 1.000	$J_\nu n^\nu=-i\sigma^{\mu\nu}J_\mu J_\nu=0$	$J_\nu n^\nu = -i\sigma^{\mu\nu}J_\mu J_\nu = 0$	since $J_\nu \bar{n}^\nu = -i\sigma^{\mu\nu}J_\mu J_\nu = 0$. Thus, $-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$ is in-
1510.06699_F00875	47	■ 1.000	$-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$	$-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$	since $J_\nu n^\nu = -i\sigma^{\mu\nu}J_\mu J_\nu = 0$. Thus, $-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$ is in-
1510.06699_F00876	47	■ 1.000	$x^\mu(\lambda)$	$x^\mu(\lambda)$	the particle is described by a worldline $x^\mu(\lambda)$, together
1510.06699_F00877	47	■ 1.000	$R(\lambda)$	$R(\lambda)$	with a ‘rotor’ $R(\lambda)$, which is a Dirac spinor defined along
1510.06699_F00878	47	■ 1.000	$\bar{R}R=1$	$\bar{R}R = 1$	$\bar{R}R = 1$ and $\bar{R}i\gamma^5R = 0$, and contains information about
1510.06699_F00879	47	■ 1.000	$\bar{R}i\gamma^5R=0$	$\bar{R}i\gamma^5R = 0$	$\bar{R}R = 1$ and $\bar{R}i\gamma^5R = 0$, and contains information about
1510.06699_F00880	47	■ 1.000	R	R	one replaces ψ by R in (A6), restricts derivatives to lie
1510.06699_F00881	47	■ 1.000	$\dot{x}^\mu\equiv dx^\mu/d\lambda$	$\dot{x}^\mu \equiv dx^\mu/d\lambda$	locity $\dot{x}^\mu \equiv dx^\mu/d\lambda$ with $\bar{R}\gamma^\mu R$. The last identification
1510.06699_F00882	47	■ 1.000	$\bar{R}\gamma^\mu R$	$\bar{R}\gamma^\mu R$	locity $\dot{x}^\mu \equiv dx^\mu/d\lambda$ with $\bar{R}\gamma^\mu R$. The last identification
1510.06699_F00883	47	■ 1.000	$p_\mu(\lambda)$	$p_\mu(\lambda)$	multiplier $p_\mu(\lambda)$, which is identified with the momentum
1510.06699_F00884	47	■ 1.000	p_μ	p_μ	of the particle; note that, in general, p_μ and $\dot{x}^\mu$ are not
1510.06699_F00885	47	■ 1.000	$\dot{x}^\mu$	$\dot{x}^\mu$	of the particle; note that, in general, p_μ and $\dot{x}^\mu$ are not
1510.06699_F00886	47	■ 0.999	$me(\lambda)$	$me(\lambda)$	ten as $me(\lambda)$ for later convenience) to ensure reparam-
1510.06699_F00887	47	■ 1.000	$x^\mu(\lambda),R(\lambda),p^\mu(\lambda)$	$x^\mu(\lambda), R(\lambda), p^\mu(\lambda)$	to the dynamical variables $x^\mu(\lambda)$, $R(\lambda)$, $p^\mu(\lambda)$ and $e(\lambda)$
1510.06699_F00888	47	■ 1.000	p^μ/m	p^μ/m	replacing $\bar{R}\gamma^\mu R$ by p^μ/m and then neglecting the particle
1510.06699_F00889	47	■ 1.000	$e(\lambda)\rightarrow\frac{1}{2}e(\lambda)$	$e(\lambda) \rightarrow \frac{1}{2}e(\lambda)$	einbein $e(\lambda) \rightarrow \frac{1}{2}e(\lambda)$ for later convenience, this process
1510.06699_F00890	47	■ 1.000	$x^\mu(\lambda),p^\mu(\lambda)$	$x^\mu(\lambda), p^\mu(\lambda)$	variables $x^\mu(\lambda)$, $p^\mu(\lambda)$ and $e(\lambda)$ leads to the classical
1510.06699_F00891	47	■ 1.000	$\rho=0$	$\rho = 0$	ticular, by setting $\rho = 0$ in the global Weyl transforma-
1510.06699_F00892	47	■ 1.000	$\rho(x)=0$	$\rho(x) = 0$	reduce to local Poincaré transformations if $\rho(x) = 0$. In
1510.06699_F00893	48	■ 1.000	$D_\mu\varphi$	$D_\mu\varphi$	derivative $D_\mu\varphi$. Thus, all the results given in Sec-
1510.06699_F00894	48	■ 0.981	$\rho(x)=0=$	$\rho(x) = 0 =$	tions II B and II C also hold in PGT if one sets $\rho(x) = 0 =$
1510.06699_F00895	48	■ 1.000	$B_\mu=0$	$B_\mu = 0$	valid in PGT if one sets $B_\mu = 0$ and removes all asterisks.
1510.06699_F00896	48	■ 1.000	$H_{\mu\nu}=0$	$H_{\mu\nu} = 0$	prescription of setting $B_\mu = 0$ (so that $H_{\mu\nu} = 0$) and re-
1510.06699_F00897	48	■ 0.999	$\mathcal{R}\equiv\mathcal{R}^{ab}_{ab},\Lambda$	$\mathcal{R} \equiv \mathcal{R}^{ab}_{ab}, \Lambda$	where $\mathcal{R} \equiv \mathcal{R}^{ab}_{ab}$, Λ is a cosmological constant, a is
1510.06699_F00898	48	■ 1.000	$a,\Lambda,\{\alpha_i\}$	$a, \Lambda, \{\alpha_i\}$	depends on the choice of the parameters a , Λ , $\{\alpha_i\}$ and
1510.06699_F00899	48	■ 1.000	$\{\beta_i\}$	$\{\beta_i\}$	$\{\beta_i\}$ in L_G . As an illustration, in Appendix C we give
1510.06699_F00900	48	■ 1.000	$\varphi,\partial\varphi,h$	$\varphi, \partial\varphi, h$	induce a dependence on φ , $\partial\varphi$, h and A , but one typically
1510.06699_F00901	48	■ 0.546	$\mathcal{H}_{ab},j^a,\zeta^a$	$\mathcal{H}_{ab}, j^a, \zeta^a$	tion III, by removing all asterisks and setting $\mathcal{H}_{ab}, j^a, \zeta^a$
1510.06699_F00902	48	■ 1.000	$\mu\phi\bar{\psi}\psi\rightarrow m\bar{\psi}\psi$	$\mu\phi\bar{\psi}\psi \rightarrow m\bar{\psi}\psi$	$\mu\phi\bar{\psi}\psi \rightarrow m\bar{\psi}\psi$ for the Dirac field. Thus, after this change
1510.06699_F00903	48	■ 1.000	$\phi=1$	$\phi = 1$	isks and setting $B_\mu = 0$ and $\phi = 1$, all the results in these
1510.06699_F00904	48	■ 0.992	${}^0\mathcal{T}^a_{bc}\equiv 0$	${}^0\mathcal{T}^a_{bc} \equiv 0$	tically, ${}^0\mathcal{T}^a_{bc} \equiv 0$, only the reduced ‘rotational’ field
1510.06699_F00905	48	■ 1.000	${}^0\mathcal{R}^{ab}_{cd}(h,\partial h,\partial^2h)$	${}^0\mathcal{R}^{ab}_{cd}(h, \partial h, \partial^2 h)$	strength ${}^0\mathcal{R}^{ab}_{cd}(h, \partial h, \partial^2 h)$ remains, which depends on
1510.06699_F00906	49	■ 1.000	${}^0\mathcal{D}_a\mathcal{T}^a=h\partial_\mu(h^{-1}h_a{}^\mu\mathcal{T}^a)$	${}^0\mathcal{D}_a\mathcal{T}^a = h\partial_\mu(h^{-1}h_a{}^\mu\mathcal{T}^a)$	terms of a total derivative as ${}^0\mathcal{D}_a\mathcal{T}^a = h\partial_\mu(h^{-1}h_a{}^\mu\mathcal{T}^a)$,
1510.06699_F00907	49	■ 0.980	$\underline{\{ \}^{105}}$	$\underline{105}$	whereas this is not possible in WGT ¹⁰⁵ .
1510.06699_F00908	49	■ 0.935	$L_0\mathcal{R}^2$	$L_0\mathcal{R}^2$	where $L_0\mathcal{R}^2$ is based on (2) with $\mathcal{R}^{ab}_{cd}$ replaced by

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00909	49	■ 0.606	$L_{\{0^{\{\mathrm{R}\}^{\{2\}}\}}$	$L_{0\mathcal{R}^2}$	tinct terms in $L_{0\mathcal{R}^2}$ is reduced to
1510.06699_F00910	49	■ 0.999	$\mathrm{R}_{\{a\ b\ c\ d\}} \rightarrow^{\{0\}} \mathrm{R}_{\{a\ b\ c\ d\}}$	$\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}$	$\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}_{abcd}$. Thus, one can set to zero any one of the
1510.06699_F00911	49	■ 1.000	$\alpha_3=0$	$\alpha_3 = 0$	—
1510.06699_F00912	49	■ 0.999	$\alpha_1=\alpha_2=0$	$\alpha_1 = \alpha_2 = 0$	h , then one must set $\alpha_1 = \alpha_2 = 0$ (at least in $D \leq 4$
1510.06699_F00913	49	■ 1.000	$a=\frac{1}{2}$	$a = \frac{1}{2}$	spacetime dimensions), and on choosing $a = \frac{1}{2}$ the re-
1510.06699_F00914	49	■ 0.999	$a=0=\Lambda, \alpha_1=-\frac{1}{3}\alpha_2$	$a = 0 = \Lambda, \alpha_1 = -\frac{1}{3}\alpha_2$	is $a = 0 = \Lambda, \alpha_1 = -\frac{1}{3}\alpha_2$, which corresponds to the free
1510.06699_F00915	49	■ 1.000	U_4	U_4	that the spacetime has, in general, a Riemann–Cartan U_4
1510.06699_F00916	49	■ 1.000	$\{ \}^{\{0\}} R^{\{\rho\}\{ \}_{\{\sigma\ \mu\ \nu\}}}$	${}^0R^\rho{}_{\sigma\mu\nu}$	leads one to recognise ${}^0R^\rho{}_{\sigma\mu\nu}$ as the standard curvature
1510.06699_F00917	49	■ 1.000	$R \equiv R^{\{\mu\ \nu\}\{ \}_{\{\mu\ \nu\}}}$	$R \equiv R^{\mu\nu}{}_{\mu\nu}$	which relates the Ricci–Cartan scalar $R \equiv R^{\mu\nu}{}_{\mu\nu}$ in
1510.06699_F00918	49	■ 0.999	$\{ \}^{\{0\}} R \equiv \{ \}^{\{0\}} R^{\{\mu\ \nu\}\{ \}_{\{\mu\ \nu\}}}$	${}^0R \equiv {}^0R^{\mu\nu}{}_{\mu\nu}$	U_4 spacetime to the Ricci scalar ${}^0R \equiv {}^0R^{\mu\nu}{}_{\mu\nu}$ in V_4
1510.06699_F00919	49	■ 1.000	$L_{\{\mathrm{R}\{G\}}}=R/(2\ \kappa)$	$L_G = -R/(2\kappa)$	taken to be $L_G = -R/(2\kappa)$ and the theory is interpreted
1510.06699_F00920	49	■ 1.000	$T_{\{\mu\ \nu\}\{\sigma\}}$	$T_{\mu\nu\sigma}$	that includes a tensor field $T_{\mu\nu\sigma}$ (obeying the appropri-
1510.06699_F00921	49	■ 1.000	$T_{\{\mu\ \nu\}\{\lambda\}}$	$T_{\mu\nu\lambda}$	Lagrangian is given by the terms containing $T_{\mu\nu\lambda}$ in (B6)
1510.06699_F00922	49	■ 1.000	$\mathrm{R}=\mathrm{R}^{\{a\ b\}\{ \}_{\{a\ b\}}}$	$\mathcal{R} = \mathcal{R}^{ab}{}_{ab}$	$\mathcal{R} = \mathcal{R}^{ab}{}_{ab}$. Thus, in Kibble’s original theory, one has
1510.06699_F00923	49	■ 1.000	$\Lambda=\alpha_i=\beta_i=0$	$\Lambda = \alpha_i = \beta_i = 0$	which corresponds to setting $a = \frac{1}{2}$ and $\Lambda = \alpha_i = \beta_i = 0$
1510.06699_F00924	49	■ 0.983	$\{ \}^{\{\underline{106}\}}$	106	theory ¹⁰⁶ , which, when re-interpreted geometrically, is a
1510.06699_F00925	50	■ 1.000	$\mathrm{T}_{\{a\}} \equiv \mathrm{T}^{\{c\}\{ \}_{\{a\ c\}}}$	$\mathcal{T}_a \equiv \mathcal{T}^c{}_{ac}$	where $\mathcal{T}_a \equiv \mathcal{T}^c{}_{ac}$, as defined in (66). In particular, by
1510.06699_F00926	50	■ 0.999	$\mathrm{A}^{\{a\ b\}\{ \}_{\{c\}}}=h_{\{c\}\{ \}^{\{\mu\}} A^{\{a\ b\}\{ \}_{\{\mu\}}}$	$\mathcal{A}^{ab}{}_c = h_c{}^\mu A^{ab}{}_\mu$	its fully Lorentz-index counterpart $\mathcal{A}^{ab}{}_c = h_c{}^\mu A^{ab}{}_\mu$. In
1510.06699_F00927	50	■ 1.000	$\sigma_{\{a\ b\}\{ \}^{\{c\}}} \neq 0$	$\sigma_{ab}{}^c \neq 0$	isks). In the general case, in which $\sigma_{ab}{}^c \neq 0$, this is not,
1510.06699_F00928	50	■ 1.000	$\sigma_{\{a\ b\ c\}}$	σ_{abc}	derivatives of the matter field(s), then σ_{abc} is indepen-
1510.06699_F00929	50	■ 1.000	$\mathrm{T}^{\{c\}\{ \}_{\{a\ b\}}}$	$\mathcal{T}^c{}_{ab}$	field strength $\mathcal{T}^c{}_{ab}$, and so the h -field itself must satisfy
1510.06699_F00930	50	■ 0.997	$\mu\ \phi \equiv m$	$\mu\phi \equiv m$	Lagrangian density is given by (70) with $\mu\phi \equiv m$, sev-
1510.06699_F00931	50	■ 1.000	$\mathrm{T}^{\{c\}\{ \}_{\{a\ b\}}}=2\ \kappa\ h\ \sigma_{\{a\ b\}\{ \}^{\{c\}}}$	$\mathcal{T}^c{}_{ab} = 2\kappa h\sigma_{ab}{}^c$	$\mathcal{T}^c{}_{ab} = 2\kappa h\sigma_{ab}{}^c$. More interestingly, the condition (C9)
1510.06699_F00932	50	■ 1.000	$R_{\{\mu\ \nu\}} \equiv R^{\{\lambda\}\{\lambda\}\{\mu\ \nu\}}$	$R_{\mu\nu} \equiv R^\lambda{}_{\mu\lambda\nu}$	where $R_{\mu\nu} \equiv R^\lambda{}_{\mu\lambda\nu}$ and $R \equiv R^\mu{}_\mu$ have their usual forms,
1510.06699_F00933	50	■ 1.000	$R \equiv R^{\{\mu\}\{ \}_{\{\mu\}}}$	$R \equiv R^\mu{}_\mu$	where $R_{\mu\nu} \equiv R^\lambda{}_{\mu\lambda\nu}$ and $R \equiv R^\mu{}_\mu$ have their usual forms,
1510.06699_F00934	50	■ 1.000	$\Gamma^{\{\lambda\}\{\lambda\}\{\mu\ \nu\}}$	$\Gamma^\lambda{}_{\mu\nu}$	but one must remember that the affine connection $\Gamma^\lambda{}_{\mu\nu}$
1510.06699_F00935	51	■ 1.000	$\tau_{\{\mu\ \nu\}}=$	$\tau_{\mu\nu} =$	is not, in general, symmetric in μ and ν . Also, $\tau_{\mu\nu} =$
1510.06699_F00936	51	■ 1.000	$e^{\{a\}\{ \}_{\{\mu\}}} e^{\{b\}\{ \}_{\{\nu\}}} \tau_{\{a\ b\}}$	$e^a{}_\mu e^b{}_\nu \tau_{ab}$	$e^a{}_\mu e^b{}_\nu \tau_{ab}$ and $\sigma_{\mu\nu}{}^\lambda = e^a{}_\mu e^b{}_\nu e_c{}^\lambda \sigma_{ab}{}^c$, where the former
1510.06699_F00937	51	■ 1.000	$\sigma_{\{\mu\ \nu\}\{ \}^{\{\lambda\}}} = e^{\{a\}\{ \}_{\{\lambda\}}} e^{\{b\}\{ \}_{\{\mu\}}} e^{\{c\}\{ \}_{\{\nu\}}} e_{\{c\}\{ \}^{\{\lambda\}}} \sigma_{\{a\ b\}\{ \}^{\{c\}}}$	$\sigma_{\mu\nu}{}^\lambda = e^a{}_\mu e^b{}_\nu e_c{}^\lambda \sigma_{ab}{}^c$	$e^a{}_\mu e^b{}_\nu \tau_{ab}$ and $\sigma_{\mu\nu}{}^\lambda = e^a{}_\mu e^b{}_\nu e_c{}^\lambda \sigma_{ab}{}^c$, where the former
1510.06699_F00938	51	■ 1.000	$\sigma_{\{\mu\ \nu\}\{ \}^{\{\lambda\}}}$	$\sigma_{\mu\nu}{}^\lambda$	angular-momentum $\sigma_{\mu\nu}{}^\lambda$ of the matter field vanishes,
1510.06699_F00939	51	■ 1.000	$T^{\{\lambda\}\{\lambda\}\{\mu\ \nu\}}$	$T^\lambda{}_{\mu\nu}$	then so too does the torsion $T^\lambda{}_{\mu\nu}$ and hence the affine
1510.06699_F00940	51	■ 1.000	$R_{\{\mu\ \nu\}}$	$R_{\mu\nu}$	case, $R_{\mu\nu}$ reduces to the standard symmetric Ricci ten-
1510.06699_F00941	51	■ 1.000	$\{ \}^{\{0\}} R_{\{\mu\ \nu\}} \equiv R_{\{\mu\ \nu\}}\left(V_4\right)$	${}^0R_{\mu\nu} \equiv R_{\mu\nu}(V_4)$	sor ${}^0R_{\mu\nu} \equiv R_{\mu\nu}(V_4)$ in a Riemann spacetime, expressed
1510.06699_F00942	51	■ 0.979	$\{ \}^{\{\underline{108}\}}$	108	grangian suggested by Dirac has the simple form ¹⁰⁸
1510.06699_F00943	51	■ 1.000	$-\frac{1}{4}$	$-\frac{1}{4}$	sionless constant (usually positive); the factor of $-\frac{1}{4}$ is
1510.06699_F00944	51	■ 1.000	ϕ^4	ϕ^4	ϕ^4 , and a term proportional to $\phi^2\mathcal{R}$ that non-minimally
1510.06699_F00945	51	■ 1.000	$L_{\{\varphi\}}(\varphi, \partial\varphi, h, A, B, \phi)$	$L_\varphi(\varphi, \partial\varphi, h, A, B, \phi)$	for the matter field φ are $L_\varphi(\varphi, \partial\varphi, h, A, B, \phi)$; note that
1510.06699_F00946	51	■ 1.000	$\mathcal{L}_T=h^{-1}L_T$	$\mathcal{L}_T = h^{-1}L_T$	The total Lagrangian density is then $\mathcal{L}_T = h^{-1}L_T$,
1510.06699_F00947	51	■ 1.000	$L_{\{T\}}$	L_T	where L_T has the form
1510.06699_F00948	51	■ 0.993	(a, ξ, ν, λ)	(a, ξ, ν, λ)	(a, ξ, ν, λ) are dimensionless (and usually positive). In
1510.06699_F00949	51	■ 1.000	ϕ^{-2}	ϕ^{-2}	plied ϕ^{-2} should replace Newton’s gravitational constant
1510.06699_F00950	51	■ 0.967	$h_{\{a\}\{ \}^{\{\mu\}}}, A^{\{a\ b\}\{ \}_{\{\mu\}}}, B_{\{\mu\}}$	$h_a{}^\mu, A^{ab}{}_\mu, B_\mu$	independent fields $h_a{}^\mu, A^{ab}{}_\mu, B_\mu$ and ϕ respectively ⁵⁰ ,
1510.06699_F00951	51	■ 1.000	$B_{\{\mu\}}, \phi$	B_μ, ϕ	with respect to B_μ, ϕ and the spacetime metric $g_{\mu\nu}$. Our
1510.06699_F00952	52	■ 1.000	$\left(\tau_{\{\varphi\}}\right)^{\{a\}\{ \}_{\{b\}}}$	$(\tau_\varphi)^a{}_b$	where $(\tau_\varphi)^a{}_b$ is the energy-momentum tensor of the mat-
1510.06699_F00953	52	■ 1.000	$\left(\sigma_{\{\varphi\}}\right)_{\{a\ b\}\{ \}^{\{c\}}}$	$(\sigma_\varphi)_{ab}{}^c$	where $(\sigma_\varphi)_{ab}{}^c$ is the spin-angular-momentum tensor of

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00954	52	■ 1.000	$\mathcal{D}_{\{a\}}^* \rightarrow \mathcal{D}_{\{a\}}, \mathcal{T}^* \rightarrow \mathcal{T}$	$\mathcal{D}_a^* \rightarrow \mathcal{D}_a, \mathcal{T}^{*c}_{ab} \rightarrow \mathcal{T}^c_{ab}$	if one makes the replacements $\mathcal{D}_a^* \rightarrow \mathcal{D}_a, \mathcal{T}^{*c}_{ab} \rightarrow \mathcal{T}^c_{ab}$
1510.06699_F00955	52	■ 1.000	$\mathcal{T}_{\{a\}}^* \rightarrow \mathcal{T}_{\{a\}}$	$\mathcal{T}_a^* \rightarrow \mathcal{T}_a$	and $\mathcal{T}_a^* \rightarrow \mathcal{T}_a$ to PGT quantities, although the resulting
1510.06699_F00956	52	■ 1.000	$\left(\zeta_{\varphi}\right)^c$	$(\zeta_{\varphi})^c$	where $(\zeta_{\varphi})^c$ is the dilation current of the φ matter field.
1510.06699_F00957	52	■ 1.000	$\left.\mathcal{D}_{\{a\}}^*\right _{\phi=\phi_0}=-\mathcal{B}_{\{a\}}$	$\mathcal{D}_a^* _{\phi=\phi_0} = -\phi_0 \mathcal{B}_a$	replacement $\mathcal{D}_a^* _{\phi=\phi_0} = -\phi_0 \mathcal{B}_a$ throughout. It is also
1510.06699_F00958	52	■ 1.000	$a=1/\left(\kappa\phi_0^2\right)$	$a = 1/(\kappa\phi_0^2)$	convenient to set $a = 1/(\kappa\phi_0^2)$, where κ is Einstein’s grav-
1510.06699_F00959	52	■ 1.000	$\Lambda=\lambda\phi_0^2/a$	$\Lambda = \lambda\phi_0^2/a = \kappa\lambda$	where $\Lambda = \lambda\phi_0^2/a = \kappa\lambda$ has the correct dimensions of
1510.06699_F00960	52	■ 1.000	$m^2=\nu\phi_0^2/\xi$	$m^2 = \nu\phi_0^2/\xi$	vector field (or Proca field) of mass-squared $m^2 = \nu\phi_0^2/\xi$.
1510.06699_F00961	53	■ 1.000	$\mathcal{R}^{ab}_{cd}, \mathcal{T}^a_{bc} = \mathcal{T}^{*a}_{bc} - \delta^a_c \mathcal{B}_b + \delta^a_b \mathcal{B}_c$	$\mathcal{R}^{ab}_{cd}, \mathcal{T}^a_{bc} = \mathcal{T}^{*a}_{bc} - \delta^a_c \mathcal{B}_b + \delta^a_b \mathcal{B}_c$	tensors $\mathcal{R}^{ab}_{cd}, \mathcal{T}^a_{bc} = \mathcal{T}^{*a}_{bc} - \delta^a_c \mathcal{B}_b + \delta^a_b \mathcal{B}_c$ and $\mathcal{H}_{ab}$
1510.06699_F00962	53	■ 1.000	$\left(\zeta_{\psi}\right)^{\mu}=0$	$(\zeta_{\psi})^{\mu} = 0$	does not appear in (D12), the dilation current $(\zeta_{\psi})^{\mu} = 0$
1510.06699_F00963	53	■ 1.000	$\mathcal{T}_{\{a\}}=0$	$\mathcal{T}_a = 0$	theory, from (C10), we again have $\mathcal{T}_a = 0$. Thus, the
1510.06699_F00964	53	■ 1.000	$\tau_{\psi}=h^{-1}\mu\phi\bar{\psi}\psi$	$\tau_{\psi} = h^{-1}\mu\phi\bar{\psi}\psi$	of the Dirac field is easily found to be $\tau_{\psi} = h^{-1}\mu\phi\bar{\psi}\psi$.
1510.06699_F00965	53	■ 1.000	$m^2\equiv\nu\phi_0^2/\xi$	$m^2 \equiv \nu\phi_0^2/\xi$	where we have defined $m^2 \equiv \nu\phi_0^2/\xi$ and $\sigma_{ab}{}^c$ is the spin-
1510.06699_F00966	53	■ 1.000	$\mathcal{B}_{\{a\}}$	$\mathcal{B}_a$	field $\mathcal{B}_a$ has a non-minimal coupling to the gravitational
1510.06699_F00967	53	■ 1.000	$D_{\{a\}}^*=D_{\{a\}}+wB_{\{a\}}$	$D_{\mu}^* = D_{\mu} + wB_{\mu}$	variant derivative (20), namely $D_{\mu}^* = D_{\mu} + wB_{\mu}$, the
1510.06699_F00968	54	■ 1.000	$P_{\{a\}}\equiv\partial_{\{a\}}\rho$	$P_{\mu} \equiv \partial_{\mu}\rho$	where $P_{\mu} \equiv \partial_{\mu}\rho$. This transformation for V_{μ} implies
1510.06699_F00969	54	■ 1.000	$(w=0)$	$(w = 0)$	variant ($w = 0$) under extended local dilations. One
1510.06699_F00970	54	■ 1.000	$\mathcal{R}^{\dagger ab}_{cd}\equiv h^{\mu}_ch^{\nu}_dR^{\dagger ab}_{\mu\nu}$	$\mathcal{R}^{\dagger ab}_{cd} \equiv h_c{}^{\mu}h_d{}^{\nu}R^{\dagger ab}_{\mu\nu}$	can then define $\mathcal{R}^{\dagger ab}_{cd} \equiv h_c{}^{\mu}h_d{}^{\nu}R^{\dagger ab}_{\mu\nu}$ as before, which
1510.06699_F00971	54	■ 0.997	$D_{\{a\}}^{\natural}\varphi$	$D_{\mu}^{\natural}\varphi$	ever, one quickly finds that $D_{\mu}^{\natural}\varphi$ does <i>not</i> transform co-
1510.06699_F00972	54	■ 1.000	$D_{\{a\}}^{\natural}$	$D_{\mu}^{\natural}$	ing our reluctance thus far to call $D_{\mu}^{\natural}$ (or $\mathcal{D}_{\mu}^{\natural}$) a covariant
1510.06699_F00973	54	■ 1.000	$\nu=\beta_1=\beta_2=0$	$\nu = \beta_1 = \beta_2 = 0$	gravitational Lagrangian L_G and $\nu = \beta_1 = \beta_2 = 0$ in the
1510.06699_F00974	54	■ 0.994	$\xi=\nu=\beta_1=\beta_2=0$	$\xi = \nu = \beta_1 = \beta_2 = 0$	in Section IV A, but with $\xi = \nu = \beta_1 = \beta_2 = 0$. These
1510.06699_F00975	54	■ 1.000	$\mathcal{D}_{\{a\}}^{\natural}\chi$	$\mathcal{D}_a^{\natural}\chi$	to $\mathcal{D}_a^{\natural}\chi$ to assemble the appropriate <i>covariant</i> derivative
1510.06699_F00976	54	■ 1.000	$\mathcal{D}_{\{a\}}^{\dagger}\chi=h^{\mu}_aD_{\mu}^{\dagger}\chi$	$\mathcal{D}_a^{\dagger}\chi = h_a{}^{\mu}D_{\mu}^{\dagger}\chi$	$\mathcal{D}_a^{\dagger}\chi$. In each case, one finds that $\mathcal{D}_a^{\dagger}\chi = h_a{}^{\mu}D_{\mu}^{\dagger}\chi$, where
1510.06699_F00977	54	■ 0.991	$D_{\{a\}}^{\dagger}$	$D_{\mu}^{\dagger}$	$D_{\mu}^{\dagger}$ has the form (143), but with the appropriate value of
1510.06699_F00978	54	■ 0.997	$\left(U_4\right)$	(U_4)	respectively, of a Riemann–Cartan (U_4) spacetime.
1510.06699_F00979	54	■ 1.000	$\delta\varphi$	$\delta\varphi$	given action, for example, then the ‘total’ variation $\delta\varphi$ and the
1510.06699_F00980	54	■ 1.000	$\left[\mathcal{D}_{\{c\}},\mathcal{D}_{\{d\}}\right]\varphi\equiv\frac{1}{2}\mathcal{R}^{ab}_{cd}\Sigma_{cd}\varphi-\mathcal{T}^a_{cd}\mathcal{D}_a\varphi,\mathcal{R}^a_b\equiv\mathcal{R}^{ac}_{bc}$	$[\mathcal{D}_c, \mathcal{D}_d]\varphi \equiv \frac{1}{2}\mathcal{R}^{ab}_{cd}\Sigma_{cd}\varphi - \mathcal{T}^a_{cd}\mathcal{D}_a\varphi, \mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}_{bc}$	signature, $[\mathcal{D}_c, \mathcal{D}_d]\varphi \equiv \frac{1}{2}\mathcal{R}^{ab}_{cd}\Sigma_{cd}\varphi - \mathcal{T}^a_{cd}\mathcal{D}_a\varphi, \mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}_{bc}$
1510.06699_F00981	54	■ 1.000	$\mathcal{T}_{\{a\}}\equiv\mathcal{T}^b_{ab}$	$\mathcal{T}_a \equiv \mathcal{T}^b{}_{ab}$	and $\mathcal{T}_a \equiv \mathcal{T}^b{}_{ab}$.
1510.06699_F00982	55	■ 1.000	$\widehat{\mathcal{D}}_{\{a\}}^*$	$\widehat{\mathcal{D}}_a^*$	(Weyl weight $w = 0$) versions of the quantities listed below are denoted with a caret, for example $\widehat{h}_a{}^{\mu}$ or $\widehat{\mathcal{D}}_a^*$ or $\widehat{\mathcal{R}}^{ab}_{cd}$; see
1510.06699_F00983	55	■ 1.000	$\widehat{\mathcal{R}}^{ab}_{cd}$	$\widehat{\mathcal{R}}^{ab}_{cd}$	(Weyl weight $w = 0$) versions of the quantities listed below are denoted with a caret, for example $\widehat{h}_a{}^{\mu}$ or $\widehat{\mathcal{D}}_a^*$ or $\widehat{\mathcal{R}}^{ab}_{cd}$; see

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09984 [conf 0.000]	55	0.000	$A^{\dagger\{a\}b}_{\mu}\}_{\mu}\equiv A^{\{a\}b}_{\mu}\}_{\mu}+2\eta^{\{a\}b\}_{\mu}h_{c\}^{\nu}\}_{\nu}V_{\nu}$	$A^{\dagger ab}_{\mu}\equiv A^{ab}_{\mu}+2\eta^{c[a}b^b]_{\mu}h_{c\}^{\nu}V_{\nu}$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu\equiv A^{ab}{}_\mu+2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field (A^1-field) Derivative operators $\mathcal{D}_a\equiv h_a{}^\mu D_\mu\equiv h_a{}^\mu(\partial_\mu+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^*\equiv\partial_\mu+wB_\mu$ $\mathcal{D}_\mu^*\equiv h_a{}^\mu D_\mu^*\equiv h_a{}^\mu(\partial_\mu^*+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^1\equiv\partial_\mu-w(V_\mu+\frac{1}{3}T_\nu)$ $\mathcal{D}_\mu^1\equiv h_a{}^\mu D_\mu^1\equiv h_a{}^\mu(\partial_\mu^1+\frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ $\mathcal{D}_\mu^2\equiv h_a{}^\mu D_\mu^2\equiv h_a{}^\mu(\partial_\mu+\frac{1}{2}A^{\dagger 1ab}{}_\mu\Sigma_{ab})$ PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}{}_{cd}=\mathcal{R}^{*ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]}+A^{\dagger a}{}_{c[\mu}A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^*b^a{}_{\nu]}$ $\mathcal{T}^{1a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^1b^a{}_{\nu]}$ $\mathcal{T}^{2a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^2b^a{}_{\nu]}$ $\mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ $\mathcal{H}_{ab}^1=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu+\frac{1}{3}T_{\nu]})$ PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Reduced A-fields and related quantities $c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ $c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^*b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^{\dagger}b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger a}{}^b)$ PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc}\equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc}\equiv 0$) eWGT reduced A¹-field (when $\mathcal{T}^{1a}{}_{bc}\equiv 0$) Currents and related quantities $\mathcal{L}\equiv h^{-1}L$ $t^a{}_b\equiv t^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ $s_{ab}{}^c\equiv s_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ $j^a\equiv j^a b^b{}_\mu\equiv b^b{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ $t^{\dagger a}{}_b\equiv t^a{}_b+2(s_{cb}{}^aV^c-s^{ac}{}_cV_b)$ $j^{\dagger a}\equiv j^a-2s^{ab}{}_b$ $\tau^a{}_b\equiv\tau^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$ $\sigma_{ab}{}^c\equiv\sigma_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$ $\zeta^a\equiv\zeta^\mu b^a{}_\mu\equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$ $\tau^{\dagger a}{}_b\equiv\tau^a{}_b+2(\sigma_{cb}{}^aV^c-\sigma^{ac}{}_cV_b)$ $\zeta^{\dagger a}\equiv\zeta^a-2s^{ab}{}_b$ Lagrangian Lagrangian density gravitational sector energy-momentum tensor^e gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^e matter sector energy-momentum tensor^e matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09985 [conf 0.000]	55	0.000	$\mathcal{D}_{\{a\}} \equiv h_{\{a\}}^{\mu} \partial_{\mu} \equiv h_a^{\mu} \partial_{\mu} + \frac{1}{2} A^{ab}_{\mu} \Sigma_{ab}$	$\mathcal{D}_a \equiv h_a^{\mu} D_{\mu} \equiv h_a^{\mu} \left(\partial_{\mu} + \frac{1}{2} A^{ab}_{\mu} \Sigma_{ab} \right)$	Gravitational gauge field variables $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[a} b^{\mu]}{}_{\mu} h_c{}^{\nu} V_{\nu}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^* \equiv \partial_{\mu} + w B_{\mu}$ $\mathcal{D}_{\mu}^* \equiv h_a{}^{\mu} D_{\mu}^* \equiv h_a{}^{\mu} (\partial_{\mu}^* + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^{\dagger} \equiv \partial_{\mu} - w(V_{\mu} + \frac{1}{3} T_{\nu})$ $\mathcal{D}_{\mu}^{\dagger} \equiv h_a{}^{\mu} D_{\mu}^{\dagger} \equiv h_a{}^{\mu} (\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ $\mathcal{D}_{\mu}^{\natural} \equiv h_a{}^{\mu} D_{\mu}^{\natural} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\dagger} b^a{}_{\nu]}$ $\mathcal{T}^{\natural ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\natural} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \frac{1}{3} T_{\nu])}$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$ $c^{*ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$ ${}^0 A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$ $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ $j^a \equiv j^a{}_{\mu} b^{\mu} \equiv b^{\mu}{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$ $\zeta^a \equiv \zeta^a{}_{\mu} b^{\mu} \equiv b^{\mu}{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*ab}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger ab}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09986 [conf 0.000]	55	0.000	$\partial_{\mu}^{\wedge{*}} \equiv \partial_{\mu} + w B_{\mu}$	$\partial_{\mu}^* \equiv \partial_{\mu} + w B_{\mu}$	Gravitational gauge field variables $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[a}b^{\dagger b]}{}_{\mu} h_c{}^{\nu} V_{\nu}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^* \equiv \partial_{\mu} + w B_{\mu}$ $\mathcal{D}_{\mu}^* \equiv h_a{}^{\mu} D_{\mu}^* \equiv h_a{}^{\mu} (\partial_{\mu}^* + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^{\dagger} \equiv \partial_{\mu} - w(V_{\mu} + \frac{1}{3} T_{\nu})$ $\mathcal{D}_{\mu}^{\dagger} \equiv h_a{}^{\mu} D_{\mu}^{\dagger} \equiv h_a{}^{\mu} (\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ $\mathcal{D}_{\mu}^{\natural} \equiv h_a{}^{\mu} D_{\mu}^{\natural} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\dagger} b^a{}_{\nu]}$ $\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\natural} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \frac{1}{3} T_{\nu])}$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$ $c^{*ab}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0 A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$ $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ $j^a \equiv j^a{}_{\mu} b^{\mu} \equiv b^{\mu}{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$ $\zeta^a \equiv \zeta^a{}_{\mu} b^{\mu} \equiv b^{\mu}{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09987 [conf 0.000]	55	0.000	$\mathcal{D}_{\mathfrak{a}}^*\equiv h_{\mathfrak{a}}^{\mu}\mathcal{D}_{\mu}^*\equiv h_{\mathfrak{a}}^{\mu}\left(\partial_{\mu}^*+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab}\right)\equiv h_{\mathfrak{a}}^{\mu}\mathcal{D}_{\mu}^*\backslash\left(\partial_{\mu}^*+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab}\right)$	$\mathcal{D}_a^*\equiv h_a{}^\mu D_\mu^*\equiv h_a{}^\mu\left(\partial_\mu^*+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab}\right)$	Gravitational gauge field variables $h_a{}^\mu$ translational gauge field (h-field) h determinant of $h_a{}^\mu$ $b^a{}_\mu$ inverse h-field $A^{ab}{}_\mu$ rotational gauge field (A-field) B_μ WGT dilational gauge field (B-field) V_μ eWGT dilational gauge field (V-field) $A^{\dagger ab}{}_\mu\equiv A^{ab}{}_\mu+2\eta^{c[ab]}{}_\mu h_c{}^\nu V_\nu$ eWGT extended rotational gauge field ($A^\dagger$-field) Derivative operators $\mathcal{D}_a\equiv h_a{}^\mu D_\mu\equiv h_a{}^\mu(\partial_\mu+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ PGT (generalised) covariant derivative^a $\partial_\mu^*\equiv\partial_\mu+wB_\mu$ WGT 'augmented' partial derivative $\mathcal{D}_\mu^*\equiv h_a{}^\mu D_\mu^*\equiv h_a{}^\mu(\partial_\mu^*+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ WGT (generalised) covariant derivative^a $\partial_\mu^\dagger\equiv\partial_\mu-w(V_\mu+\frac{1}{3}T_\nu)$ eWGT 'augmented' partial derivative $\mathcal{D}_a^\dagger\equiv h_a{}^\mu D_\mu^\dagger\equiv h_a{}^\mu(\partial_\mu^\dagger+\frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ eWGT (generalised) covariant derivative^a $\mathcal{D}_\mu^\dagger\equiv h_a{}^\mu D_\mu^\dagger\equiv h_a{}^\mu(\partial_\mu^\dagger+\frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}{}_{cd}=\mathcal{R}^{*ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]}+A_{c[\mu}^a A^{cb}{}_{\nu]}+A_{c[\mu}^a A^{cb}{}_{\nu]})$ PGT and WGT rotational gauge field strength^{a, b} $\mathcal{R}^{\dagger ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]}+A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ eWGT rotational gauge field strength^{a, b} $\mathcal{T}^{ab}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}b^{a}{}_{\nu]}$ PGT translational gauge field strength^b $\mathcal{T}^{*a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^*b^a{}_{\nu]}$ WGT translational gauge field strength^b $\mathcal{T}^{\dagger a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ eWGT translational gauge field strength^b $\mathcal{T}^{\mathfrak{a}}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ eWGT translational gauge 'semi' field strength^{a, b} $\mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ WGT dilational gauge field strength $\mathcal{H}_{ab}^\dagger=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu+\frac{1}{3}T_{\nu]})$ eWGT dilational gauge field strength Reduced A-fields and related quantities $c^{ab}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ PGT Ricci rotation coefficients $c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^*b^a{}_{\nu]}$ WGT Ricci rotation coefficients $c^{\dagger a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^\dagger b^a{}_{\nu]}$ eWGT Ricci rotation coefficients ${}^0A^{ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^a{}^b)$ PGT reduced A-field (when $\mathcal{T}^a{}_{bc}\equiv 0$) ${}^0A^{*ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c{}^a{}^b)$ WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc}\equiv 0$) ${}^0A^{\dagger ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger a}{}^b)$ eWGT reduced A[†]-field (when $\mathcal{T}^{\dagger a}{}_{bc}\equiv 0$) Currents and related quantities $\mathcal{L}\equiv h^{-1}L$ Lagrangian $t^a{}_b\equiv t^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ Lagrangian density $s_{ab}{}^c\equiv s_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ gravitational sector energy-momentum tensor^c $j^a\equiv j^a{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ gravitational sector spin-angular-momentum tensor^c $t^{\dagger a}{}_b\equiv t^a{}_b+2(s_{cb}{}^a\mathcal{V}^c-s^{ac}{}_c\mathcal{V}_b)$ gravitational sector dilation current^c $j^{\dagger a}\equiv j^a-2s^{ab}{}_b$ gravitational sector eWGT covariant energy-momentum tensor^c $\tau^a{}_b\equiv\tau^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$ matter sector energy-momentum tensor^c $\sigma_{ab}{}^c\equiv\sigma_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$ gravitational sector eWGT covariant dilation current^c $\zeta^a\equiv\zeta^a{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_M/\delta B_\mu$ matter sector spin-angular-momentum tensor^c $\tau^{\dagger a}{}_b\equiv\tau^a{}_b+2(\sigma_{cb}{}^a\mathcal{V}^c-\sigma^{ac}{}_c\mathcal{V}_b)$ matter sector dilation current^c $\zeta^{\dagger a}\equiv\zeta^a-2s^{ab}{}_b$ gravitational sector eWGT covariant energy-momentum tensor^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09988 [conf 0.000]	55	0.000	$\partial_{\mu}^{\dagger}\equiv\partial_{\mu}-w\left(V_{\mu}+\frac{1}{3}T_{\mu}\right)$	$\partial_{\mu}^{\dagger}\equiv\partial_{\mu}-w\left(V_{\mu}+\frac{1}{3}T_{\mu}\right)$	<i>Gravitational gauge field variables</i> $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu}\equiv A^{ab}{}_{\mu}+2\eta^{c[ab]}{}_{\mu}h_c{}^{\nu}V_{\nu}$ <i>Derivative operators</i> $\mathcal{D}_a\equiv h_a{}^{\mu}D_{\mu}\equiv h_a{}^{\mu}(\partial_{\mu}+\tfrac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab})$ $\partial_{\mu}^a\equiv\partial_{\mu}+wB_{\mu}$ $\mathcal{D}_{\mu}^a\equiv h_a{}^{\mu}D_{\mu}^a\equiv h_a{}^{\mu}(\partial_{\mu}^a+\tfrac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab})$ $\partial_{\mu}^{\dagger}\equiv\partial_{\mu}-w(V_{\mu}+\tfrac{1}{3}T_{\mu})$ $\mathcal{D}_{\mu}^{\dagger}\equiv h_a{}^{\mu}D_{\mu}^{\dagger}\equiv h_a{}^{\mu}(\partial_{\mu}^{\dagger}+\tfrac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab})$ $\mathcal{D}_{\mu}^a\equiv h_a{}^{\mu}D_{\mu}^a\equiv h_a{}^{\mu}(\partial_{\mu}+\tfrac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab})$ <i>Gauge field strengths</i> $\mathcal{R}^{ab}{}_{cd}=\mathcal{R}^{*ab}{}_{cd}=2h_a{}^{\mu}h_b{}^{\nu}(\partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd}=2h_a{}^{\mu}h_b{}^{\nu}(\partial_{[\mu}A^{\dagger ab}{}_{\nu]}+A^{\dagger a}{}_{c[\mu}A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc}=2h_b{}^{\mu}h_c{}^{\nu}D_{[\mu}b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc}=2h_b{}^{\mu}h_c{}^{\nu}D_{[\mu}^*b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc}=2h_b{}^{\mu}h_c{}^{\nu}D_{[\mu}^{\dagger}b^a{}_{\nu]}$ $\mathcal{T}^a{}_{bc}=2h_b{}^{\mu}h_c{}^{\nu}D_{[\mu}^2b^a{}_{\nu]}$ $\mathcal{H}_{ab}=2h_a{}^{\mu}h_b{}^{\nu}\partial_{[\mu}B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger}=2h_a{}^{\mu}h_b{}^{\nu}\partial_{[\mu}(V_{\nu}+\tfrac{1}{3}T_{\nu])}$ <i>Reduced A-fields and related quantities</i> $c^a{}_{bc}\equiv 2h_b{}^{\mu}h_c{}^{\nu}\partial_{[\mu}b^a{}_{\nu]}$ $c^{*a}{}_{bc}\equiv 2h_b{}^{\mu}h_c{}^{\nu}\partial_{[\mu}^*b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc}\equiv 2h_b{}^{\mu}h_c{}^{\nu}\partial_{[\mu}^{\dagger}b^a{}_{\nu]}$ ${}^0A^{ab}{}_{\mu}\equiv\tfrac{1}{2}b^c{}_{\mu}(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^{ab})$ ${}^0A^{*ab}{}_{\mu}\equiv\tfrac{1}{2}b^c{}_{\mu}(c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c{}^{*ab})$ ${}^0A^{\dagger ab}{}_{\mu}\equiv\tfrac{1}{2}b^c{}_{\mu}(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger ab})$ <i>Currents and related quantities</i> $\mathcal{L}\equiv h^{-1}L$ $t^a{}_b\equiv t^a{}_{\mu}h_b{}^{\mu}\equiv h_b{}^{\mu}\delta\mathcal{L}_G/\delta h_a{}^{\mu}$ $s_{ab}{}^c\equiv s_{ab}{}^{\mu}b^c{}_{\mu}\equiv b^c{}_{\mu}\delta\mathcal{L}_G/\delta A^{ab}{}_{\mu}$ $j^a\equiv j^a{}_{\mu}b^{\mu}{}_{\nu}\equiv b^{\nu}{}_{\mu}\delta\mathcal{L}_G/\delta B_{\mu}$ $\hat{t}^{\dagger a}{}_b\equiv \hat{t}^a{}_b+2(s_{cb}{}^a\mathcal{V}^c-s^{ac}{}_c\mathcal{V}_b)$ $\hat{j}^{\dagger a}\equiv j^a-2s^{ab}{}_b$ $\tau^a{}_b\equiv\tau^a{}_{\mu}h_b{}^{\mu}\equiv h_b{}^{\mu}\delta\mathcal{L}_M/\delta h_a{}^{\mu}$ $\sigma_{ab}{}^c\equiv\sigma_{ab}{}^{\mu}b^c{}_{\mu}\equiv b^c{}_{\mu}\delta\mathcal{L}_M/\delta A^{ab}{}_{\mu}$ $\zeta^a\equiv\zeta^{\mu}b^a{}_{\mu}\equiv b^a{}_{\mu}\delta\mathcal{L}_M/\delta B_{\mu}$ $\tau^{\dagger a}{}_b\equiv\tau^a{}_b+2(\sigma_{cb}{}^a\mathcal{V}^c-\sigma^{ac}{}_c\mathcal{V}_b)$ $\zeta^{\dagger a}\equiv\zeta^a-2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc}\equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc}\equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc}\equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09989 [conf 0.000]	55	0.000	$\mathcal{D}_a^{\dagger} \equiv h_a{}^{\mu} \mathcal{D}_{\mu}^{\dagger} \equiv h_a{}^{\mu} \left(\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab} \right)$	$\mathcal{D}_a^{\dagger} \equiv h_a{}^{\mu} D_{\mu}^{\dagger} \equiv h_a{}^{\mu} \left(\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab} \right)$	Gravitational gauge field variables $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[ab]}{}_{\mu} h_c{}^{\nu} V_{\nu}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^a \equiv \partial_{\mu} + w B_{\mu}$ $\mathcal{D}_{\mu}^a \equiv h_a{}^{\mu} D_{\mu}^a \equiv h_a{}^{\mu} (\partial_{\mu}^a + \frac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^{\dagger} \equiv \partial_{\mu} - w(V_{\mu} + \frac{1}{3} T_{\nu})$ $\mathcal{D}_{\mu}^{\dagger} \equiv h_a{}^{\mu} D_{\mu}^{\dagger} \equiv h_a{}^{\mu} (\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ $\mathcal{D}_{\mu}^a \equiv h_a{}^{\mu} D_{\mu}^a \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{[a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^a b^a{}_{\nu]}$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^a b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \frac{1}{3} T_{\nu])}$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_{\mu} + c^b{}_{\mu}{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{*ab}{}_{\mu} + c^{*b}{}_{\mu}{}^a - c_c^{*ab})$ ${}^0 A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_{\mu} + c^{\dagger b}{}_{\mu}{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$ $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ $j^a \equiv j^a{}_{\mu} b^{\mu} \equiv b^{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$ $\zeta^a \equiv \zeta^a{}_{\mu} b^{\mu} \equiv b^{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09990 [conf 0.000]	55	0.000	$\mathrm{D}_{\mathrm{a}}^{\mathrm{a}} \equiv h_{\mathrm{a}}^{\mathrm{a}} \mathrm{D}_{\mathrm{a}}^{\mathrm{a}} \equiv h_{\mathrm{a}}^{\mathrm{a}} \left(\partial_{\mathrm{a}} + \frac{1}{2} A^{\dagger \mathrm{a} \mathrm{b}} \Sigma_{\mathrm{a} \mathrm{b}} \right)$	$\mathcal{D}_a^{\mathfrak{a}} \equiv h_a{}^\mu D_\mu^{\mathfrak{a}} \equiv h_a{}^\mu \left(\partial_\mu + \frac{1}{2} A^{\dagger ab} \Sigma_{ab} \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^{\mathfrak{a}} \equiv h_a{}^\mu D_\mu^{\mathfrak{a}} \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}_{cd}^{ab} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09991 [conf 0.000]	55	0.000	a	a	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09993 [conf 0.000]	55	0.000	$\{ \}^{\mathrm{a}}\{ \}^{\mathrm{b}}$	ab	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3}T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_\nu \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_\nu \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09994 [conf 0.000]	55	0.000	$\{ \}^{\wedge}\{\backslash\text{text {b }}\}$	b	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\mu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F08995 [conf 0.000]	55	0.000	$\mathrm{\mathfrak{b}}$	b	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09997 [conf 0.000]	55	0.000	$c^{\{*\ a\}\}_{\{b\ c\}} \equiv 2\ h_{\{b\}\{ \}^{\{\mu\}}}\ h_{\{c\}\{ \}^{\{\nu\}}}\ \partial_{[\mu}^{\{*\}}\ b^{\{a\}\}_{\}_{\nu]}}$	$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2}A^{\dagger\ ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger\ ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger\ ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger\ ab}{}_{\nu]} + A^{\dagger\ a}{}_{c[\mu} A^{\dagger\ cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathcal{A}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathcal{A}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1}L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{} b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09998 [conf 0.000]	55	0.000	$c^{\dagger a}_{\{ }_{\{b\} c} \equiv h_{\{b\} \{ }^{\mu} h_{\{c\} \{ }^{\nu} h_{\{a\} \{ }^{\dagger} b_{\{ }_{\{ \nu \} \} } \partial_{[\mu} b^a_{\{ }_{\{ \nu \} \}]}$	$c^{\dagger a}_{bc} \equiv 2 h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[ab}b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathcal{A}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathcal{A}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^a{}^{*b})$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F09999 [conf 0.000]	55	0.000	$\{ \}^{\{0\}} A^{\{a\} b\} \{ \}_{\mu} \equiv \frac{1}{2} b^{\{c\} \{ \}_{\mu}} \left(c^{\{a\} b\} \{ \}_{c} + c^{\{b\} \{ \}_{c}} \{ \}^{\{a\} - c\} \{ \}_{b} \right)$	${}^0A^{ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} \left(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab} \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\mu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{[a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\perp} b^a{}_{\nu]}$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\perp} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0A^{\perp ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\perp ab}{}_c + c^{\perp b}{}_c{}^a - c_c^{\perp ab})$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{} b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01000 [conf 0.000]	55	0.000	$\{ \}^{\{0\}} A^{\{* \text{ a b}\} \{ \}_{\mu}} \equiv \frac{1}{2} b^{\{c\} \{ \}_{\mu}} \left(c^{\{* \text{ a b}\} \{ \}_{c} + c^{\{* \text{ b}\} \{ \}_{c}} - c_c^{*ab} \{ \}^{\{a\} - c_{\{c\} \{ * \text{ a b}\} \right)}$	${}^0 A^{*ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} \left(c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab} \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}_{cd}^{ab} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A_{c[\mu}^a A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{3a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^3 b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c_c^{b\ a} - c_c^{*ab})$ ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c_c^{*b\ a} - c_c^{ab})$ ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c_c^{\dagger b\ a} - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field (<i>A</i>¹-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced <i>A</i>-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced <i>A</i>-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced <i>A</i>¹-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01001 [conf 0.000]	55	0.000	$\{ \}^{\wedge\{0\}} A^{\{\dagger a \ b\}\{ \}_{\mu}} \equiv \frac{1}{2} b^{\{c\}\{ \}_{\mu}} \left(c^{\{\dagger a \ b\}\{ \}_{c}} + c^{\{\dagger b \ a \}_{c}} - c_c^{\{\dagger ab\}\{ \}_{c}} \right) \{ \}_{c}\{ \}^{\{a\}-c_{\{c\}}^{\{\dagger a \ b\}\right)}$	${}^0A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} \left(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab} \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}_{cd}^{ab} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A_{c[\mu}^a A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ $c^{\mathfrak{a}}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c_c^{b\ a} - c_c^{a\ b})$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c_c^{*b\ a} - c_c^{*a\ b})$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c_c^{\dagger b\ a} - c_c^{\dagger a\ b})$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field (<i>A</i>¹-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced <i>A</i>-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced <i>A</i>-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced <i>A</i>¹-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01002 [conf 0.000]	55	0.000	$\mathrm{cal}^{\mathrm{T}}\{a\}\{ \}_{b\ c} \equiv 0$	$\mathcal{T}^a{}_{bc} \equiv 0$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathcal{A}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathcal{A}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}^{\dagger}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1}L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^b{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^b{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01003 [conf 0.000]	55	0.000	$\mathrm{cal{T}}^{\ast\mathrm{a}}_{\mathrm{bc}}\equiv 0$	$\mathcal{T}^{\ast a}_{bc}\equiv 0$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu\equiv A^{ab}{}_\mu+2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a\equiv h_a{}^\mu D_\mu\equiv h_a{}^\mu(\partial_\mu+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^\ast\equiv\partial_\mu+wB_\mu$ $\mathcal{D}_\mu^\ast\equiv h_a{}^\mu D_\mu^\ast\equiv h_a{}^\mu(\partial_\mu^\ast+\frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^\dagger\equiv\partial_\mu-w(V_\mu+\frac{1}{3}T_\nu)$ $\mathcal{D}_a^\dagger\equiv h_a{}^\mu D_\mu^\dagger\equiv h_a{}^\mu(\partial_\mu^\dagger+\frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ $\mathcal{D}_\mu^\dagger\equiv h_a{}^\mu D_\mu^\dagger\equiv h_a{}^\mu(\partial_\mu+\frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd}=\mathcal{R}^{\ast ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]}+A^{\dagger a}{}_{c[\mu}A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ $\mathcal{T}^{\ast a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^\ast b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\ast\dagger a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\ast\dagger} b^a{}_{\nu]}$ $\mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ $\mathcal{H}^{\dagger}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu+\frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ $c^{\ast a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^\ast b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^a{}^b)$ ${}^0A^{\ast ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{\ast ab}{}_c+c^{\ast b}{}_c{}^a-c_c{}^{\ast a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu\equiv\frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c{}^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L}\equiv h^{-1}L$ $t^a{}_b\equiv t^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ $s_{ab}{}^c\equiv s_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ $j^a\equiv j^a{}^\mu b^b{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ $t^{\dagger a}{}_b\equiv t^{\dagger a}{}_b+2(s_{cb}{}^a\mathcal{V}^c-s^{\ast c}{}_c\mathcal{V}_b)$ $j^{\dagger a}\equiv j^a-2s^{\ast a}{}_b$ $\tau^a{}_b\equiv\tau^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$ $\sigma_{ab}{}^c\equiv\sigma_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$ $\zeta^a\equiv\zeta^a{}^\mu b^b{}_\mu\equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$ $\tau^{\dagger a}{}_b\equiv\tau^a{}_b+2(\sigma_{cb}{}^a\mathcal{V}^c-\sigma^{\ast c}{}_c\mathcal{V}_b)$ $\zeta^{\dagger a}\equiv\zeta^a-2s^{\ast a}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01004 [conf 0.000]	55	0.000	$\mathrm{\mathcal{T}}^{\dagger a}{}_{bc} \equiv 0$	$\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1}L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^b{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^b{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01005 [conf 0.000]	55	0.000	L	L	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{A}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{A}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01086 [conf 0.000]	55	0.000	$\mathrm{\mathcal{L}} \equiv h^{-1} L$	$\mathcal{L} \equiv h^{-1} L$	Gravitational gauge field variables $h_a{}^\mu$ translational gauge field (h-field) h determinant of $h_a{}^\mu$ $b^a{}_\mu$ inverse h-field $A^{ab}{}_\mu$ rotational gauge field (A-field) B_μ WGT dilational gauge field (B-field) V_μ eWGT dilational gauge field (V-field) $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{\dagger b]}{}_\mu h_c{}^\nu V_\nu$ eWGT extended rotational gauge field ($A^\dagger$-field) Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ PGT (generalised) covariant derivative^a $\partial_\mu^a \equiv \partial_\mu + w B_\mu$ WGT 'augmented' partial derivative $\mathcal{D}_\mu^a \equiv h_a{}^\mu D_\mu^a \equiv h_a{}^\mu (\partial_\mu^a + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ WGT (generalised) covariant derivative^a $\partial_\mu^{\dagger a} \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ eWGT 'augmented' partial derivative $\mathcal{D}_a^{\dagger a} \equiv h_a{}^\mu D_\mu^{\dagger a} \equiv h_a{}^\mu (\partial_\mu^{\dagger a} + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ eWGT (generalised) covariant derivative^a $\mathcal{D}_\mu^{\dagger a} \equiv h_a{}^\mu D_\mu^{\dagger a} \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ eWGT (generalised) semi-covariant derivative^a

Gauge field strengths

$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$
PGT and WGT rotational gauge field strength^{a, b}

 $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$
eWGT rotational gauge field strength^{a, b}
 $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$
PGT translational gauge field strength^b
 $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$
WGT translational gauge field strength^b
 $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\dagger a} b^a{}_{\nu]}$
eWGT translational gauge field strength^b
 $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\dagger a} b^a{}_{\nu]}$
eWGT translational gauge 'semi' field strength^{a, b}
 $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$
WGT dilational gauge field strength

 $\mathcal{H}_{ab}^{\dagger a} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$
eWGT dilational gauge field strength

Reduced A-fields and related quantities

$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$
PGT Ricci rotation coefficients

 $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$
WGT Ricci rotation coefficients

 $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger a} b^a{}_{\nu]}$
eWGT Ricci rotation coefficients

 ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$
PGT reduced A -field (when $\mathcal{T}^a{}_{bc} \equiv 0$)

 ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$
WGT reduced A -field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)

 ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$
eWGT reduced $A^\dagger$ -field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)

Currents and related quantities

$\mathcal{L}$
Lagrangian

 $\mathcal{L} \equiv h^{-1} L$
Lagrangian density

 $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$
gravitational sector energy-momentum tensor^c
 $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$
gravitational sector spin-angular-momentum tensor^c
 $j^a \equiv j^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$
gravitational sector dilation current^c
 $t^{\dagger a}{}_b \equiv t^{\dagger a}{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac} \mathcal{V}_b)$
gravitational sector eWGT covariant energy-momentum tensor^c
 $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$
gravitational sector eWGT covariant dilation current^c
 $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$
matter sector energy-momentum tensor^c
 $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$
matter sector spin-angular-momentum tensor^c
 $\zeta^a \equiv \zeta^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$
matter sector dilation current^c
 $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac} \mathcal{V}_b)$
gravitational sector eWGT covariant energy-momentum tensor^c
 $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$
gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01008 [conf 0.000]	55	0.000	$s_{\{a\} \{b\} \{c\}} \equiv s_{\{a\} \{b\} \{\mu\}} b^{\{c\} \{ \}_\mu} \equiv b^{\{c\} \{ \}_\mu} \delta \mathcal{L}_G / \delta A^{ab}_\mu$	$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}_\mu$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_\mu A^{ab}{}_\nu - A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_\mu A^{\dagger ab}{}_\nu + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $T^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $T^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $T^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^2 b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_{\nu]})$ PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced A[†]-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^{\mu}{}_\nu \equiv b^{\mu}{}_\nu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^{\dagger a}{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^{\mu}{}_\nu \equiv b^{\mu}{}_\nu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01009 [conf 0.000]	55	0.000	$j^{\mathbf{a}} \equiv j^{\mathbf{a}}_{\mu} \quad b^{\mathbf{a}} \equiv b^{\mathbf{a}}_{\mu} \quad \delta \mathcal{L}_G / \delta B_{\mu}$	$j^a \equiv j^{\mu} b^a_{\mu} \equiv b^a_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$	Gravitational gauge field variables $h_a{}^{\mu}$ h $b^a{}_{\mu}$ $A^{ab}{}_{\mu}$ B_{μ} V_{μ} $A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[a}b^b]{}_{\mu} h_c{}^{\nu} V_{\nu}$ Derivative operators $\mathcal{D}_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^* \equiv \partial_{\mu} + w B_{\mu}$ $\mathcal{D}_{\mu}^* \equiv h_a{}^{\mu} D_{\mu}^* \equiv h_a{}^{\mu} (\partial_{\mu}^* + \tfrac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab})$ $\partial_{\mu}^{\dagger} \equiv \partial_{\mu} - w(V_{\mu} + \tfrac{1}{3} T_{\nu})$ $\mathcal{D}_{\mu}^{\dagger} \equiv h_a{}^{\mu} D_{\mu}^{\dagger} \equiv h_a{}^{\mu} (\partial_{\mu}^{\dagger} + \tfrac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ $\mathcal{D}_{\mu}^{\natural} \equiv h_a{}^{\mu} D_{\mu}^{\natural} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\dagger} b^a{}_{\nu]}$ $\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu}^{\natural} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \tfrac{1}{3} T_{\nu])}$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0 A^{*ab}{}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0 A^{\dagger ab}{}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$ $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$ $j^a \equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$ $\zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^{\mu}$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01010 [conf 0.000]	55	0.000	$t^{\dagger}\{\backslash dagger\ a\}\{ \}_{_b}\ \equiv t^{\wedge}\{a\}\{ \}_{_b}+2\left(s_{\{c\ b\}\{ \}^{\wedge}\{a\}\ \backslash mathcal\{V\}^{\wedge}\{c\}\cdot s^{\wedge}\{a\ c\}\{ \}_{_c}\ \backslash mathcal\{V\}_{_b}\right)$	$t^{\dagger}{}_b \equiv t^a{}_b + 2\left(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b\right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \tfrac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{A}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{A}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \tfrac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}^b)$ ${}^0A^{*ab}{}_\mu \equiv \tfrac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}^b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}^b)$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (<i>h</i>-field) determinant of $h_a{}^\mu$ inverse <i>h</i>-field rotational gauge field (<i>A</i>-field) WGT dilational gauge field (<i>B</i>-field) eWGT dilational gauge field (<i>V</i>-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced <i>A</i>-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced <i>A</i>-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01011 [conf 0.000]	55	0.000	$\text{j}^{\dagger a} \equiv j^a - 2s^a{}_b \text{ }_{\text{ }_b}$	$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ $c^{\mathfrak{a}}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}_b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}_b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}_b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\mathfrak{a}}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\mathfrak{a}} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01014 [conf 0.000]	55	0.000	$\backslash\mathrm{zeta}^{\mathrm{a}}\backslash\mathrm{equiv}\backslash\mathrm{zeta}^{\mathrm{\mu}}\backslash\mathrm{b}^{\mathrm{a}}\{\}_{\backslash\mathrm{\mu}}\backslash\mathrm{equiv}\backslash\mathrm{b}^{\mathrm{a}}\{\}_{\backslash\mathrm{\mu}}\backslash\mathrm{delta}\backslash\mathrm{mathcal}\{\mathrm{L}\}_{\backslash\mathrm{M}}\backslash\mathrm{delta}\backslash\mathrm{B}_{\backslash\mathrm{\mu}}$	$\zeta^a\equiv\zeta^\mu b^a{}_\mu\equiv b^a{}_\mu\delta\mathcal{L}_\mathrm{M}/\delta B_\mu$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu\equiv A^{ab}{}_\mu+2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a\equiv h_a{}^\mu D_\mu\equiv h_a{}^\mu(\partial_\mu+\tfrac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^*\equiv\partial_\mu+wB_\mu$ $\mathcal{D}_\mu^*\equiv h_a{}^\mu D_\mu^*\equiv h_a{}^\mu(\partial_\mu^*+\tfrac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ $\partial_\mu^\perp\equiv\partial_\mu-w(V_\mu+\tfrac{1}{3}T_\nu)$ $\mathcal{D}_a^\dagger\equiv h_a{}^\mu D_\mu^\dagger\equiv h_a{}^\mu(\partial_\mu^\dagger+\tfrac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ $\mathcal{D}_\mu^\dagger\equiv h_a{}^\mu D_\mu^\dagger\equiv h_a{}^\mu(\partial_\mu+\tfrac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd}=\mathcal{R}^{*ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]}+A^a{}_{c[\mu}A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd}=2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]}+A^{\dagger a}{}_{c[\mu}A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^*b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathrm{a}}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathrm{a}}b^a{}_{\nu]}$ $\mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ $\mathcal{H}_{ab}^\dagger=2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu+\tfrac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ $c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^*b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^\dagger b^a{}_{\nu]}$ $c^{\mathrm{a}}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}^{\mathrm{a}}b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu\equiv\tfrac{1}{2}b^c{}_\mu(c^{ab}{}_c+c^b{}_c{}^a-c_c{}^{ab})$ ${}^0A^{*ab}{}_\mu\equiv\tfrac{1}{2}b^c{}_\mu(c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c^{*ab})$ ${}^0A^{\dagger ab}{}_\mu\equiv\tfrac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L}\equiv h^{-1}L$ $t^a{}_b\equiv t^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ $s_{ab}{}^c\equiv s_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ $j^a\equiv j^a{}_\mu b^b{}_\mu\equiv b^b{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ $\hat{t}^{\dagger a}{}_b\equiv t^a{}_b+2(s_{cb}{}^a\mathcal{V}^c-s^{ac}{}_c\mathcal{V}_b)$ $\hat{j}^{\dagger a}\equiv j^a-2s^{ab}{}_b$ $\tau^a{}_b\equiv\tau^a{}_\mu h_b{}^\mu\equiv h_b{}^\mu\delta\mathcal{L}_\mathrm{M}/\delta h_a{}^\mu$ $\sigma_{ab}{}^c\equiv\sigma_{ab}{}^\mu b^c{}_\mu\equiv b^c{}_\mu\delta\mathcal{L}_\mathrm{M}/\delta A^{ab}{}_\mu$ $\zeta^a\equiv\zeta^\mu b^a{}_\mu\equiv b^a{}_\mu\delta\mathcal{L}_\mathrm{M}/\delta B_\mu$ $\tau^{\mathrm{a}}{}_b\equiv\tau^a{}_b+2(\sigma_{cb}{}^a\mathcal{V}^c-\sigma^{ac}{}_c\mathcal{V}_b)$ $\zeta^{\mathrm{a}}\equiv\zeta^a-2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc}\equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc}\equiv 0$) eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc}\equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01015 [conf 0.000]	55	0.000	$\tau^{\dagger a}{}_{\{b\}}{}_{\{c\}} \equiv \tau^a{}_{\{a\}}{}_{\{b\}} + 2 \left(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_{\{c\}}{}_{\{b\}} \mathcal{V}_{\{b\}} \right)$	$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2 \left(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b \right)$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \tfrac{1}{3} T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0A^{*ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0A^{\dagger ab}{}_\mu \equiv \tfrac{1}{2} b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $\hat{j}^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}_\mu b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*ab}{}_{bc} \equiv 0$) eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger ab}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01016 [conf 0.000]	55	0.000	$\backslash\mathrm{zeta}^{\wedge}\{\backslash\mathrm{dagger}\ a\}\ \backslash\mathrm{equiv}\ \backslash\mathrm{zeta}^{\wedge}\{a\}\!-\!2\ s^{\wedge}\{a\ b\}\{ \}_{\!-\}\{b\}$	$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\perp \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ $\mathcal{D}_a^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu^\perp + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\perp \equiv h_a{}^\mu D_\mu^\perp \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\perp a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\perp b^a{}_{\nu]}$ $\mathcal{T}^{\mathcal{A}a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathcal{A}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\perp = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\perp a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\perp b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^a{}_b)$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*a}{}_b)$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger a}{}_b)$ Currents and related quantities $\mathcal{L} \equiv h^{-1}L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\perp$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) eWGT reduced $A^\perp$-field (when $\mathcal{T}^{\perp a}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F01017 [conf 0.000]	55	0.000	c	c	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^b]{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\nu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3}T_{\nu]})$ Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^\dagger b^a{}_{\nu]}$ $c^{\mathfrak{a}}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]}$ ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$ ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L} \equiv h^{-1}L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a b^b{}_\mu \equiv b^b{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $\hat{t}^{\dagger a}{}_b \equiv \hat{t}^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^\dagger$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F01018 [conf 0.000]	55	0.000	$\{ \}^{\text{c}}$	c	Gravitational gauge field variables $h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ Derivative operators $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^a \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_\mu^a \equiv h_a{}^\mu D_\mu^a \equiv h_a{}^\mu (\partial_\mu^a + \frac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^{\dagger a} \equiv \partial_\mu - w(V_\mu + \frac{1}{2} T_\mu)$ $\mathcal{D}_\mu^{\dagger a} \equiv h_a{}^\mu D_\mu^{\dagger a} \equiv h_a{}^\mu (\partial_\mu^{\dagger a} + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_\mu^a \equiv h_a{}^\mu D_\mu^a \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ Gauge field strengths $\mathcal{R}_{cd}^{ab} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_\mu A^{ab}{}_\nu - A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_\mu A^{\dagger ab}{}_\nu + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\dagger a} b^a{}_{\nu]}$ $\mathcal{T}^{ab}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^a b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^{\dagger} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{2} T_{\nu]})$ Reduced A-fields and related quantities $c^{ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$ $c^{*ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$ $c^{\dagger ab}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger a} b^a{}_{\nu]}$ ${}^0 A^{ab}{}_\mu \equiv \frac{1}{2} b^a{}_\mu (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}{}_\mu \equiv \frac{1}{2} b^a{}_\mu (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$ ${}^0 A^{\dagger ab}{}_\mu \equiv \frac{1}{2} b^a{}_\mu (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ Currents and related quantities $\mathcal{L}$ $\mathcal{L} \equiv h^{-1} L$ $t^a{}_\mu \equiv t^a{}_\mu h_a{}^\mu \equiv h_a{}^\mu \delta \mathcal{L}_G / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$ $j^a \equiv j^a{}^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_G / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac} \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_M / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta \mathcal{L}_M / \delta A^{ab}{}_\mu$ $\zeta^a \equiv \zeta^a{}^\mu b^a{}_\mu \equiv b^a{}_\mu \delta \mathcal{L}_M / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac} \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ translational gauge field (h-field) determinant of $h_a{}^\mu$ inverse h-field rotational gauge field (A-field) WGT dilational gauge field (B-field) eWGT dilational gauge field (V-field) eWGT extended rotational gauge field ($A^{\dagger}$-field) PGT (generalised) covariant derivative^a WGT 'augmented' partial derivative WGT (generalised) covariant derivative^a eWGT 'augmented' partial derivative eWGT (generalised) covariant derivative^a eWGT (generalised) semi-covariant derivative^a PGT and WGT rotational gauge field strength^{a, b} eWGT rotational gauge field strength^{a, b} PGT translational gauge field strength^b WGT translational gauge field strength^b eWGT translational gauge field strength^b eWGT translational gauge 'semi' field strength^{a, b} WGT dilational gauge field strength eWGT dilational gauge field strength PGT Ricci rotation coefficients WGT Ricci rotation coefficients eWGT Ricci rotation coefficients PGT reduced A-field (when $\mathcal{T}^{ab}{}_{bc} \equiv 0$) WGT reduced A-field (when $\mathcal{T}^{*ab}{}_{bc} \equiv 0$) eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger ab}{}_{bc} \equiv 0$) Lagrangian Lagrangian density gravitational sector energy-momentum tensor^c gravitational sector spin-angular-momentum tensor^c gravitational sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c matter sector energy-momentum tensor^c matter sector spin-angular-momentum tensor^c matter sector dilation current^c gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c
1510.06699_F01019	55	0.999	$\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}, \mathcal{R} \equiv \mathcal{R}^a{}_a$		^b Contractions are defined by $\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}, \mathcal{R} \equiv \mathcal{R}^a{}_a$ and $\mathcal{T}_a = \mathcal{T}^b{}_{ab}$, and similarly for other versions.
1510.06699_F01020	55	0.999	$\mathcal{T}_a = \mathcal{T}^b{}_{ab}$		^b Contractions are defined by $\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}, \mathcal{R} \equiv \mathcal{R}^a{}_a$ and $\mathcal{T}_a = \mathcal{T}^b{}_{ab}$, and similarly for other versions.
1510.06699_F01021	56	1.000	$SL(2, C)$	$SL(2, C)$	Lorentz transformations by Λ , rather than $SL(2, C)$ elements.
1510.06699_F01022	56	1.000	$h^{\mu}{}_a J_{\mu} \neq \mathcal{J}_a$	$h^{\mu}{}_a J_{\mu} \neq \mathcal{J}_a$	$\mathcal{J}$, so that, for example, $h^{\mu}{}_a J_{\mu} \neq \mathcal{J}_a$. We will not, however,
1510.06699_F01023	56	1.000	$\{ \}^{\{0\}} \mathcal{A}_{abc}$	${}^0 \mathcal{A}_{abc}$	${}^0 \mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0 \mathcal{A}^{*ab}{}_c =$
1510.06699_F01024	56	1.000	$\mathcal{K}_{abc}$	$\mathcal{K}_{abc}$	${}^0 \mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0 \mathcal{A}^{*ab}{}_c =$
1510.06699_F01025	56	1.000	$\{ \}^{\{0\}} \mathcal{A}^{*ab}{}_c =$	${}^0 \mathcal{A}^{*ab}{}_c =$	${}^0 \mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0 \mathcal{A}^{*ab}{}_c =$
1510.06699_F01026	56	1.000	$\{ \}^{\{0\}} \mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$	${}^0 \mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$	${}^0 \mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$ and $\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$, so that the
1510.06699_F01027	56	1.000	$\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$	$\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$	${}^0 \mathcal{A}^{ab}{}_c + \delta_c^a \mathcal{B}^b - \delta_c^b \mathcal{B}^a$ and $\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c - \delta_c^a \mathcal{B}^b + \delta_c^b \mathcal{B}^a$, so that the
1510.06699_F01028	56	1.000	$t^a{}_\mu \equiv h \delta \mathcal{L}_G / \delta h_a{}^\mu, s_{ab}{}^\mu \equiv$	$t^a{}_\mu \equiv h \delta \mathcal{L}_G / \delta h_a{}^\mu, s_{ab}{}^\mu \equiv$	⁷² One could alternatively define $t^a{}_\mu \equiv h \delta \mathcal{L}_G / \delta h_a{}^\mu$, $s_{ab}{}^\mu \equiv$
1510.06699_F01029	56	1.000	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$, and $j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$, and similarly for the mat-
1510.06699_F01030	56	1.000	$j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$	$j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$	$h \delta \mathcal{L}_G / \delta A^{ab}{}_\mu$, and $j^\mu \equiv h \delta \mathcal{L}_G / \delta B_\mu$, and similarly for the mat-
1510.06699_F01031	56	1.000	$\dot{R} \equiv$	$\dot{R} \equiv$	point particle one must, in addition, make the replacement $\dot{R} \equiv$
1510.06699_F01032	56	1.000	$\dot{x}^\mu \partial_\mu R \rightarrow \dot{v}^a \mathcal{D}_a^* R$	$\dot{x}^\mu \partial_\mu R \rightarrow \dot{v}^a \mathcal{D}_a^* R$	$\dot{x}^\mu \partial_\mu R \rightarrow \dot{v}^a \mathcal{D}_a^* R$.
1510.06699_F01033	56	0.994	$\{ \}^{\{77,78\}}$	77,78	theory ^{77,78} . In the standard interpretation, the scalar field in
1510.06699_F01034	56	0.999	$\bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a^* \psi = \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi$	$\bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a^* \psi = \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi$	⁸¹ It is easily shown that $\bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a^* \psi = \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi$ (which does not
1510.06699_F01035	56	1.000	$\theta(x)$	$\theta(x)$	a field $\theta(x)$. Provided it has Weyl weight $w = 0$, one would ob-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F01036	57	■ 1.000	$D_{\mu}^{\dagger} \varphi =$	$D_{\mu}^{\dagger} \varphi =$	ticular, we note that the eWGT covariant derivative $D_{\mu}^{\dagger} \varphi =$
1510.06699_F01037	57	■ 1.000	$\left[D_{\mu} + \frac{1}{2} (\mathcal{V}^a b^b{}_{\mu} - \mathcal{V}^b b^a{}_{\mu}) \Sigma_{ab} - w V_{\mu} - \frac{1}{3} w T_{\mu}\right] \varphi$	$\left[D_{\mu} + \frac{1}{2} (\mathcal{V}^a b^b{}_{\mu} - \mathcal{V}^b b^a{}_{\mu}) \Sigma_{ab} - w V_{\mu} - \frac{1}{3} w T_{\mu}\right] \varphi$	$[D_{\mu} + \frac{1}{2} (\mathcal{V}^a b^b{}_{\mu} - \mathcal{V}^b b^a{}_{\mu}) \Sigma_{ab} - w V_{\mu} - \frac{1}{3} w T_{\mu}] \varphi$ represents a far
1510.06699_F01038	57	■ 1.000	$D_{\mu}^{\ast} \varphi = \left(D_{\mu} + w B_{\mu}\right) \varphi$	$D_{\mu}^{\ast} \varphi = (D_{\mu} + w B_{\mu}) \varphi$	derivative $D_{\mu}^{\ast} \varphi = (D_{\mu} + w B_{\mu}) \varphi$
1510.06699_F01039	57	■ 0.995	$\{ \}^{\{0\}} \mathcal{M}^{\{A\} \dagger} \{a\} \{b\} \{ \}_c = \{ \}^{\{0\}} \mathcal{M}^{\{A\} \dagger} \{a\} \{b\} \{ \}_c - \delta_{\{a\} \{b\}} \left(\mathcal{M}^{\{V\} \dagger} \{b\} + \frac{1}{3} \mathcal{M}^{\{T\} \dagger} \{b\} \right) + \delta_{\{c\} \{b\}} \left(\mathcal{M}^{\{V\} \dagger} \{a\} + \frac{1}{3} \mathcal{M}^{\{T\} \dagger} \{a\} \right)$	${}^0 \mathcal{A}^{\dagger ab}{}_c = {}^0 \mathcal{A}^{ab}{}_c - \delta_c^a \left(\mathcal{V}^b + \frac{1}{3} \mathcal{T}^b \right) + \delta_c^b (\mathcal{V}^a +$	pendix B), one has ${}^0 \mathcal{A}^{\dagger ab}{}_c = {}^0 \mathcal{A}^{ab}{}_c - \delta_c^a (\mathcal{V}^b + \frac{1}{3} \mathcal{T}^b) + \delta_c^b (\mathcal{V}^a +$
1510.06699_F01040	57	■ 0.600	$\frac{1}{3} \mathcal{T}^a$	$\frac{1}{3} \mathcal{T}^a$	$\frac{1}{3} \mathcal{T}^a$) and $\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$, so confirming that
1510.06699_F01041	57	■ 0.600	$\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$	$\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$	$\frac{1}{3} \mathcal{T}^a$) and $\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3} \delta_c^a \mathcal{T}^b - \frac{1}{3} \delta_c^b \mathcal{T}^a$, so confirming that
1510.06699_F01042	57	■ 0.965	$\bar{\psi} \gamma^a \mathcal{D}_a^{\dagger} \psi = \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi$	$\bar{\psi} \gamma^a \mathcal{D}_a^{\dagger} \psi = \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi$	⁹⁴ It is easily shown that $\bar{\psi} \gamma^a \mathcal{D}_a^{\dagger} \psi = \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi$ (which does not
1510.06699_F01043	57	■ 1.000	$\nabla_{\sigma} g_{\mu \nu} =$	$\nabla_{\sigma} g_{\mu \nu} =$	that given in (124) for WGT can both be written as $\nabla_{\sigma} g_{\mu \nu} =$
1510.06699_F01044	57	■ 1.000	$-\frac{1}{2} (\Gamma^{\lambda}{}_{\lambda \sigma} - {}^0 \Gamma^{\lambda}{}_{\lambda \sigma}) g_{\mu \nu}$	$-\frac{1}{2} (\Gamma^{\lambda}{}_{\lambda \sigma} - {}^0 \Gamma^{\lambda}{}_{\lambda \sigma}) g_{\mu \nu}$	$-\frac{1}{2} (\Gamma^{\lambda}{}_{\lambda \sigma} - {}^0 \Gamma^{\lambda}{}_{\lambda \sigma}) g_{\mu \nu}$, where ${}^0 \Gamma^{\lambda}{}_{\mu \nu}$ is the metric connection;
1510.06699_F01045	57	■ 1.000	$L_{\mathcal{R}^2} =$	$L_{\mathcal{R}^2} =$	which its symmetries are more manifest, namely $L_{\mathcal{R}^2} =$
1510.06699_F01046	57	■ 1.000	$\alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2' \mathcal{R}_{(ab)}^{\dagger} \mathcal{R}^{\dagger (ab)} + \alpha_3' \mathcal{R}_{[ab]}^{\dagger} \mathcal{R}^{\dagger [ab]} + \alpha_4' \mathcal{R}_{a(bc)d}^{\dagger} \mathcal{R}^{\dagger a(bc)d} +$	$\alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2' \mathcal{R}_{(ab)}^{\dagger} \mathcal{R}^{\dagger (ab)} + \alpha_3' \mathcal{R}_{[ab]}^{\dagger} \mathcal{R}^{\dagger [ab]} + \alpha_4' \mathcal{R}_{a(bc)d}^{\dagger} \mathcal{R}^{\dagger a(bc)d} +$	$\alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2' \mathcal{R}_{(ab)}^{\dagger} \mathcal{R}^{\dagger (ab)} + \alpha_3' \mathcal{R}_{[ab]}^{\dagger} \mathcal{R}^{\dagger [ab]} + \alpha_4' \mathcal{R}_{a(bc)d}^{\dagger} \mathcal{R}^{\dagger a(bc)d} +$
1510.06699_F01047	57	■ 1.000	$\alpha_5' \mathcal{R}_{a[bc]d}^{\dagger} \mathcal{R}^{\dagger a[bc]d} + \alpha_6 \mathcal{R}_{abcd}^{\dagger} \mathcal{R}^{\dagger cdab}$	$\alpha_5' \mathcal{R}_{a[bc]d}^{\dagger} \mathcal{R}^{\dagger a[bc]d} + \alpha_6 \mathcal{R}_{abcd}^{\dagger} \mathcal{R}^{\dagger cdab}$	$\alpha_5' \mathcal{R}_{a[bc]d}^{\dagger} \mathcal{R}^{\dagger a[bc]d} + \alpha_6 \mathcal{R}_{abcd}^{\dagger} \mathcal{R}^{\dagger cdab}$, where the new coefficients
1510.06699_F01048	57	■ 0.994	$\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$	$\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$	are $\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$ and $\alpha_5' = \alpha_4 - \alpha_5$.
1510.06699_F01049	57	■ 0.994	$\alpha_5' = \alpha_4 - \alpha_5$	$\alpha_5' = \alpha_4 - \alpha_5$	are $\alpha_2' = \alpha_2 + \alpha_3, \alpha_3' = \alpha_2 - \alpha_3, \alpha_4' = \alpha_4 + \alpha_5$ and $\alpha_5' = \alpha_4 - \alpha_5$.
1510.06699_F01050	57	■ 1.000	$\alpha_2' = \alpha_3'$	$\alpha_2' = \alpha_3'$	allows one to set α_1 or α_6 to zero as before, or set $\alpha_2' = \alpha_3'$.
1510.06699_F01051	57	■ 0.999	$L_{\mathcal{T}^2} = \beta_1' \mathcal{T}_{(ab)c}^{\dagger} \mathcal{T}^{\dagger (ab)c} + \beta_2' \mathcal{T}_{[ab]c}^{\dagger} \mathcal{T}^{\dagger [ab]c}$	$L_{\mathcal{T}^2} = \beta_1' \mathcal{T}_{(ab)c}^{\dagger} \mathcal{T}^{\dagger (ab)c} + \beta_2' \mathcal{T}_{[ab]c}^{\dagger} \mathcal{T}^{\dagger [ab]c}$	symmetric form $L_{\mathcal{T}^2} = \beta_1' \mathcal{T}_{(ab)c}^{\dagger} \mathcal{T}^{\dagger (ab)c} + \beta_2' \mathcal{T}_{[ab]c}^{\dagger} \mathcal{T}^{\dagger [ab]c}$ where
1510.06699_F01052	57	■ 0.998	$\beta_1' = \beta_1 + \beta_2$	$\beta_1' = \beta_1 + \beta_2$	$\beta_1' = \beta_1 + \beta_2$ and $\beta_2' = \beta_1 - \beta_2$.
1510.06699_F01053	57	■ 0.998	$\beta_2' = \beta_1 - \beta_2$	$\beta_2' = \beta_1 - \beta_2$	$\beta_1' = \beta_1 + \beta_2$ and $\beta_2' = \beta_1 - \beta_2$.
1510.06699_F01054	57	■ 1.000	$S_{\mathrm{T}} = S_{\mathrm{G}} + S_{\mathrm{M}}$	$S_{\mathrm{T}} = S_{\mathrm{G}} + S_{\mathrm{M}}$	lations, but the total action $S_{\mathrm{T}} \equiv S_{\mathrm{G}} + S_{\mathrm{M}}$ is so. We will not
1510.06699_F01055	57	■ 0.999	${}^0 \mathcal{R}^{*ab}{}_{cd} = {}^0 \mathcal{R}^{ab}{}_{cd} - 2 \delta_d^{[b} ({}^0 \mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^{a]} +$	${}^0 \mathcal{R}^{*ab}{}_{cd} = {}^0 \mathcal{R}^{ab}{}_{cd} - 2 \delta_d^{[b} ({}^0 \mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^{a]} +$	quantity ${}^0 \mathcal{R}^{*ab}{}_{cd}$ by ${}^0 \mathcal{R}^{*ab}{}_{cd} = {}^0 \mathcal{R}^{ab}{}_{cd} - 2 \delta_d^{[b} ({}^0 \mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^{a]} +$
1510.06699_F01056	57	■ 1.000	$2 \delta_c^{[b} ({}^0 \mathcal{D}_d - \mathcal{B}_d) \mathcal{B}^{a]} - 2 \mathcal{B}^e \mathcal{B}_e \delta_c^{[a} \delta_d^{b]}$	$2 \delta_c^{[b} ({}^0 \mathcal{D}_d - \mathcal{B}_d) \mathcal{B}^{a]} - 2 \mathcal{B}^e \mathcal{B}_e \delta_c^{[a} \delta_d^{b]}$	$2 \delta_c^{[b} ({}^0 \mathcal{D}_d - \mathcal{B}_d) \mathcal{B}^{a]} - 2 \mathcal{B}^e \mathcal{B}_e \delta_c^{[a} \delta_d^{b]}$, from which one finds that their
1510.06699_F01057	57	■ 1.000	${}^0 \mathcal{R}^{*a}{}_c = {}^0 \mathcal{R}^a{}_c - 2 ({}^0 \mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^a -$	${}^0 \mathcal{R}^{*a}{}_c = {}^0 \mathcal{R}^a{}_c - 2 ({}^0 \mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^a -$	contractions are related by ${}^0 \mathcal{R}^{*a}{}_c = {}^0 \mathcal{R}^a{}_c - 2 ({}^0 \mathcal{D}_c - \mathcal{B}_c) \mathcal{B}^a -$
1510.06699_F01058	57	■ 1.000	$({}^0 \mathcal{D}_e + 2 \mathcal{B}_e) \mathcal{B}^e \delta_c^a$	$({}^0 \mathcal{D}_e + 2 \mathcal{B}_e) \mathcal{B}^e \delta_c^a$	$({}^0 \mathcal{D}_e + 2 \mathcal{B}_e) \mathcal{B}^e \delta_c^a$ and ${}^0 \mathcal{R}^* = {}^0 \mathcal{R} - 6 ({}^0 \mathcal{D}_e + \mathcal{B}_e) \mathcal{B}^e$. Similarly,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F01059	57	■ 1.000	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}^{\{*\}} = \{ \}^{\{0\}} \backslash \mathrm{mathcal{R}} - 6 \backslash \mathrm{left}(\{ \}^{\{0\}} \backslash \mathrm{mathcal{D}}_{\{e\}} + \backslash \mathrm{mathcal{B}}_{\{e\}} \backslash \mathrm{right}) \backslash \mathrm{mathcal{B}}^{\{e\}}$	${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6 \left({}^0\mathcal{D}_e + \mathcal{B}_e \right) \mathcal{B}^e$	$({}^0\mathcal{D}_e + 2\mathcal{B}_e)\mathcal{B}^e\delta_c^a$ and ${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6({}^0\mathcal{D}_e + \mathcal{B}_e)\mathcal{B}^e$. Similarly,
1510.06699_F01060	57	■ 0.996	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}^{\{* \text{ a b} \} \{ \}_{\{c \text{ d}\}} \backslash \mathrm{rightarrow} \{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}^{\{\dagger \text{ a b} \} \{ \}_{\{c \text{ d}\}}}$	${}^0\mathcal{R}^{*ab}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}_{cd}$	placements ${}^0\mathcal{R}^{*ab}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}_{cd}$ and $\mathcal{B}_a \rightarrow -(\mathcal{V}_a + \frac{1}{3}\mathcal{T}_a)$.
1510.06699_F01061	57	■ 0.996	$\backslash \mathrm{mathcal{B}}_{\{a\}} \backslash \mathrm{rightarrow} - \backslash \mathrm{left}(\backslash \mathrm{mathcal{V}}_{\{a\}} + \frac{1}{3} \backslash \mathrm{mathcal{T}}_{\{a\}} \backslash \mathrm{right})$	$\mathcal{B}_a \rightarrow - \left(\mathcal{V}_a + \frac{1}{3} \mathcal{T}_a \right)$	placements ${}^0\mathcal{R}^{*ab}_{cd} \rightarrow {}^0\mathcal{R}^{\dagger ab}_{cd}$ and $\mathcal{B}_a \rightarrow -(\mathcal{V}_a + \frac{1}{3}\mathcal{T}_a)$.
1510.06699_F01062	57	■ 1.000	$\{ \}^{\{0\}} \backslash \mathrm{mathcal{R}}^{\{*\}}$	${}^0\mathcal{R}^*$	‘reduced’ versions ${}^0\mathcal{D}_a^*$ and ${}^0\mathcal{R}^*$, respectively, as described in